Science Monk · Presents

NEET Physics

Previous Year Question Papers

2016 – 2026 · Eleven Years · 515 Questions

> *The important thing*
> *is not to stop questioning*
>
> — Albert Einstein

every question retyped *every diagram redrawn*



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Founder & Physics Mentor

Science Monk

"Appa Deepo Bhava"

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About this book. The book holds the Physics section of the main NEET (UG) paper of each year from 2016 to 2026: 45 questions (2016–2020, 2025, 2026) or 50 questions (2021–2024, Section A 35 and Section B 15, of which any 10 were to be attempted), four marks each with one mark deducted for a wrong answer. For 2026 the re-examination of 21 June is used, because the 3 May examination was cancelled. Questions keep their original numbers and option labels, so they match the official answer key of the test booklet code named above. Every question has been retyped and every diagram redrawn from the English version of the paper. Every question was checked one by one against the four 2026-27 NCERT books (Class XI Part 1 & 2, Class XII Part 1 & 2) and carries a tag: Direct (answer stated in the textbook, an NCERT solved Example or Exercise, or a single printed formula), Applied (NCERT concept used in several steps or a new situation), or Deleted Portion (topic removed from the rationalised syllabus; not analysed). Direct and Applied tags give the NCERT page with a tappable link to that chapter's official PDF. Each paper is followed by its answer key (the official NTA final key for 2021, 2022, 2023, 2025 and 2026, checked by the editor; solved by the editor for the other years) and a one-page analysis. The book ends with an eleven-year summary, the 15 most repeated topics and a preparation plan.

Prepared by Sumeet Birajadar for Science Monk online classes.

NEET 2026
Physics Question Paper

Re-Examination · 21 June 2026 · Test Booklet Code 50 · Physics Q 1–45 · 180 Marks

The NEET (UG) 2026 examination of 3 May 2026 was cancelled by NTA; the result was declared on the re-examination of 21 June 2026, given here.

1. A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is:

APPLIED — Class XI Part 1 · Ch 5 · **Page 75** · work by variable force (area under F-x) Open PDF ↗

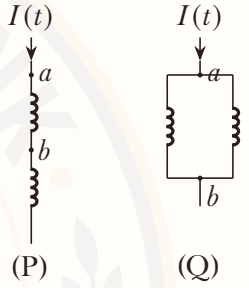
(1) 3 **(2)** 1 **(3)** $\frac{1}{3}$ **(4)** $\frac{1}{2}$

2. The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, the correct option is:

DIRECT — Class XI Part 2 · Ch 12 · **Page 254** · l=1/(sqrt2 n pi d^2) Open PDF ↗

(1) $n_A = n_B$ **(2)** $n_A = 2n_B$
(3) $n_A = \frac{1}{4}n_B$ **(4)** $n_A = \frac{1}{2}n_B$

3. Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is: [Neglect the effect of mutual inductance.]



APPLIED — Class XII Part 1 · Ch 6 · **Page 168** Open PDF ↗

(1) $\frac{1}{4}$ **(2)** $\frac{1}{2}$ **(3)** 1 **(4)** 2

4. For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is

APPLIED — Class XI Part 2 · Ch 14 · **Page 292** Open PDF ↗

(1) $\frac{5}{9}$ **(2)** $\frac{9}{5}$ **(3)** 1 **(4)** $\frac{3}{5}$

5. Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is:

DIRECT — Class XII Part 1 · Ch 6 · **Page 168** · L=mu0 n^2 A l Open PDF ↗

(1) $\mu_0\pi n^2 r^2 l$ **(2)** $\mu_0 n^2 r^2 l$
(3) $(\mu_0/2\pi)n^2 r^2 l$ **(4)** $2\mu_0\pi n^2 r^2 l$

6. Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1\,\text{m s}^{-2}$, $\beta = 6\,\text{m s}^{-1}$ and $\gamma = 5\,\text{m}$. The average speed of the particle, in m s^{-1} from $t = 0$ to $t = 6\,\text{s}$ is:

APPLIED Class XI Part 1 · Ch 2 · **Page 21** Open PDF ↗

(1) 12 **(2)** 6 **(3)** 3 **(4)** 0

7. Consider the following nuclear reaction

$$^{238}\text{U} \longrightarrow {}^{234}\text{Th} + {}^{4}\text{He}$$

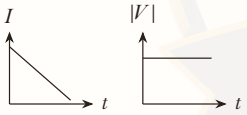
Take masses of ^{238}U, ^{234}Th and ^{4}He as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is: [Given: 1 u = 931.5 MeV c^{-2}]

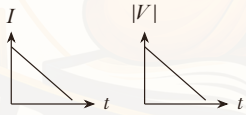
DIRECT Class XII Part 2 · Ch 13 · **Page 319** · Q=(sum mi - sum mf)c^2 Open PDF ↗

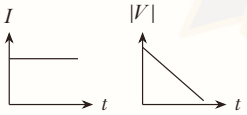
(1) 3726 **(2)** 3730 **(3)** 3736 **(4)** 3740

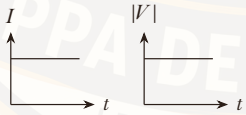
8. A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by:

DIRECT Class XII Part 2 · Ch 11 · **Page 278** · Figs 11.2, 11.3 Open PDF ↗

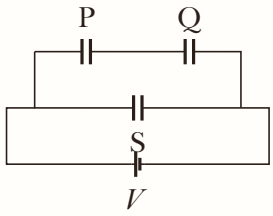
(1) 

(2) 

(3) 

(4) 

9. Three identical capacitors, P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\dfrac{U_P}{U_T}$ is:



APPLIED Class XII Part 1 · Ch 2 · **Page 72** Open PDF ↗

(1) 2/3 **(2)** 1/3 **(3)** 1/2 **(4)** 1/6

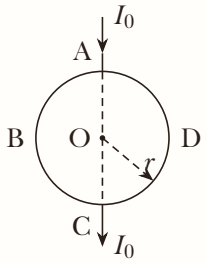
10. Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is $15\,\text{N m}^{-2}$. The area of cross-section at P and Q are $40\,\text{cm}^2$ and $20\,\text{cm}^2$, respectively. The rate of flow of water through the pipe, in $\text{cm}^3\,\text{s}^{-1}$, is: [Take density of water = $1000\,\text{kg m}^{-3}$]



APPLIED Class XI Part 2 · Ch 9 · **Page 188** Open PDF ↗

(1) 100 **(2)** 200 **(3)** 300 **(4)** 400

11. A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the centre O of the loop is:

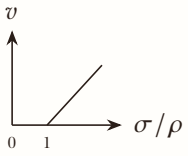


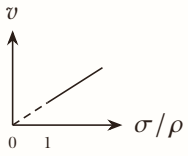
APPLIED Class XII Part 1 · Ch 4 · **Page 116** Open PDF ↗

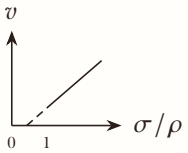
(1) $\dfrac{\mu_0 I_0}{12r}$ **(2)** $\dfrac{\mu_0 I_0}{4r}$ **(3)** $\dfrac{\mu_0 I_0}{2r}$ **(4)** $\dfrac{\mu_0 I_0}{2\pi r}$

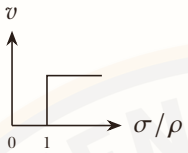
12. In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by:

DIRECT · Class XI Part 2 · Ch 9 · **Page 192** · vt=2a^2(rho-sigma)g/9eta Open PDF ↗

(1) 

(2) 

(3) 

(4) 

13. Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2 = \dfrac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\dfrac{v_2}{v_1}$ is:

DIRECT · Class XI Part 1 · Ch 7 · **Page 136** · v=sqrt(2GM/R) Open PDF ↗

(1) $\dfrac{1}{\sqrt{2}}$ (2) 1 (3) $\sqrt{2}$ (4) 2

14. In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to:

DIRECT · Class XI Part 1 · Ch 7 · **Page 129** · T^2 proportional to a^3 Open PDF ↗

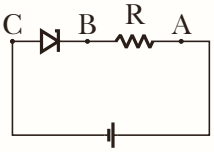
(1) $R^{1/2}$ (2) $R^{3/2}$ (3) R^2 (4) R^3

15. Two infinitely long parallel conducting wires A and B carry currents I and $2I$, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is: [g is the acceleration due to gravity and μ_0 is the permeability of free space.]

APPLIED · Class XII Part 1 · Ch 4 · **Page 123** Open PDF ↗

(1) $\dfrac{\mu_0 I^2}{2\pi\lambda g}$ (2) $\dfrac{\mu_0 I^2}{\pi\lambda g}$
(3) $\dfrac{2\mu_0 I^2}{\pi\lambda g}$ (4) $\dfrac{4\mu_0 I^2}{\pi\lambda g}$

16. An ideal Zener diode with breakdown voltage of -3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between point B and A is:



DELETED PORTION · Zener diode · removed from the 2026-27 NCERT; not analysed

(1) 3 V (2) 2 V (3) 1 V (4) 0 V

17. In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60 K to 50 K. The work done by the gas in the process is: (Take the universal gas constant as $R = 8.3\,\text{J mol}^{-1}\text{K}^{-1}$)

DIRECT · Class XI Part 2 · Ch 11 · **Page 235** · W=muR(T1-T2)/(gamma-1) Open PDF ↗

(1) 41.5 J (2) 83 J
(3) 124.5 J (4) 166 J

18. A ray of light with wavelength λ is incident on three different photoelectric cells namely 1, 2 and 3. The threshold wavelength of these photoelectric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ and $\lambda_3 \gg \lambda$. The correct option is:

APPLIED · Class XII Part 2 · Ch 11 · **Page 281** Open PDF ↗

(1) $V_1 = 0, V_2 < V_3$ (2) $V_1 = 0, V_2 > V_3$
(3) $V_1 > V_2, V_3 = 0$ (4) $V_1 < V_2, V_3 = 0$

19. A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\dfrac{p_e}{p_{Ph}}$ is: [Take speed of light = $3 \times 10^8\,\text{m s}^{-1}$, charge of electron = -1.6×10^{-19} C and mass of electron = 9×10^{-31} kg]

APPLIED · Class XII Part 2 · Ch 11 · **Page 283** Open PDF ↗

(1) $\dfrac{2}{450}$ (2) $\dfrac{1}{250}$ (3) 225 (4) 275

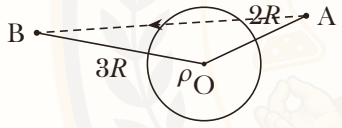
20. Which of the following measurements require 'index correction'?

APPLIED

Class XI Part 1 · Ch 1 · **Page 6** · Lab manual: index correction on the optical bench Open PDF ↗

(1) Measurement of resistance of a wire using meter bridge
(2) Measurement of gravitational acceleration using simple pendulum
(3) Measurement of focal length of lenses using optical bench
(4) Measurement of speed of sound using resonance tube

21. A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distance $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\varepsilon_0}$. The value of n is: (ε_0 is the permittivity of vacuum)

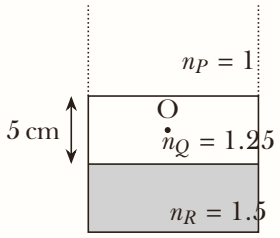


APPLIED

Class XII Part 1 · Ch 2 · **Page 52** Open PDF ↗

(1) 2 **(2)** 6 **(3)** 9 **(4)** 18

22. Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5 respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the centre of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm, is:

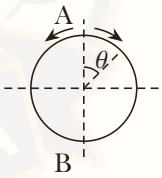


APPLIED

Class XII Part 2 · Ch 9 · **Page 229** Open PDF ↗

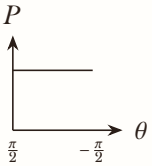
(1) 0 **(2)** 1 **(3)** 2 **(4)** 3

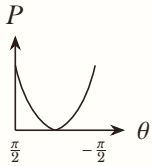
23. A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point A ($\theta = \frac{\pi}{2}$) with identical uniform angular speeds in opposite directions, and meet again at point B ($\theta = -\frac{\pi}{2}$). During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ?

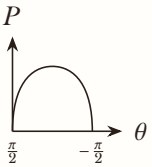


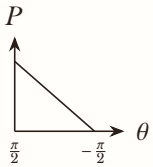
APPLIED

Class XI Part 1 · Ch 4 · **Page 53** Open PDF ↗

(1)


(2)


(3)


(4)


24. The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is:

DIRECT — Class XI Part 2 · Ch 10 · **Page 207** · alpha_V=3 alpha_1 Open PDF ↗

(1) $2\pi R^3\alpha\Delta T$ **(2)** $3\pi R^3\alpha\Delta T$
(3) $4\pi R^3\alpha\Delta T$ **(4)** $6\pi R^3\alpha\Delta T$

25. A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period $2T$. The value of ρ_2/ρ_1 is:

DIRECT — Class XI Part 2 · Ch 13 · **Page 277** · Exercise 13.17 Open PDF ↗

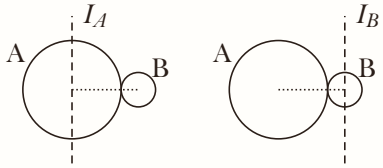
(1) 4 **(2)** 2 **(3)** 1/2 **(4)** 1/4

26. One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of the wire to be 1 cm, the actual length of the wire is:

APPLIED — Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: vernier zero error Open PDF ↗

(1) 0.60 cm **(2)** 0.96 cm
(3) 1.00 cm **(4)** 1.04 cm

27. A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is:



DELETED PORTION — parallel axis theorem · removed from the 2026-27 NCERT; not analysed

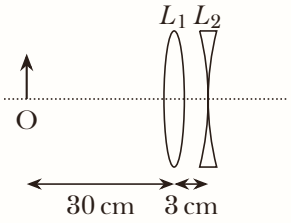
(1) $(M-m)(R+r)^2$ **(2)** $(m-M)(R+r)^2$
(3) $(m-M)(R-r)^2$ **(4)** 0

28. Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is k N m^{-1}. At a given instant, the extension of the spring is x metre and the speed of the particle is v m s^{-1}. On the x–v plane, if the graph of v as a function of x is a circle, then the correct option is:

APPLIED — Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $k = \dfrac{1}{m}$ **(2)** $k = m$
(3) $k = m^2$ **(4)** $k = \sqrt{m}$

29. The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and −10 cm, respectively. The position of the image formed is:



APPLIED — Class XII Part 2 · Ch 9 · **Page 238** Open PDF ↗

(1) 20 cm to the left of the concave lens
(2) 60 cm to the left of the concave lens
(3) 30 cm to the right of the concave lens
(4) 60 cm to the right of the concave lens

30. An ac voltage $V = 220\sin(2\times10^3 t)$ volt is applied to a series LCR circuit. Then the current amplitude in this circuit is: (Given: $L = 10$ mH, $C = 25\ \mu$F, $R = 100\ \Omega$)

APPLIED Class XII Part 1 · Ch 7 · **Page 188** Open PDF ↗

(1) 2.2 A **(2)** 5.5 A
(3) 11.0 A **(4)** 22.0 A

31. A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2\times OA$. The value of $\frac{L_A}{L_B}$ is:

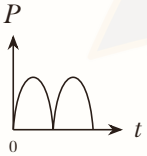


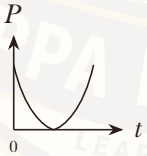
APPLIED Class XI Part 1 · Ch 6 · **Page 108** Open PDF ↗

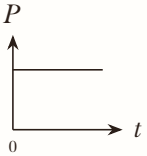
(1) $\frac{1}{4}$ **(2)** $\frac{1}{2}$ **(3)** 1 **(4)** 2

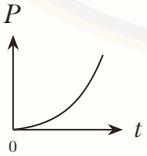
32. A conducting loop of finite resistance lies on the x–y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1+\sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is:

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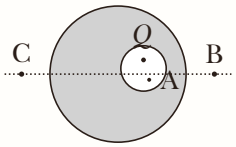
(1) 

(2) 

(3) 

(4) 

33. A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B, E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is:



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(1) $E_A = 0, E_B = E_C$ **(2)** $E_A \neq 0, E_B = E_C$
(3) $E_A = 0, E_B > E_C$ **(4)** $E_A \neq 0, E_B < E_C$

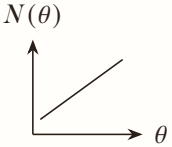
34. Consider a fixed uniformly charged insulating sphere with radius R and total charge $+Q$. A point charge $-q$ ($q \ll Q$) with mass m is released from rest at a distance of $3R$ from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is: (ε_0 is the permittivity of vacuum, neglect gravitational forces).

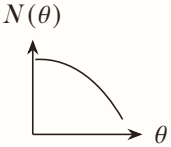
APPLIED Class XII Part 1 · Ch 2 · **Page 55** Open PDF ↗

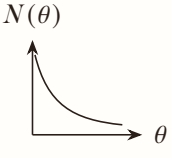
(1) $\sqrt{\frac{3Qq}{4\pi\varepsilon_0 mR}}$ **(2)** $\sqrt{\frac{2Qq}{3\pi\varepsilon_0 mR}}$
(3) $\sqrt{\frac{Qq}{3\pi\varepsilon_0 mR}}$ **(4)** $\sqrt{\frac{Qq}{4\pi\varepsilon_0 mR}}$

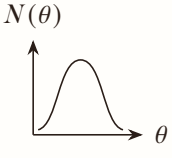
35. In Geiger–Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the correct plot? ($N(\theta)$ on a logarithmic scale, θ from 0° to 180°.)

DIRECT Class XII Part 2 · Ch 12 · **Page 293** · Fig 12.3 Open PDF ↗

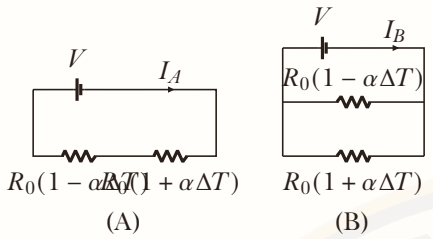
(1) 

(2) 

(3) 

(4) 

36. Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is:



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(1) I_A remains constant while I_B increases
(2) I_A decreases while I_B increases
(3) I_A increases while I_B decreases
(4) Both I_A and I_B remain constant

37. Consider that σ_s, k_B, b represents Stefan–Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is

APPLIED Class XI Part 1 · Ch 1 · **Page 7** Open PDF ↗

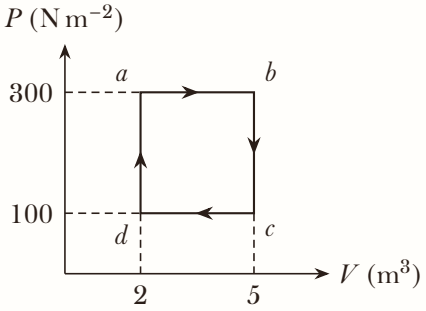
(1) $[L^{-1}T^{-1}K^{-2}]$ **(2)** $[L^{-1}K^{-2}]$
(3) $[L^{-1}T^{-1}K^{-3}]$ **(4)** $[L^{-1}T^{-1}K^{-4}]$

38. An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant $\varepsilon_r = 9$, has the electric field $E_x = E_0 \sin(kz - 2\pi \times 10^6 t)\,\text{V m}^{-1}$ where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is

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(1) The speed of the electromagnetic wave inside the medium is $10^8\,\text{m s}^{-1}$
(2) The wavelength of the electromagnetic wave inside the medium is 300 m
(3) The magnetic field is given by the relation $B_y = \frac{B_0}{v}\sin(kz - 2\pi \times 10^6 t)$ where v is the speed of the electromagnetic wave inside the medium
(4) The direction of propagation of the electromagnetic wave is along $+z$

39. One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is:



DIRECT Class XI Part 2 · Ch 11 · **Page 236** · cyclic process: Q=W=area Open PDF ↗

(1) 400 J **(2)** 500 J **(3)** 600 J **(4)** 800 J

40. Consider that an electron is revolving in an excited state of hydrogen atom with velocity $\sqrt{25.6} \times 10^5\,\text{m s}^{-1}$. The radius of the orbit is $x \times 10^{-9}$ m. The value of x is: [Take the mass of electron to be 9×10^{-31} kg, charge of electron = -1.6×10^{-19} C and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\,\text{N m}^2\text{C}^{-2}$]

DIRECT Class XII Part 2 · Ch 12 · **Page 295** · r=e^2/(4 pi eps0 m v^2) Open PDF ↗

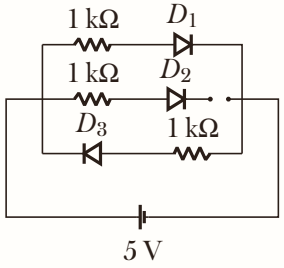
(1) 4 **(2)** 3 **(3)** 2 **(4)** 1

41. A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed $10\,\text{m s}^{-1}$, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be $10\,\text{m s}^{-2}$, the value of θ is:

DIRECT Class XI Part 1 · Ch 4 · **Page 64** · v0=sqrt(Rg tan theta) Open PDF ↗

(1) $\tan^{-1}\left(\frac{1}{5}\right)$ **(2)** $\tan^{-1}\left(\frac{2}{5}\right)$
(3) $\tan^{-1}(\sqrt{3}/2)$ **(4)** $\tan^{-1}(2\sqrt{3})$

42. Three identical p-n junction diodes D_1, D_2, and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is:



APPLIED Class XII Part 2 · Ch 14 · **Page 335** Open PDF ↗

(1) $W_1 > W_2 > W_3$ **(2)** $W_3 = W_1 > W_2$
(3) $W_3 > W_2 > W_1$ **(4)** $W_2 > W_1 = W_3$

43. The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

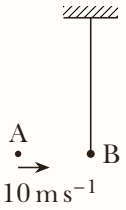
Part of the spectrum	*Application*
P. Microwave	I. For purifying the water
Q. UV rays	II. For warming the food
R. Gamma rays	III. For AM and FM communication systems
S. Radio wave	IV. For treating the cancer cells

The correct option is:

DIRECT Class XII Part 1 · Ch 8 · **Page 209** · uses listed for each band Open PDF ↗

(1) P-I, Q-II, R-III, S-IV
(2) P-I, Q-IV, R-II, S-III
(3) P-II, Q-I, R-IV, S-III
(4) P-II, Q-IV, R-III, S-I

44. Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed $10\,\text{m s}^{-1}$ hits bob B elastically. The bob B rises h metre after the collision. Taking the acceleration due to gravity $g = 10\,\text{m s}^{-2}$ and neglecting the size of the bob, the value of h is:



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(1) 8 **(2)** 7 **(3)** 5 **(4)** 2.5

45. An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_P/C_V of the gas is 8/7, then the value of f is:

DIRECT Class XI Part 2 · Ch 12 · **Page 253** · Cv=(3+f)R Open PDF ↗

(1) 4 **(2)** 3 **(3)** 2 **(4)** 1

NEET 2026 · Answer Key

1 (4)	**2** (4)	**3** (4)	**4** (3)	**5** (1)	**6** (3)	**7** (1)	**8** (1)	**9** (4)	**10** (4)
11 (1)	**12** (1)	**13** (3)	**14** (2)	**15** (2)	16 (2)†	**17** (3)	**18** (1)	**19** (3)	**20** (3)
21 (4)	**22** (2)	**23** (3)	**24** (3)	**25** (4)	**26** –*	27 (2)†	**28** (2)	**29** (2)	**30** (1)
31 (3)	**32** (2)	**33** (2)	**34** (3)	**35** (3)	**36** (1)	**37** (1)	**38** (2)	**39** (3)	**40** (4)
41 (1)	**42** (3)	**43** (3)	**44** (3)	**45** (1)					

Official NTA final key (Test Booklet Code 50); every answer was also solved and checked by the editor. (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). –: question dropped by NTA (marks given to all). *: see note.

Q26. Dropped by NTA. The zero error is −0.06 cm, so the actual length is 1.06 cm, which is not among the options.

NEET 2026 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 43 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**16**	37%
text, summary, table or figure	15	
NCERT solved Example	0	
NCERT Exercise	1	
Applied (concept in NCERT)	**27**	63%
Analysed	43	
Deleted portion (not analysed)	2	
Total	**45**	

Direct: Q 2, 5, 7, 8, 12, 13, 14, 17, 24, 25, 35, 39, 40, 41, 43, 45.

From NCERT Exercises: Q 25 (Exercise 13.17).

2. Difficulty level

Easy	16
Moderate	27
Hard	0

The re-examination was moderate: most questions need one or two steps from a printed NCERT formula, and many test reading of graphs and figures. Class XI: 21 questions, Class XII: 22 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	3	0	3
Motion in a Straight Line	XI	1	0	1
Laws of Motion	XI	2	1	1
Work, Energy & Power	XI	2	0	2
Rotational Motion	XI	1	0	1
Gravitation	XI	2	2	0
Mech. Prop. of Fluids	XI	2	1	1
Thermal Prop. of Matter	XI	1	1	0
Thermodynamics	XI	2	2	0
Kinetic Theory	XI	2	2	0
Oscillations	XI	2	1	1
Waves	XI	1	0	1
Electrostatic Potential & Capacitance	XII	4	0	4
Current Electricity	XII	1	0	1
Moving Charges & Magnetism	XII	2	0	2
Electromagnetic Induction	XII	3	1	2
Alternating Current	XII	1	0	1
Electromagnetic Waves	XII	2	1	1
Ray Optics	XII	2	0	2
Dual Nature	XII	3	1	2
Atoms	XII	2	2	0
Nuclei	XII	1	1	0
Semiconductor Electronics	XII	1	0	1

4. Teaching takeaways

- Thermodynamics and kinetic theory gave quick direct marks: adiabatic work, cyclic process, mean free path and $C_v = (3 + f)R$ are printed formulas.
- Exercise 13.17 (floating cork in SHM) was asked almost unchanged; NCERT exercises must be solved fully.
- Two practical-skill questions (index correction, vernier zero error) need the Lab Manual; the vernier question was dropped by NTA.
- Graph-reading questions (photocurrent vs time, terminal velocity vs σ/ρ, $N(\theta)$ in Geiger–Marsden) come straight from NCERT figures.

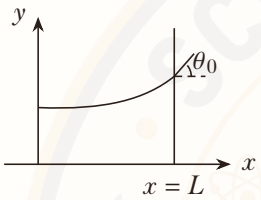
5. Deleted portions

2 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: Zener diode (Q 16); parallel axis theorem (Q 27).

NEET 2025
Physics Question Paper

4 May 2025 · Test Booklet Code 45 · Physics Q 1–45 · 180 Marks

1. Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z direction. When filled with a liquid of surface tension S and density ρ, the liquid surface makes angle θ_0 ($\theta_0 \ll 1$) with the x-axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is: (take $\theta(x) = \sin\theta(x) = \tan\theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity)



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(1) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}x$ **(2)** $\frac{d^2y}{dx^2} = \frac{\rho g}{S}y$

(3) $\frac{d^2y}{dx^2} = \sqrt{\frac{\rho g}{S}}$ **(4)** $\frac{dy}{dx} = \sqrt{\frac{\rho g}{S}}x$

2. A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is

DIRECT — Class XII Part 2 · Ch 9 · **Page 244** · m=(L/fo)(D/fe) Open PDF ↗

(1) 100 **(2)** 125 **(3)** 150 **(4)** 250

3. An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) moving with speed $c/100$ (c = speed of light) is injected into a magnetic field $\vec{B}$ of magnitude 9×10^{-4} T perpendicular to its direction of motion. We wish to apply a uniform electric field $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then (speed of light $c = 3 \times 10^8$ m s^{-1})

DELETED PORTION — velocity selector (crossed E and B) · removed from the 2026-27 NCERT; not analysed

(1) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^4 V m^{-1}

(2) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^2 V m^{-1}

(3) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^2 V m^{-1}

(4) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^4 V m^{-1}

4. There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The coefficient of kinetic friction (μ_k) between the object and the rough surface is close to

APPLIED — Class XI Part 1 · Ch 4 · **Page 61** Open PDF ↗

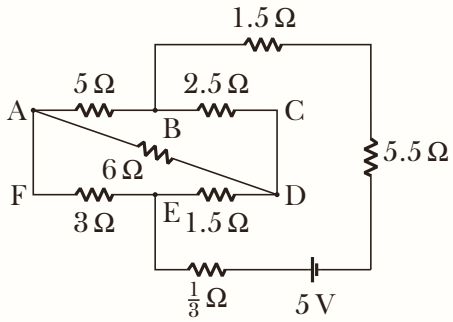
(1) 0.25 **(2)** 0.40 **(3)** 0.5 **(4)** 0.75

5. The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying brakes, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the brakes on cars A and B, respectively, then the ratio F_A/F_B is

APPLIED — Class XI Part 1 · Ch 5 · **Page 73** Open PDF ↗

(1) $\frac{3}{2}$ **(2)** $\frac{2}{3}$ **(3)** $\frac{1}{3}$ **(4)** $\frac{1}{2}$

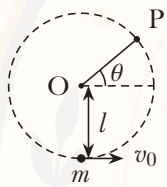
6. The current passing through the battery in the given circuit, is:



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(1) 2.0 A **(2)** 0.5 A **(3)** 2.5 A **(4)** 1.5 A

7. A bob of heavy mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:

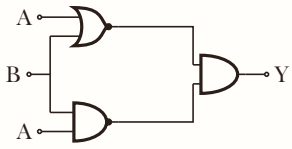


APPLIED Class XI Part 1 · Ch 5 · **Page 79** · Example 5.7 setting Open PDF ↗

(1) $(\sin\theta)^{1/2}$ **(2)** $\left(\frac{1}{2+3\sin\theta}\right)^{1/2}$

(3) $\left(\frac{\cos\theta}{2+3\sin\theta}\right)^{1/2}$ **(4)** $\left(\frac{\sin\theta}{2+3\sin\theta}\right)^{1/2}$

8. The output (Y) of the given logic implementation is similar to the output of an/a ——— gate.



DELETED PORTION digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) AND **(2)** NAND
(3) OR **(4)** NOR

9. The electric field in a plane electromagnetic wave is given by $E_z = 60\cos(5x + 1.5 \times 10^9 t)$ V/m. Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field):

DIRECT Class XII Part 1 · Ch 8 · **Page 208** · Example 8.2 Open PDF ↗

(1) $B_y = 2 \times 10^{-7}\cos(5x + 1.5 \times 10^9 t)$ T
(2) $B_x = 2 \times 10^{-7}\cos(5x + 1.5 \times 10^9 t)$ T
(3) $B_z = 60\cos(5x + 1.5 \times 10^9 t)$ T
(4) $B_y = 60\sin(5x + 1.5 \times 10^9 t)$ T

10. A ball of mass 0.5 kg is dropped from a height of 40 m. The ball hits the ground and rises to a height of 10 m. The impulse imparted to the ball during its collision with the ground is (Take $g = 9.8\ \text{m/s}^2$)

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(1) 21 N s **(2)** 7 N s **(3)** 0 **(4)** 84 N s

11. AB is a part of an electrical circuit (see figure). The potential difference $V_A - V_B$, at the instant when current $i = 2$ A and is increasing at a rate of 1 A per second is:



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(1) 5 volt **(2)** 6 volt
(3) 9 volt **(4)** 10 volt

12. A 2 A current is flowing through two different small circular copper coils having radii ratio 1 : 2. The ratio of their respective magnetic moments will be

DIRECT Class XII Part 1 · Ch 4 · **Page 125** · m=NIA Open PDF ↗

(1) 1:4 **(2)** 1:2 **(3)** 2:1 **(4)** 4:1

13. In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively

DIRECT Class XII Part 2 · Ch 9 · **Page 238** · P=sum Pi, m=m1 m2 m3 Open PDF ↗

(1) $4p$ and $4m$ **(2)** p^4 and $4m$
(3) $4p$ and m^4 **(4)** p^4 and m^4

14. An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperature 27°C. The mass of the oxygen withdrawn from the cylinder is nearly equal to: [Given, $R = \frac{100}{12}\,\text{J mol}^{-1}\text{K}^{-1}$, and molecular mass of $O_2 = 32$, 1 atm pressure $= 1.01 \times 10^5\ \text{N/m}^2$]

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(1) 0.125 kg **(2)** 0.144 kg
(3) 0.116 kg **(4)** 0.156 kg

15. In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

APPLIED Class XI Part 1 · Ch 2 · **Page 15** Open PDF ↗

(1) $-\frac{2}{(x+2)^3}$ **(2)** $-\frac{2}{(2x+1)^3}$
(3) $+\frac{2}{(x+1)^3}$ **(4)** $+\frac{2}{2x+1}$

16. To an ac power supply of 220 V at 50 Hz, a resistor of 20 Ω, a capacitor of reactance 25 Ω and an inductor of reactance 45 Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively

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(1) 7.8 A and 30° **(2)** 7.8 A and 45°
(3) 15.6 A and 30° **(4)** 15.6 A and 45°

17. The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

APPLIED Class XI Part 1 · Ch 6 · **Page 122** Open PDF ↗

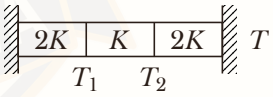
(1) 100 days **(2)** 105 days
(3) 115 days **(4)** 108 days

18. A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

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(1) $\frac{he}{\pi m}$ **(2)** $\frac{he}{2\pi m}$ **(3)** $\frac{heB}{\pi m}$ **(4)** $\frac{heB}{2\pi m}$

19. Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K. The left end of the combination is maintained at temperature $3T$ and the right end at T. The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2. The ratio T_1/T_2 is



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(1) $\frac{3}{2}$ **(2)** $\frac{4}{3}$ **(3)** $\frac{5}{3}$ **(4)** $\frac{5}{4}$

20. The plates of a parallel plate capacitor are separated by d. Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates. If $K_1 = 1.25K_2$, the value of K_1 is:

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(1) 2.66 **(2)** 2.33 **(3)** 1.60 **(4)** 1.33

21. Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

APPLIED Class XI Part 1 · Ch 3 · **Page 46** · relative velocity (Points to ponder) Open PDF ↗

(1) 9 min, 40 km/h
(2) 25 min, 100 km/h
(3) 10 min, 90 km/h
(4) 15 min, 120 km/h

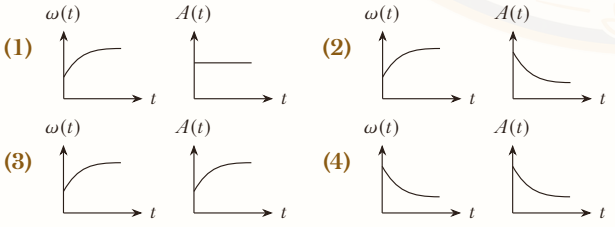
22. A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (take $g = 10\ \text{m/s}^2$)

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(1) 100 N **(2)** $100\sqrt{3}$ N
(3) 200 N **(4)** $200\sqrt{3}$ N

23. In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t. Which one of the following options schematically depicts these changes correctly?

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24. A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A. It is filled with a gas of density ρ and takes a spherical shape of radius R. When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T. If the speed $v(r)$ of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$
(2) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$
(3) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$
(4) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

25. Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed. If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is

APPLIED Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: vernier zero error Open PDF ↗

(1) 5.18 cm **(2)** 5.08 cm
(3) 4.98 cm **(4)** 5.00 cm

26. A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is:

DIRECT Class XII Part 1 · Ch 8 · **Page 204** · B between the plates Open PDF ↗

(1) zero at all places
(2) constant between the plates and zero outside the plates
(3) non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates
(4) zero between the plates and non-zero outside

27. An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) reflected light is completely polarized and the angle of reflection is close to 60°
(2) reflected light is partially polarized and the angle of reflection is close to 30°
(3) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30°, respectively
(4) transmitted light is completely polarized with angle of refraction close to 30°

28. Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F. A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as:

APPLIED — Class XII Part 1 · Ch 1 · **Page 10** Open PDF ↗

(1) $\frac{3F}{5}$ **(2)** $\frac{2F}{3}$ **(3)** $\frac{F}{2}$ **(4)** $\frac{3F}{8}$

29. A container has two chambers of volumes V_1 = 2 litres and V_2 = 3 litres separated by a partition made of a thermal insulator. The chambers contain $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures p_1 = 1 atm and p_2 = 2 atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of:

APPLIED — Class XI Part 2 · Ch 12 · **Page 247** Open PDF ↗

(1) 1.3 atm **(2)** 1.6 atm
(3) 1.4 atm **(4)** 1.8 atm

30. A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius r of the nth orbit and the particle's speed v in the orbit depend on n as

APPLIED — Class XII Part 2 · Ch 12 · **Page 299** Open PDF ↗

(1) $r \propto n^{1/3}$; $v \propto n^{1/3}$
(2) $r \propto n^{1/3}$; $v \propto n^{2/3}$
(3) $r \propto n^{2/3}$; $v \propto n^{1/3}$
(4) $r \propto n^{4/3}$; $v \propto n^{-1/3}$

31. The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of 1 year on Mercury?

DIRECT — Class XI Part 1 · Ch 7 · **Page 129** · T^2 proportional to a^3 Open PDF ↗

(1) 88 earth days **(2)** 225 earth days
(3) 172 earth days **(4)** 124 earth days

32. A body weighs 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is:

DIRECT — Class XI Part 1 · Ch 7 · **Page 133** · g(h)=GM/(R+h)^2 Open PDF ↗

(1) 16 N **(2)** 27 N **(3)** 32 N **(4)** 36 N

33. A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is:

APPLIED — Class XII Part 1 · Ch 3 · **Page 98** Open PDF ↗

(1) $\frac{R}{64}$ **(2)** $\frac{R}{32}$ **(3)** $\frac{R}{16}$ **(4)** $\frac{R}{8}$

34. De-Broglie wavelength of an electron orbiting in the n = 2 state of hydrogen atom is close to (Given Bohr radius = 0.052 nm)

APPLIED — Class XII Part 2 · Ch 12 · **Page 301** Open PDF ↗

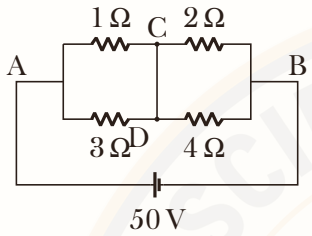
(1) 0.067 nm **(2)** 0.67 nm
(3) 1.67 nm **(4)** 2.67 nm

35. An electric dipole with dipole moment 5×10^{-6} C m is aligned with the direction of a uniform electric field of magnitude 4×10^{5} N/C. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is:

DIRECT Class XII Part 1 · Ch 2 · **Page 61** · W=pE(cos theta0 - cos theta1) Open PDF ↗

(1) 0.8 J **(2)** 1.0 J **(3)** 1.2 J **(4)** 1.5 J

36. A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is:



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(1) 1.5 A **(2)** 2.0 A **(3)** 2.5 A **(4)** 3.0 A

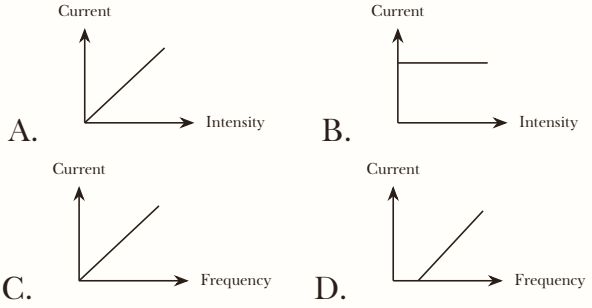
37. A photon and an electron (mass m) have the same energy E. The ratio $(\lambda_{\text{photon}}/\lambda_{\text{electron}})$ of their de Broglie wavelengths is: (c is the speed of light)

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(1) $\sqrt{\dfrac{E}{2m}}$ **(2)** $c\sqrt{2mE}$

(3) $c\sqrt{\dfrac{2m}{E}}$ **(4)** $\dfrac{1}{c}\sqrt{\dfrac{E}{2m}}$

38. Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?

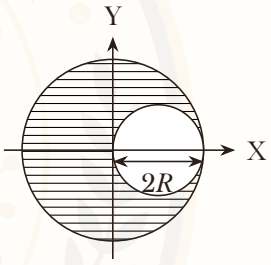


DIRECT Class XII Part 2 · Ch 11 · **Page 278** · Figs 11.2 and 11.4 Open PDF ↗

(1) A only **(2)** A and C

(3) A and D **(4)** B and D

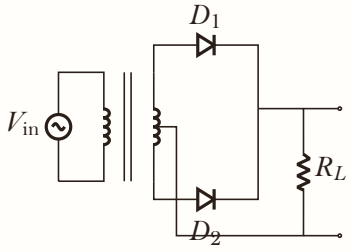
39. A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is:



DELETED PORTION parallel axis theorem · removed from the 2026-27 NCERT; not analysed

(1) $\dfrac{7}{8}$ **(2)** $\dfrac{7}{40}$ **(3)** $\dfrac{7}{57}$ **(4)** $\dfrac{7}{64}$

40. A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{\text{in}} = 220\sin(100\pi t)$ volt, then at $t = 15$ ms



APPLIED Class XII Part 2 · Ch 14 · **Page 339** Open PDF ↗

(1) D_1 is forward biased, D_2 is reverse biased

(2) D_1 is reverse biased, D_2 is forward biased

(3) D_1 and D_2 both are forward biased

(4) D_1 and D_2 both are reverse biased

41. Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B, respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm, respectively. If the change in their internal energy is the same, then the ratio r_A/r_B is equal to

APPLIED Class XI Part 2 · Ch 11 · **Page 230** Open PDF ↗

(1) $\frac{4}{3}$ **(2)** $\frac{3}{4}$ **(3)** $\frac{2}{\sqrt{3}}$ **(4)** $\frac{\sqrt{3}}{2}$

42. A physical quantity P is related to four observations a, b, c and d as follows: $P = a^3b^2/c\sqrt{d}$. The percentage errors of measurement in a, b, c and d are 1%, 3%, 2%, and 4% respectively. The percentage error in the quantity P is

DIRECT Class XI Part 1 · Ch 1 · **Page 6** · relative errors add Open PDF ↗

(1) 10% **(2)** 2% **(3)** 13% **(4)** 15%

43. The intensity of transmitted light when a polaroid sheet, placed between two crossed polaroids at 22.5° from the polarization axis of one of the polaroid, is (I_0 is the intensity of polarised light after passing through the first polaroid):

DIRECT Class XII Part 2 · Ch 10 · **Page 271** · Example 10.2 Open PDF ↗

(1) $\frac{I_0}{2}$ **(2)** $\frac{I_0}{4}$ **(3)** $\frac{I_0}{8}$ **(4)** $\frac{I_0}{16}$

44. Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2, respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q/A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is:

APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $\frac{k_2}{k_1}$ **(2)** $\frac{k_1}{k_2}$ **(3)** $\sqrt{\frac{k_2}{k_1}}$ **(4)** $\sqrt{\frac{k_1}{k_2}}$

45. A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to:

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(1) $\frac{f}{2}$ **(2)** f **(3)** $\frac{3f}{2}$ **(4)** $2f$

NEET 2025 · Answer Key

1 (2)	**2** (2)	3 (2)†	**4** (4)	**5** (2)	**6** (2)	**7** (4)	8 (4)†	**9** (1)	**10** (1)
11 (4)	**12** (1)	**13** (3)	**14** (3)	**15** (2)	**16** (2)	**17** (4)	**18** (2)	**19** (3)	**20** (1)
21 (1)	**22** (2)	**23** (2)	**24** (3)	**25** (3)	**26** (3)	27 (1)†	**28** (4)	**29** (2)	**30** (3)
31 (1)	**32** (2)	**33** (3)	**34** (2)	**35** (2)	**36** (2)	**37** (3)	**38** (1)	39 (3)†	**40** (1)*
41 (2)	**42** (3)	**43** (3)	**44** (4)	**45** (2)					

Official NTA final key (Test Booklet Code 45); every answer was also solved and checked by the editor. (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). *: see note.

Q40. The official NTA key accepts both (1) and (2).

NEET 2025 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 41 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**11**	27%
text, summary, table or figure	9	
NCERT solved Example	2	
NCERT Exercise	0	
Applied (concept in NCERT)	**30**	73%
Analysed	41	
Deleted portion (not analysed)	4	
Total	**45**	

Direct: Q 2, 9, 12, 13, 26, 31, 32, 35, 38, 42, 43. From NCERT Examples: Q 9 (Example 8.2), 43 (Example 10.2).

2. Difficulty level

Easy	13
Moderate	23
Hard	5

Hard: Q 1, 7, 18, 24, 30. The 2025 paper was the most demanding of the recent papers: several questions (Q1, Q7, Q18, Q24, Q30) need a model to be built, not just a formula. Class XI: 21 questions, Class XII: 20 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	3	1	2
Motion in a Straight Line	XI	1	0	1
Motion in a Plane	XI	1	0	1
Laws of Motion	XI	2	0	2
Work, Energy & Power	XI	2	0	2
Rotational Motion	XI	2	0	2
Gravitation	XI	2	2	0
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	1	0	1
Thermodynamics	XI	1	0	1
Kinetic Theory	XI	2	0	2
Oscillations	XI	2	0	2
Waves	XI	1	0	1
Electric Charges & Fields	XII	1	0	1
Electrostatic Potential & Capacitance	XII	2	1	1
Current Electricity	XII	3	0	3
Moving Charges & Magnetism	XII	2	1	1
Electromagnetic Induction	XII	1	0	1
Alternating Current	XII	1	0	1
Electromagnetic Waves	XII	2	2	0
Ray Optics	XII	2	2	0
Wave Optics	XII	1	1	0
Dual Nature	XII	2	1	1
Atoms	XII	2	0	2
Semiconductor Electronics	XII	1	0	1

4. Teaching takeaways

- NCERT Examples 8.2 (EM wave fields) and 10.2 (polaroid between crossed polaroids) were asked with new numbers.
- Kepler's third law and g at a height were straight one-line questions.
- Three questions were on deleted topics (velocity selector, logic gates, Brewster's law) and one needed the parallel-axis theorem.
- Vernier zero-error correction again came from the Lab Manual.

5. Deleted portions

4 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: velocity selector (crossed E and B) (Q 3); digital electronics (logic gates) (Q 8); Brewster's law (Q 27); parallel axis theorem (Q 39).

NEET 2024
Physics Question Paper

5 May 2024 · Test Booklet Code S6 · Physics Q 1–50 (Section A: 1–35, Section B: 36–50, any 10 of 15) · 180 Marks

1. Match List I with List II.

List I (spectral lines of hydrogen for transitions from)		*List II* (wavelength, nm)	
A.	$n_2 = 3$ to $n_1 = 2$	I.	410.2
B.	$n_2 = 4$ to $n_1 = 2$	II.	434.1
C.	$n_2 = 5$ to $n_1 = 2$	III.	656.3
D.	$n_2 = 6$ to $n_1 = 2$	IV.	486.1

Choose the correct answer from the options given below:

APPLIED — Class XII Part 2 · Ch 12 · **Page 300** · lines computed from En=-13.6/n^2 eV Open PDF ↗

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-I, D-II

2. Match List-I with List-II.

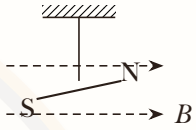
List-I (Material)		*List-II* (Susceptibility χ)	
A.	Diamagnetic	I.	$\chi = 0$
B.	Ferromagnetic	II.	$0 > \chi \geq -1$
C.	Paramagnetic	III.	$\chi \gg 1$
D.	Non-magnetic	IV.	$0 < \chi < \varepsilon$ (a small positive number)

Choose the correct answer from the options given below:

DIRECT — Class XII Part 1 · Ch 5 · **Page 147** · Table 5.2 Open PDF ↗

(1) A-IV, B-III, C-II, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-II, C-I, D-IV

3. In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is 9.8×10^{-6} kg m^2. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}$ A m^2, then the value of x is:



DELETED PORTION — oscillation period of a magnetic needle · removed from the 2026-27 NCERT; not analysed

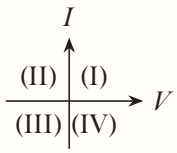
(1) $1280\pi^2$ **(2)** $5\pi^2$
(3) $128\pi^2$ **(4)** $50\pi^2$

4. An unpolarised light beam strikes a glass surface at Brewster's angle. Then

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) the reflected light will be completely polarised but the refracted light will be partially polarised.
(2) the reflected light will be partially polarised.
(3) the refracted light will be completely polarised.
(4) both the reflected and refracted light will be completely polarised.

5. Consider the following statements A and B and identify the correct answer:

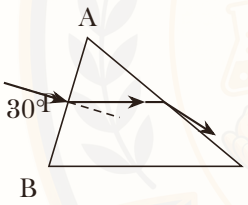


A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.

DELETED PORTION — solar cell (special purpose diodes) · removed from the 2026-27 NCERT; not analysed

(1) Both A and B are incorrect.
(2) A is correct but B is incorrect.
(3) A is incorrect but B is correct.
(4) Both A and B are correct.

6. A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is:



APPLIED — Class XII Part 2 · Ch 9 · **Page 240** Open PDF ↗

(1) $\frac{\sqrt{3}}{2}$ **(2)** $\frac{\sqrt{5}}{4}$ **(3)** $\frac{\sqrt{5}}{2}$ **(4)** $\frac{\sqrt{3}}{4}$

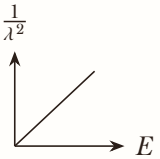
7. A particle moving with uniform speed in a circular path maintains:

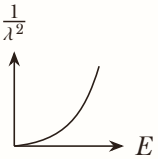
DIRECT — Class XI Part 1 · Ch 3 · **Page 42** · acceleration of UCM is not a constant vector Open PDF ↗

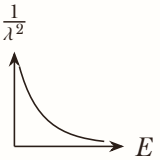
(1) varying velocity and varying acceleration.
(2) constant velocity.
(3) constant acceleration.
(4) constant velocity but varying acceleration.

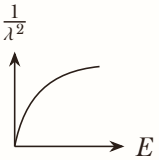
8. The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle):

APPLIED — Class XII Part 2 · Ch 11 · **Page 285** Open PDF ↗

(1)


(2)


(3)


(4)


9. A wire of length 'l' and resistance 100 Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

APPLIED — Class XII Part 1 · Ch 3 · **Page 98** Open PDF ↗

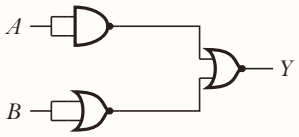
(1) 60 Ω **(2)** 26 Ω **(3)** 52 Ω **(4)** 55 Ω

10. The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is 2400 g cm^2. The length of the 400 g rod is nearly:

DIRECT — Class XI Part 1 · Ch 6 · **Page 116** · Table 6.1 Open PDF ↗

(1) 72.0 cm **(2)** 8.5 cm
(3) 17.5 cm **(4)** 20.7 cm

11. The output (Y) of the given logic gate is similar to the output of an/a:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) AND gate **(2)** NAND gate
(3) NOR gate **(4)** OR gate

12. In a vernier calipers, $(N+1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:

APPLIED — Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: vernier constant Open PDF ↗

(1) $10(N+1)$ **(2)** $\dfrac{1}{10N}$

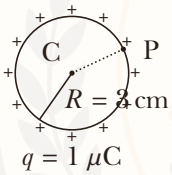
(3) $\dfrac{1}{100(N+1)}$ **(4)** $100N$

13. At any instant of time t, the displacement of any particle is given by $2t - 1$ (SI unit) under the influence of force of 5 N. The value of instantaneous power is (in SI unit):

DIRECT — Class XI Part 1 · Ch 5 · **Page 83** · P=F.v Open PDF ↗

(1) 6 **(2)** 10 **(3)** 5 **(4)** 7

14. A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is: (Take $\dfrac{1}{4\pi\varepsilon_0} = 9\times10^9$ SI units)



DIRECT — Class XII Part 1 · Ch 2 · **Page 52** · V inside a shell is constant Open PDF ↗

(1) zero **(2)** 3×10^5

(3) 1×10^5 **(4)** 0.5×10^5

15. A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 N m^{-1}, then the excess force required to take it away from the surface is:

APPLIED — Class XI Part 2 · Ch 9 · **Page 195** Open PDF ↗

(1) 99 N **(2)** 19.8 mN

(3) 198 N **(4)** 1.98 mN

16. A logic circuit provides the output Y as per the following truth table:

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is:

DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) B **(2)** $A\cdot B+\bar{A}$

(3) $A\cdot\bar{B}+\bar{A}$ **(4)** $\bar{B}$

17. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude 4×10^{-6} C m, is $\pm9\times10^3$ V. (Take $\dfrac{1}{4\pi\varepsilon_0} = 9\times10^9$ SI units)

Reason R: $V = \pm\dfrac{2P}{4\pi\varepsilon_0 r^2}$, where r is the distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the *correct* answer from the options given below:

APPLIED — Class XII Part 1 · Ch 2 · **Page 50** Open PDF ↗

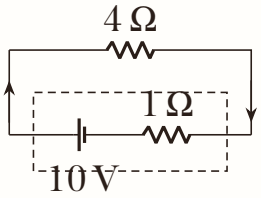
(1) A is false but R is true.

(2) Both A and R are true and R is the correct explanation of A.

(3) Both A and R are true and R is NOT the correct explanation of A.

(4) A is true but R is false.

18. The terminal voltage of the battery, whose emf is 10 V and internal resistance 1 Ω, when connected through an external resistance of 4 Ω as shown in the figure is:



DIRECT Class XII Part 1 · Ch 3 · **Page 94** · V=E-Ir Open PDF ↗

(1) 10 V **(2)** 4 V **(3)** 6 V **(4)** 8 V

19.



In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

APPLIED Class XII Part 1 · Ch 6 · **Page 160** Open PDF ↗

(1) BA and DC **(2)** AB and DC
(3) BA and CD **(4)** AB and CD

20. In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{2}$. The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning):

DIRECT Class XII Part 1 · Ch 7 · **Page 195** · Vs/Vp=Ns/Np Open PDF ↗

(1) 1:4 **(2)** 1:2 **(3)** 2:1 **(4)** 1:1

21. A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

DIRECT Class XII Part 1 · Ch 4 · **Page 117** · Example 4.6 Open PDF ↗

(1) 44 T **(2)** 44 mT
(3) 4.4 T **(4)** 4.4 mT

22. Given below are two statements:
Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.
Statement II: Atoms of each element are stable and emit their characteristic spectrum.
In the light of the above statements, choose the *most appropriate* answer from the options given below:

DIRECT Class XII Part 2 · Ch 12 · **Page 290** Open PDF ↗

(1) Statement I is incorrect but Statement II is correct.
(2) Both Statement I and Statement II are correct.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is correct but Statement II is incorrect.

23. The quantities which have the same dimensions as those of solid angle are:

DIRECT Class XI Part 1 · Ch 1 · **Page 7** · dimensionless quantities Open PDF ↗

(1) angular speed and stress
(2) strain and angle
(3) stress and angle
(4) strain and arc

24. If c is the velocity of light in free space, the correct statements about photon among the following are:
A. The energy of a photon is $E = h\nu$.
B. The velocity of a photon is c.
C. The momentum of a photon, $p = \frac{h\nu}{c}$.
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses positive charge.
Choose the correct answer from the options given below:

DIRECT Class XII Part 2 · Ch 11 · **Page 283** · photon properties Open PDF ↗

(1) A, B, D and E only
(2) A and B only
(3) A, B, C and D only
(4) A, C and D only

25. A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2ω while keeping the same radius, the tension in the string becomes:

APPLIED — Class XI Part 1 · Ch 4 · **Page 63** Open PDF ↗

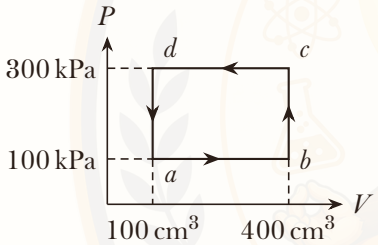
(1) $\sqrt{2}T$ **(2)** T **(3)** $4T$ **(4)** $\frac{T}{4}$

26. If $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ m represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are:

DIRECT — Class XI Part 2 · Ch 13 · **Page 264** · T=2 pi/omega Open PDF ↗

(1) 5 m, 1 s **(2)** 5 cm, 2 s
(3) 5 m, 2 s **(4)** 5 cm, 1 s

27. A thermodynamic system is taken through the cycle $abcda$. The work done by the gas along the path bc is:



DIRECT — Class XI Part 2 · Ch 11 · **Page 235** · isochoric: W=0 Open PDF ↗

(1) −60 J **(2)** zero **(3)** 30 J **(4)** −90 J

28. $^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$. In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are:

DELETED PORTION — alpha- and beta-decay · removed from the 2026-27 NCERT; not analysed

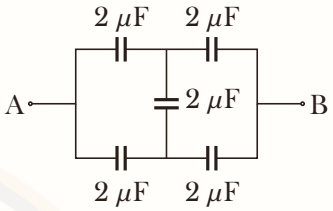
(1) 286, 81 **(2)** 280, 81
(3) 286, 80 **(4)** 288, 82

29. Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is:

DIRECT — Class XI Part 1 · Ch 5 · **Page 84** · completely inelastic collision Open PDF ↗

(1) 1:4 **(2)** 1:2 **(3)** 2:1 **(4)** 4:1

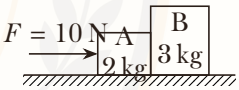
30. In the following circuit, the equivalent capacitance between terminal A and terminal B is:



APPLIED — Class XII Part 1 · Ch 2 · **Page 72** Open PDF ↗

(1) 4 μF **(2)** 2 μF
(3) 1 μF **(4)** 0.5 μF

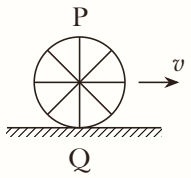
31. A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is:



APPLIED — Class XI Part 1 · Ch 4 · **Page 54** Open PDF ↗

(1) 10 N **(2)** zero **(3)** 4 N **(4)** 6 N

32. A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?



APPLIED Class XI Part 1 · Ch 6 · **Page 94** · rolling (qualitative) Open PDF ↗

(1) Point P has zero speed.
(2) Point P moves slower than point Q.
(3) Point P moves faster than point Q.
(4) Both the points P and Q move with equal speed.

33. The mass of a planet is $\frac{1}{10}$th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is:

APPLIED Class XI Part 1 · Ch 7 · **Page 133** Open PDF ↗

(1) $3.92\ \text{m s}^{-2}$ **(2)** $19.6\ \text{m s}^{-2}$
(3) $9.8\ \text{m s}^{-2}$ **(4)** $4.9\ \text{m s}^{-2}$

34. The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are $8 \times 10^8\ \text{N m}^{-2}$ and $2 \times 10^{11}\ \text{N m}^{-2}$, is:

DIRECT Class XI Part 2 · Ch 8 · **Page 170** · Y=FL/(A dL) Open PDF ↗

(1) 8 mm **(2)** 4 mm
(3) 0.4 mm **(4)** 40 mm

35. If the monochromatic source in Young's double slit experiment is replaced by white light, then

APPLIED Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

(1) all bright fringes will be of equal width.
(2) interference pattern will disappear.
(3) there will be a central dark fringe surrounded by a few coloured fringes.
(4) there will be a central bright white fringe surrounded by a few coloured fringes.

36. A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is:

DIRECT Class XII Part 2 · Ch 9 · **Page 253** · Exercise 9.27 Open PDF ↗

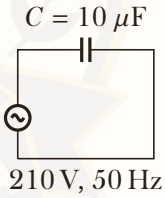
(1) 32 **(2)** 34 **(3)** 28 **(4)** 17

37. If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then
A. the charge stored in it, increases.
B. the energy stored in it, decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.
Choose the most appropriate answer from the options given below:

APPLIED Class XII Part 1 · Ch 2 · **Page 69** Open PDF ↗

(1) A, B and C only **(2)** A, B and E only
(3) A, C and E only **(4)** B, D and E only

38. A 10 μF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$):

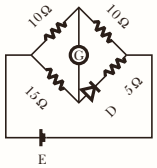


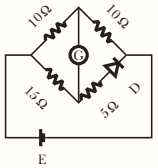
DIRECT Class XII Part 1 · Ch 7 · **Page 185** · Example 7.4 Open PDF ↗

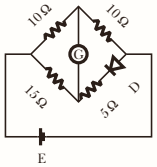
(1) 0.35 A **(2)** 0.58 A
(3) 0.93 A **(4)** 1.20 A

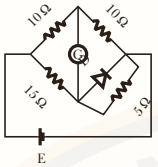
39. Choose the correct circuit which can achieve the bridge balance. (Each figure: arms 10 Ω and 10 Ω above, 15 Ω and 5 Ω below, galvanometer G across the bridge, diode D in the 5 Ω arm, cell E and key K.)

APPLIED Class XII Part 1 · Ch 3 · **Page 101** Open PDF ↗

(1) 

(2) 

(3) 

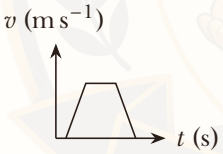
(4) 

40. Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

APPLIED Class XII Part 1 · Ch 3 · **Page 92** Open PDF ↗

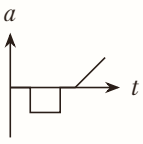
(1) 2:3 **(2)** 1:1 **(3)** 2:9 **(4)** 1:2

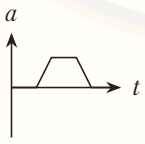
41. The velocity (v)–time (t) plot of the motion of a body is shown below:

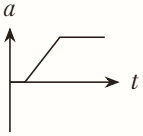


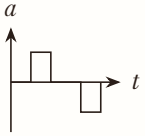
The acceleration (a)–time (t) graph that best suits this motion is:

APPLIED Class XI Part 1 · Ch 2 · **Page 16** Open PDF ↗

(1) 

(2) 

(3) 

(4) 

42. If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

DIRECT Class XI Part 2 · Ch 13 · **Page 271** · T=2 pi sqrt(L/g) Open PDF ↗

(1) 4 **(2)** $\sqrt{3}$ **(3)** $\sqrt{2}$ **(4)** $2\sqrt{3}$

43. A metallic bar of Young's modulus, 0.5×10^{11} N m^{-2} and coefficient of linear thermal expansion 10^{-5} °C^{-1}, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is:

DIRECT Class XI Part 2 · Ch 10 · **Page 207** · thermal stress Open PDF ↗

(1) 2×10^3 N **(2)** 5×10^3 N
(3) 50×10^3 N **(4)** 100×10^3 N

44. A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to:

A. hold the sheet there if it is magnetic.
B. hold the sheet there if it is non-magnetic.
C. move the sheet away from the pole with uniform velocity if it is conducting.
D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

APPLIED Class XII Part 1 · Ch 6 · **Page 160** Open PDF ↗

(1) C only **(2)** B and D only
(3) A and C only **(4)** A, C and D only

45. The property which is not of an electromagnetic wave travelling in free space is that:

DIRECT Class XII Part 1 · Ch 8 · **Page 205** · accelerated charges radiate Open PDF ↗

(1) they originate from charges moving with uniform speed.
(2) they are transverse in nature.
(3) the energy density in electric field is equal to energy density in magnetic field.
(4) they travel with a speed equal to $\frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

46. A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates:

DIRECT Class XII Part 1 · Ch 8 · **Page 204** · i_d = i Open PDF ↗

(1) displacement current of magnitude greater than I flows but can be in any direction.
(2) there is no current.
(3) displacement current of magnitude equal to I flows in the same direction as I.
(4) displacement current of magnitude equal to I flows in a direction opposite to that of I.

47. An iron bar of length L has magnetic moment M. It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is:

APPLIED Class XII Part 1 · Ch 5 · **Page 138** Open PDF ↗

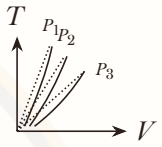
(1) $\frac{M}{\sqrt{3}}$ **(2)** M **(3)** $\frac{M}{2}$ **(4)** 2M

48. The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is:

APPLIED Class XI Part 1 · Ch 7 · **Page 138** Open PDF ↗

(1) $\frac{GmM}{3R}$ **(2)** $\frac{5GmM}{6R}$
(3) $\frac{2GmM}{3R}$ **(4)** $\frac{GmM}{2R}$

49. The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.



Then the correct relation is:

APPLIED Class XI Part 2 · Ch 12 · **Page 246** Open PDF ↗

(1) $P_1 > P_2 > P_3$ **(2)** $P_3 > P_2 > P_1$
(3) $P_1 > P_3 > P_2$ **(4)** $P_2 > P_1 > P_3$

50. A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is:

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) $\alpha\beta/t$ **(2)** $\beta t/\alpha$ **(3)** $\alpha t/\beta$ **(4)** $\alpha\beta t$

NEET 2024 · Answer Key

1 (3)	**2** (2)	**3** (1)†	**4** (1)†	**5** (2)†	**6** (3)	**7** (1)	**8** (1)	**9** (3)	**10** (2)
11 (1)†	**12** (3)	**13** (2)	**14** (1)	**15** (2)	**16** (4)†	**17** (4)	**18** (4)	**19** (2)*	**20** (3)
21 (4)	**22** (2)*	**23** (2)	**24** (3)	**25** (3)	**26** (3)	**27** (2)	**28** (1)†	**29** (3)	**30** (2)
31 (4)	**32** (3)	**33** (1)	**34** (2)	**35** (4)	**36** (3)	**37** (3)	**38** (3)	**39** (3)*	**40** (3)
41 (4)	**42** (3)	**43** (3)	**44** (3)	**45** (1)	**46** (3)	**47** (3)	**48** (2)	**49** (1)	**50** (3)

Answers solved by the editor (the official final key of this year could not be obtained from the NTA/CBSE websites). (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). *: see note.

Q19. By Lenz's law, solenoid-1 (N pole receding) and solenoid-2 (S pole approaching) both oppose the change; with the winding shown this gives A to B and D to C.

Q22. NCERT (p 290) states both: atoms are neutral and each element has a characteristic spectrum.

Q39. In (3) the diode is reverse-biased, so its arm is open and no current flows through G.

NEET 2024 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 50, 44 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**21**	48%
text, summary, table or figure	18	
NCERT solved Example	2	
NCERT Exercise	1	
Applied (concept in NCERT)	**23**	52%
Analysed	44	
Deleted portion (not analysed)	6	
Total	**50**	

Direct: Q 2, 7, 10, 13, 14, 18, 20, 21, 22, 23, 24, 26, 27, 29, 34, 36, 38, 42, 43, 45, 46.

From NCERT Examples: Q 21 (Example 4.6), 38 (Example 7.4).

From NCERT Exercises: Q 36 (Exercise 9.27).

2. Difficulty level

Easy	27
Moderate	17
Hard	0

The 2024 paper was easy to moderate, with many one-line questions from NCERT statements, tables and examples. Class XI: 20 questions, Class XII: 24 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	3	1	2
Motion in a Straight Line	XI	1	0	1
Motion in a Plane	XI	1	1	0
Laws of Motion	XI	2	0	2
Work, Energy & Power	XI	2	2	0
Rotational Motion	XI	2	1	1
Gravitation	XI	2	0	2
Mech. Prop. of Solids	XI	1	1	0
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	1	1	0
Thermodynamics	XI	1	1	0
Kinetic Theory	XI	1	0	1
Oscillations	XI	2	2	0
Electrostatic Potential & Capacitance	XII	4	1	3
Current Electricity	XII	4	1	3
Moving Charges & Magnetism	XII	1	1	0
Magnetism & Matter	XII	2	1	1
Electromagnetic Induction	XII	2	0	2
Alternating Current	XII	2	2	0
Electromagnetic Waves	XII	2	2	0
Ray Optics	XII	2	1	1
Wave Optics	XII	1	0	1
Dual Nature	XII	2	1	1
Atoms	XII	2	1	1

4. Teaching takeaways

- Exercise 9.27 (telescope, $f_o = 140\,\text{cm}$, $f_e = 5\,\text{cm}$) and Examples 4.6 and 7.4 were asked with the same or new numbers.
- Table 5.2 (susceptibility) and Table 6.1 (moments of inertia) each gave a direct question.
- Six questions were on deleted portions (logic gates twice, Brewster's law, solar cell, magnetic needle, α/β decay).
- Bridge balance with a diode (Q39) needs the idea that a reverse-biased diode opens its arm.

5. Deleted portions

6 of the 50 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: oscillation period of a magnetic needle (Q 3); Brewster's law (Q 4); solar cell (special purpose diodes) (Q 5); digital electronics (logic gates) (Q 11, 16); alpha- and beta-decay (Q 28).

NEET 2023
Physics Question Paper

7 May 2023 · Test Booklet Code G6 · Physics Q 1–50 (Section A: 1–35, Section B: 36–50, any 10 of 15) · 180 Marks

1. The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?

DIRECT — Class XII Part 2 · Ch 11 · **Page 281** · Kmax=h nu - phi0 Open PDF ↗

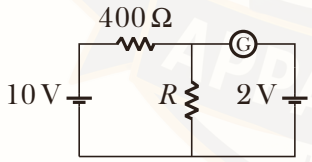
(1) Na only **(2)** Cs only
(3) Both Na and K **(4)** K only

2. The net magnetic flux through any closed surface is:

DIRECT — Class XII Part 1 · Ch 5 · **Page 142** · net flux through a closed surface is zero Open PDF ↗

(1) Negative **(2)** Zero
(3) Positive **(4)** Infinity

3. If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by:



APPLIED — Class XII Part 1 · Ch 3 · **Page 98** Open PDF ↗

(1) 400 Ω **(2)** 200 Ω **(3)** 50 Ω **(4)** 100 Ω

4. A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?

DIRECT — Class XII Part 1 · Ch 7 · **Page 195** · ideal transformer Open PDF ↗

(1) 0.37 A **(2)** 0.27 A
(3) 2.7 A **(4)** 3.7 A

5. A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output?

DIRECT — Class XII Part 2 · Ch 14 · **Page 339** · capacitor filter Open PDF ↗

(1) Load resistance
(2) A centre-tapped transformer
(3) p-n junction diodes
(4) Capacitor

6. In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude $48\ \mathrm{V\,m^{-1}}$. Then the amplitude of oscillating magnetic field is: (Speed of light in free space = $3 \times 10^8\ \mathrm{m\,s^{-1}}$)

DIRECT — Class XII Part 1 · Ch 8 · **Page 207** · B0=E0/c Open PDF ↗

(1) 1.6×10^{-6} T **(2)** 1.6×10^{-9} T
(3) 1.6×10^{-8} T **(4)** 1.6×10^{-7} T

7. A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be:

APPLIED — Class XI Part 1 · Ch 1 · **Page 6** Open PDF ↗

(1) 1.4% **(2)** 1.2% **(3)** 1.3% **(4)** 1.6%

8. Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

APPLIED — Class XII Part 2 · Ch 9 · **Page 229** Open PDF ↗

(1) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ **(2)** $\sin^{-1}\left(\frac{t_2}{t_1}\right)$
(3) $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ **(4)** $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$

9. An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 N C^{-1}. It experiences a torque equal to 4 N m. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.

DIRECT — Class XII Part 1 · Ch 1 · **Page 27** · tau=pE sin theta Open PDF ↗

(1) 2 mC **(2)** 8 mC **(3)** 6 mC **(4)** 4 mC

10. Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is:

DIRECT — Class XI Part 2 · Ch 8 · **Page 168** · stress=F/A Open PDF ↗

(1) Zero **(2)** $2W/A$
(3) W/A **(4)** $W/2A$

11. In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Brackett series is:

APPLIED — Class XII Part 2 · Ch 12 · **Page 300** · series limits from En Open PDF ↗

(1) 16λ **(2)** 2λ **(3)** 4λ **(4)** 9λ

12. The temperature of a gas is −50°C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

APPLIED — Class XI Part 2 · Ch 12 · **Page 250** Open PDF ↗

(1) 223 K **(2)** 669°C
(3) 3295°C **(4)** 3097 K

13. A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is:

APPLIED — Class XI Part 1 · Ch 4 · **Page 54** Open PDF ↗

(1) along south-west **(2)** along eastward
(3) along northward **(4)** along north-east

14. The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is:

DIRECT — Class XI Part 2 · Ch 14 · **Page 292** · open v/2L, closed v/4L Open PDF ↗

(1) 3:1 **(2)** 1:2 **(3)** 2:1 **(4)** 1:3

15. The angular acceleration of a body, moving along the circumference of a circle, is:

DIRECT — Class XI Part 1 · Ch 6 · **Page 105** · alpha along the axis Open PDF ↗

(1) along the axis of rotation
(2) along the radius, away from centre
(3) along the radius towards the centre
(4) along the tangent to its position

16. Given below are two statements:
Statement I: Photovoltaic devices can convert optical radiation into electricity.
Statement II: Zener diode is designed to operate under reverse bias in breakdown region.
In the light of the above statements, choose the *most appropriate* answer from the options given below:

DELETED PORTION — Zener diode · removed from the 2026-27 NCERT; not analysed

(1) Statement I is incorrect but Statement II is correct.
(2) Both Statement I and Statement II are correct.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is correct but Statement II is incorrect.

17. If $\oint_S \vec{E} \cdot d\vec{S} = 0$ over a surface, then:

DIRECT — Class XII Part 1 · Ch 1 · **Page 31** · zero net flux Open PDF ↗

(1) the electric field inside the surface is necessarily uniform.
(2) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.
(3) the magnitude of electric field on the surface is constant.
(4) all the charges must necessarily be inside the surface.

18. Resistance of a carbon resistor determined from colour codes is (22000 ± 5%) Ω. The colour of third band must be:

DELETED PORTION — resistor colour code · removed from the 2026-27 NCERT; not analysed

(1) Yellow **(2)** Red
(3) Green **(4)** Orange

19. The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is:

DIRECT — Class XII Part 1 · Ch 6 · **Page 169** · U=LI^2/2 Open PDF ↗

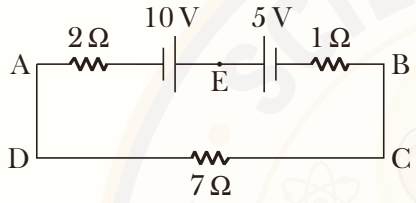
(1) 8 μJ **(2)** 4 μJ **(3)** 4 mJ **(4)** 8 mJ

20. In a series LCR circuit, the inductance L is 10 mH, capacitance C is 1 μF and resistance R is 100 Ω. The frequency at which resonance occurs is:

DIRECT — Class XII Part 1 · Ch 7 · **Page 188** · omega0=1/sqrt(LC) Open PDF ↗

(1) 1.59 kHz **(2)** 15.9 rad/s
(3) 15.9 kHz **(4)** 1.59 rad/s

21. The magnitude and direction of the current in the following circuit is



APPLIED — Class XII Part 1 · Ch 3 · **Page 95** Open PDF ↗

(1) 1.5 A from B to A through E
(2) 0.2 A from B to A through E
(3) 0.5 A from A to B through E
(4) $\frac{5}{9}$ A from A to B through E

22. The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:

DIRECT — Class XII Part 2 · Ch 11 · **Page 288** · Exercise 11.1 Open PDF ↗

(1) V^2 **(2)** $\sqrt{V}$ **(3)** $\frac{1}{V}$ **(4)** $\frac{1}{\sqrt{V}}$

23. The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are:

DELETED PORTION — types of errors · removed from the 2026-27 NCERT; not analysed

(1) Random errors
(2) Instrumental errors
(3) Personal errors
(4) Least count errors

24. For Young's double slit experiment, two statements are given below:
Statement I: If screen is moved away from the plane of slits, angular separation of the fringes remains constant.
Statement II: If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.
In the light of the above statements, choose the *correct* answer from the options given below:

APPLIED — Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

(1) Statement I is false but Statement II is true.
(2) Both Statement I and Statement II are true.
(3) Both Statement I and Statement II are false.
(4) Statement I is true but Statement II is false.

25. A bullet is fired from a gun at the speed of 280 m s^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is (g = 9.8 m s^{-2}, sin 30° = 0.5):

DIRECT — Class XI Part 1 · Ch 3 · **Page 39** · hm=(v sin theta)^2/2g Open PDF ↗

(1) 3000 m **(2)** 2800 m
(3) 2000 m **(4)** 1000 m

26. A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is:

DIRECT — Class XI Part 2 · Ch 11 · **Page 238** · eta=1-T2/T1 Open PDF ↗

(1) 200°C **(2)** 27°C
(3) 15°C **(4)** 100°C

27. The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly: (surface tension of soap solution = 0.03 N m^{-1})

APPLIED — Class XI Part 2 · Ch 9 · **Page 194** Open PDF ↗

(1) 50.1×10^{-4} J **(2)** 30.16×10^{-4} J
(3) 5.06×10^{-4} J **(4)** 3.01×10^{-4} J

28. The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{\text{th}}$ of its initial value?

DELETED PORTION — radioactive decay law (half-life) · removed from the 2026-27 NCERT; not analysed

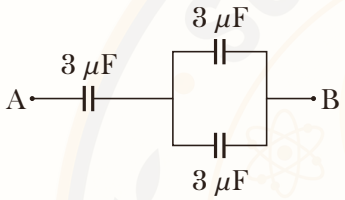
(1) 80 minutes **(2)** 20 minutes
(3) 40 minutes **(4)** 60 minutes

29. The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, potential energy stored in it will be:

DIRECT — Class XI Part 1 · Ch 5 · **Page 81** · U=kx^2/2 Open PDF ↗

(1) 16U **(2)** 2U **(3)** 4U **(4)** 8U

30. The equivalent capacitance of the system shown in the following circuit is:



DIRECT — Class XII Part 1 · Ch 2 · **Page 72** · series and parallel Open PDF ↗

(1) 9 μF **(2)** 2 μF **(3)** 3 μF **(4)** 6 μF

31. A vehicle travels half the distance with speed ϑ and the remaining distance with speed 2ϑ. Its average speed is:

APPLIED — Class XI Part 1 · Ch 2 · **Page 21** Open PDF ↗

(1) $\frac{3\vartheta}{4}$ **(2)** $\frac{\vartheta}{3}$ **(3)** $\frac{2\vartheta}{3}$ **(4)** $\frac{4\vartheta}{3}$

32. The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is:

APPLIED — Class XI Part 1 · Ch 6 · **Page 115** Open PDF ↗

(1) 5:2 **(2)** 3:5 **(3)** 5:3 **(4)** 2:5

33. Two bodies of mass m and $9m$ are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant):

APPLIED — Class XI Part 1 · Ch 7 · **Page 136** · Example 7.4 setting Open PDF ↗

(1) $-\frac{20Gm}{R}$ **(2)** $-\frac{8Gm}{R}$
(3) $-\frac{12Gm}{R}$ **(4)** $-\frac{16Gm}{R}$

34. The venturi-meter works on:

APPLIED — Class XI Part 2 · Ch 9 · **Page 188** · venturimeter (Bernoulli) Open PDF ↗

(1) The principle of perpendicular axes
(2) Huygen's principle
(3) Bernoulli's principle
(4) The principle of parallel axes

35. An ac source is connected to a capacitor C. Due to decrease in its operating frequency:

APPLIED — Class XII Part 1 · Ch 7 · **Page 185** Open PDF ↗

(1) capacitive reactance remains constant
(2) capacitive reactance decreases.
(3) displacement current increases.
(4) displacement current decreases.

36. The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom?

DIRECT — Class XII Part 2 · Ch 12 · **Page 305** · Exercise 12.7 Open PDF ↗

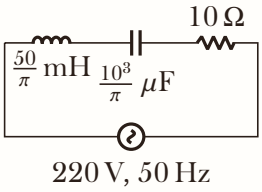
(1) 4.77 Å **(2)** 0.53 Å
(3) 1.06 Å **(4)** 1.59 Å

37. The resistance of platinum wire at 0°C is 2 Ω and 6.8 Ω at 80°C. The temperature coefficient of resistance of the wire is:

DIRECT — Class XII Part 1 · Ch 3 · **Page 90** · rho=rho0[1+alpha(T-T0)] Open PDF ↗

(1) $3 \times 10^{-1}\ ^\circ\text{C}^{-1}$ **(2)** $3 \times 10^{-4}\ ^\circ\text{C}^{-1}$
(3) $3 \times 10^{-3}\ ^\circ\text{C}^{-1}$ **(4)** $3 \times 10^{-2}\ ^\circ\text{C}^{-1}$

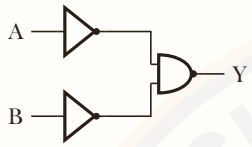
38. The net impedance of circuit (as shown in figure) will be:



APPLIED Class XII Part 1 · Ch 7 · **Page 187** Open PDF ↗

(1) 25 Ω **(2)** $10\sqrt{2}\,\Omega$
(3) 15 Ω **(4)** $5\sqrt{5}\,\Omega$

39. For the following logic circuit, the truth table is:



DELETED PORTION digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(4)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

40. 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is:

APPLIED Class XII Part 1 · Ch 3 · **Page 98** Open PDF ↗

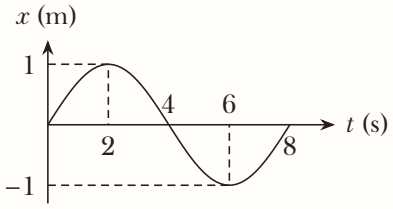
(1) 1000 **(2)** 10 **(3)** 100 **(4)** 1

41. Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10\,\text{m s}^{-2}$).

DIRECT Class XI Part 1 · Ch 4 · **Page 61** · Example 4.7 Open PDF ↗

(1) $50\,\text{m s}^{-2}$ **(2)** $1.2\,\text{m s}^{-2}$
(3) $150\,\text{m s}^{-2}$ **(4)** $1.5\,\text{m s}^{-2}$

42. The x-t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2$ s is:



APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

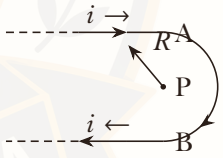
(1) $-\frac{\pi^2}{16}\,\text{m s}^{-2}$ **(2)** $\frac{\pi^2}{8}\,\text{m s}^{-2}$
(3) $-\frac{\pi^2}{8}\,\text{m s}^{-2}$ **(4)** $\frac{\pi^2}{16}\,\text{m s}^{-2}$

43. A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents:

APPLIED Class XI Part 1 · Ch 7 · **Page 137** Open PDF ↗

(1) $\sqrt{T}$ **(2)** T **(3)** T^2 **(4)** T^3

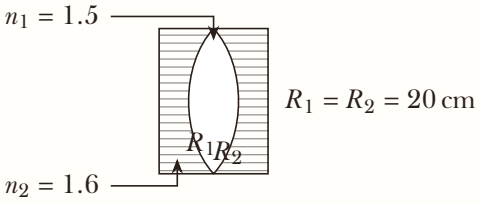
44. A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by:



DELETED PORTION magnetic field of a finite straight wire · removed from the 2026-27 NCERT; not analysed

(1) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed into the page
(2) $\frac{\mu_0 i}{4R}$ pointed into the page
(3) $\frac{\mu_0 i}{4R}$ pointed away from the page
(4) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed away from page

45. In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



APPLIED Class XII Part 2 · Ch 9 · **Page 238** Open PDF ↗

(1) −50 cm **(2)** 40 cm
(3) −40 cm **(4)** −100 cm

46. Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be:

DIRECT Class XII Part 2 · Ch 9 · **Page 237** · 1/f=1/f1+1/f2 Open PDF ↗

(1) Infinite **(2)** Zero
(3) $f/4$ **(4)** $f/2$

47. A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ T. The magnitude of the magnetic force acting on the wire is:

DIRECT Class XII Part 1 · Ch 4 · **Page 110** · F=I l x B Open PDF ↗

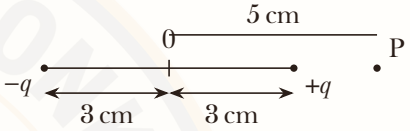
(1) $\sqrt{3}\,IL$ **(2)** $3IL$ **(3)** $\sqrt{5}\,IL$ **(4)** $5IL$

48. A bullet from a gun is fired on a rectangular wooden block with velocity u. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is:

APPLIED Class XI Part 1 · Ch 2 · **Page 17** Open PDF ↗

(1) 30 cm **(2)** 27 cm **(3)** 24 cm **(4)** 28 cm

49. An electric dipole is placed as shown in the figure. The electric potential (in 10^2 V) at point P due to the dipole is (ε_0 = permittivity of free space and $\frac{1}{4\pi\varepsilon_0} = K$):



APPLIED Class XII Part 1 · Ch 2 · **Page 51** Open PDF ↗

(1) $\left(\frac{8}{3}\right)qK$ **(2)** $\left(\frac{3}{8}\right)qK$
(3) $\left(\frac{5}{8}\right)qK$ **(4)** $\left(\frac{8}{5}\right)qK$

50. A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity 4 m s^{-1}. The ball strikes the water surface after 4 s. The height of bridge above water surface is (Take $g = 10$ m s^{-2}):

DIRECT Class XI Part 1 · Ch 2 · **Page 18** · Example 2.3 Open PDF ↗

(1) 68 m **(2)** 56 m **(3)** 60 m **(4)** 64 m

NEET 2023 · Answer Key

1 (2)	**2** (2)	**3** (4)	**4** (2)	**5** (4)	**6** (4)	**7** (4)	**8** (1)	**9** (1)	**10** (3)
11 (3)	**12** (3)	**13** (4)	**14** (3)	**15** (1)	**16** (2)†	**17** (2)	**18** (4)†	**19** (1)	**20** (1)
21 (3)	**22** (3)	**23** (1)†	**24** (4)	**25** (4)	**26** (2)	**27** (4)	**28** (1)†	**29** (1)	**30** (2)
31 (4)	**32** –*	**33** (4)	**34** (3)	**35** (4)	**36** (1)	**37** (4)	**38** (4)	**39** (3)†	**40** (3)
41 (4)	**42** (1)	**43** (3)	**44** (1)†	**45** (4)	**46** (1)	**47** (4)	**48** (2)	**49** (2)	**50** (4)

Official NTA final key (Test Booklet Code G6); every answer was also solved and checked by the editor. (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). –: question dropped by NTA (marks given to all). *: see note.

Q32. Dropped by NTA. The ratio is $\sqrt{2/5} : \sqrt{2/3} = \sqrt{3} : \sqrt{5}$, which is not among the options.

NEET 2023 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 50, 44 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**23**	52%
text, summary, table or figure	19	
NCERT solved Example	2	
NCERT Exercise	2	
Applied (concept in NCERT)	**21**	48%
Analysed	44	
Deleted portion (not analysed)	6	
Total	**50**	

Direct: Q 1, 2, 4, 5, 6, 9, 10, 14, 15, 17, 19, 20, 22, 25, 26, 29, 30, 36, 37, 41, 46, 47, 50.

From NCERT Examples: Q 41 (Example 4.7), 50 (Example 2.3).

From NCERT Exercises: Q 22 (Exercise 11.1), 36 (Exercise 12.7).

2. Difficulty level

Easy	25
Moderate	18
Hard	1

Hard: Q 45. The 2023 paper was easy: just over half the in-syllabus questions are one-step uses of NCERT formulas, examples and exercises. Class XI: 19 questions, Class XII: 25 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	1	0	1
Motion in a Straight Line	XI	3	1	2
Motion in a Plane	XI	1	1	0
Laws of Motion	XI	2	1	1
Work, Energy & Power	XI	1	1	0
Rotational Motion	XI	2	1	1
Gravitation	XI	2	0	2
Mech. Prop. of Solids	XI	1	1	0
Mech. Prop. of Fluids	XI	2	0	2
Thermodynamics	XI	1	1	0
Kinetic Theory	XI	1	0	1
Oscillations	XI	1	0	1
Waves	XI	1	1	0
Electric Charges & Fields	XII	2	2	0
Electrostatic Potential & Capacitance	XII	2	1	1
Current Electricity	XII	4	1	3
Moving Charges & Magnetism	XII	1	1	0
Magnetism & Matter	XII	1	1	0
Electromagnetic Induction	XII	1	1	0
Alternating Current	XII	4	2	2
Electromagnetic Waves	XII	1	1	0
Ray Optics	XII	3	1	2
Wave Optics	XII	1	0	1
Dual Nature	XII	2	2	0
Atoms	XII	2	1	1
Semiconductor Electronics	XII	1	1	0

4. Teaching takeaways

- NCERT Examples 2.3 and 4.7 and Exercises 11.1 and 12.7 were asked with the same idea or the same numbers.
- One-line formulas ($\omega_0 = 1/\sqrt{LC}$, $B_0 = E_0/c$, $\frac{1}{2}LI^2$, Carnot efficiency) gave quick marks.
- Six questions were on deleted portions (Zener diode, colour code, types of errors, half-life, logic gates, finite wire).
- Q32 (radius of gyration) was dropped by NTA because no option is correct.

5. Deleted portions

6 of the 50 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: Zener diode (Q 16); resistor colour code (Q 18); types of errors (Q 23); radioactive decay law (half-life) (Q 28); digital electronics (logic gates) (Q 39); magnetic field of a finite straight wire (Q 44).

NEET 2022
Physics Question Paper

17 July 2022 · Test Booklet Code R6 · Physics Q 1–50 (Section A: 1–35, Section B: 36–50, any 10 of 15) · 180 Marks

1. Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is:

DIRECT — Class XII Part 1 · Ch 3 · **Page 92** · P=V^2/R Open PDF ↗

(1) 4:1 **(2)** 1:2 **(3)** 2:1 **(4)** 1:4

2. Two hollow conducting spheres of radii R_1 and R_2 ($R_1 \gg R_2$) have equal charges. The potential would be:

DIRECT — Class XII Part 1 · Ch 2 · **Page 52** · V=kQ/R Open PDF ↗

(1) dependent on the material property of the sphere
(2) more on bigger sphere
(3) more on smaller sphere
(4) equal on both the spheres

3. When two monochromatic lights of frequency, ν and $\frac{\nu}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{V_s}{2}$ and V_s respectively. The threshold frequency for this metal is:

APPLIED — Class XII Part 2 · Ch 11 · **Page 281** Open PDF ↗

(1) $\frac{3}{2}\nu$ **(2)** 2ν **(3)** 3ν **(4)** $\frac{2}{3}\nu$

4. As the temperature increases, the electrical resistance:

DIRECT — Class XII Part 1 · Ch 3 · **Page 91** · conductors rise, semiconductors fall Open PDF ↗

(1) decreases for conductors but increases for semiconductors
(2) increases for both conductors and semiconductors
(3) decreases for both conductors and semiconductors
(4) increases for conductors but decreases for semiconductors

5. If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is:

DIRECT — Class XI Part 2 · Ch 14 · **Page 285** · v=sqrt(T/mu) Open PDF ↗

(1) 1:2 **(2)** 1:1 **(3)** $\sqrt{2}:1$ **(4)** $1:\sqrt{2}$

6. Match List-I with List-II:

List-I (EM waves)		*List-II* (λ)	
(a)	AM radio waves	(i)	10^{-10} m
(b)	Microwaves	(ii)	10^{2} m
(c)	Infrared radiations	(iii)	10^{-2} m
(d)	X-rays	(iv)	10^{-4} m

Choose the correct answer from the options given below:

DIRECT — Class XII Part 1 · Ch 8 · **Page 209** · wavelength ranges Open PDF ↗

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

7. The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is:

DIRECT — Class XI Part 1 · Ch 6 · **Page 116** · Table 6.1 (k=R/sqrt2, R/2) Open PDF ↗

(1) $1 : \sqrt{2}$ **(2)** 2:1 **(3)** $\sqrt{2} : 1$ **(4)** 4:1

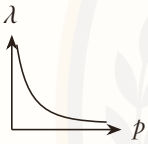
8. A biconvex lens has radii of curvature, 20 cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is:

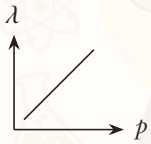
DIRECT — Class XII Part 2 · Ch 9 · **Page 234** · lens maker's formula Open PDF ↗

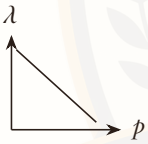
(1) infinity **(2)** +2 D
(3) +20 D **(4)** +5 D

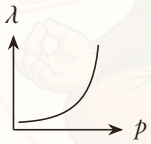
9. The graph which shows the variation of the de Broglie wavelength (λ) of a particle and its associated momentum (p) is:

DIRECT — Class XII Part 2 · Ch 11 · **Page 285** · lambda=h/p Open PDF ↗

(1) 

(2) 

(3) 

(4) 

10. When light propagates through a material medium of relative permittivity ε_r and relative permeability μ_r, the velocity of light, v is given by: (c – velocity of light in vacuum)

DIRECT — Class XII Part 1 · Ch 8 · **Page 207** · v=1/sqrt(mu eps) Open PDF ↗

(1) $v = \dfrac{c}{\sqrt{\varepsilon_r \mu_r}}$ **(2)** $v = c$
(3) $v = \sqrt{\dfrac{\mu_r}{\varepsilon_r}}$ **(4)** $v = \sqrt{\dfrac{\varepsilon_r}{\mu_r}}$

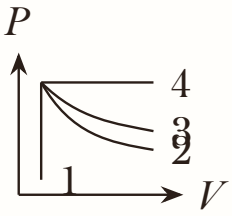
11. In the given nuclear reaction, the element X is:

$$^{22}_{11}\text{Na} \longrightarrow \text{X} + e^{+} + \nu$$

DELETED PORTION — alpha- and beta-decay · removed from the 2026-27 NCERT; not analysed

(1) $^{22}_{12}$Mg **(2)** $^{23}_{11}$Na **(3)** $^{23}_{10}$Ne **(4)** $^{22}_{10}$Ne

12. An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is:



APPLIED — Class XI Part 2 · Ch 11 · **Page 235** Open PDF ↗

(1) 4 **(2)** 1 **(3)** 2 **(4)** 3

13. The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second:

DIRECT — Class XI Part 1 · Ch 2 · **Page 20** · Example 2.5 Open PDF ↗

(1) 1:1:1:1 **(2)** 1:2:3:4
(3) 1:4:9:16 **(4)** 1:3:5:7

14. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is:

DIRECT — Class XI Part 1 · Ch 6 · **Page 96** · two-particle CM Open PDF ↗

(1) 5 m **(2)** $\dfrac{10}{3}$ m **(3)** $\dfrac{20}{3}$ m **(4)** 10 m

15. The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is:

DIRECT — Class XI Part 1 · Ch 5 · **Page 83** · energy = power x time Open PDF ↗

(1) 1×10^5 J **(2)** 36×10^7 J
(3) 36×10^4 J **(4)** 36×10^5 J

16. An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of $1.5\,m\,s^{-1}$. The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is: ($g = 10\,m\,s^{-2}$)

DIRECT Class XI Part 1 · Ch 5 · **Page 83** · Example 5.10 Open PDF ↗

(1) 23500 **(2)** 23000
(3) 20000 **(4)** 34500

17. A copper wire of length 10 m and radius $(10^{-2}/\sqrt{\pi})$ m has electrical resistance of 10 Ω. The current density in the wire for an electric field strength of 10 V/m is:

APPLIED Class XII Part 1 · Ch 3 · **Page 85** Open PDF ↗

(1) $10^5\,A/m^2$ **(2)** $10^4\,A/m^2$
(3) $10^6\,A/m^2$ **(4)** $10^{-5}\,A/m^2$

18. If a soap bubble expands, the pressure inside the bubble:

APPLIED Class XI Part 2 · Ch 9 · **Page 196** Open PDF ↗

(1) is equal to the atmospheric pressure
(2) decreases
(3) increases
(4) remains the same

19. Given below are two statements:
Statement I: Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current carrying conductor only.
Statement II: Biot-Savart's law is analogous to Coulomb's inverse square law of charge q, with the former being related to the field produced by a scalar source, Idl while the latter being produced by a vector source, q.
In light of above statements choose the most appropriate answer from the options given below:

DIRECT Class XII Part 1 · Ch 4 · **Page 114** · Biot-Savart law Open PDF ↗

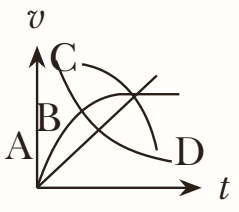
(1) Statement I is incorrect and Statement II is correct
(2) Both Statement I and Statement II are correct
(3) Both Statement I and Statement II are incorrect
(4) Statement I is correct and Statement II is incorrect

20. A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v, the speed of the third (lighter) fragment is:

APPLIED Class XI Part 1 · Ch 4 · **Page 57** Open PDF ↗

(1) $3\sqrt{2}\,v$ **(2)** v **(3)** $\sqrt{2}\,v$ **(4)** $2\sqrt{2}\,v$

21. A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball (v) as a function of time (t) is:



DIRECT Class XI Part 2 · Ch 9 · **Page 192** · terminal velocity Open PDF ↗

(1) D **(2)** A **(3)** B **(4)** C

22. The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s^2 is:

DIRECT Class XI Part 1 · Ch 6 · **Page 117** · Example 6.11 Open PDF ↗

(1) 104π **(2)** 2π **(3)** 4π **(4)** 12π

23. The peak voltage of the ac source is equal to:

DIRECT Class XII Part 1 · Ch 7 · **Page 180** · Vm=sqrt2 Vrms Open PDF ↗

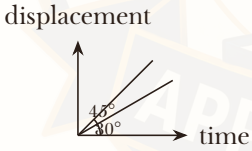
(1) $1/\sqrt{2}$ times the rms value of the ac source
(2) the value of voltage supplied to the circuit
(3) the rms value of the ac source
(4) $\sqrt{2}$ times the rms value of the ac source

24. The dimensions $[MLT^{-2}A^{-2}]$ belong to the:

DIRECT Class XII Part 1 · Ch 4 · **Page 134** · dimension table (mu0) Open PDF ↗

(1) electric permittivity
(2) magnetic flux
(3) self inductance
(4) magnetic permeability

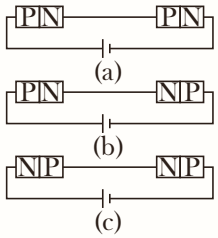
25. The displacement-time graphs of two moving particles make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is:



APPLIED Class XI Part 1 · Ch 2 · **Page 14** Open PDF ↗

(1) $1:\sqrt{3}$ **(2)** $\sqrt{3}:1$ **(3)** 1:1 **(4)** 1:2

26.



In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in:

APPLIED Class XII Part 2 · Ch 14 · **Page 335** Open PDF ↗

(1) Both circuits (a) and (c)
(2) Circuit (a) only
(3) Circuit (b) only
(4) Circuit (c) only

27. The angle between the electric lines of force and the equipotential surface is:

DIRECT Class XII Part 1 · Ch 2 · **Page 54** · E perpendicular to equipotential Open PDF ↗

(1) 180° **(2)** 0° **(3)** 45° **(4)** 90°

28. Plane angle and solid angle have:

DIRECT Class XI Part 1 · Ch 1 · **Page 2** · both are dimensionless (radian, steradian) Open PDF ↗

(1) Both units and dimensions
(2) Units but no dimensions
(3) Dimensions but no units
(4) No units and no dimensions

29. In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is:

APPLIED Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

(1) 12 **(2)** 6 **(3)** 8 **(4)** 9

30. A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60°. The angle between the refracted and reflected rays would be:

APPLIED Class XII Part 2 · Ch 9 · **Page 228** Open PDF ↗

(1) 120° **(2)** 30° **(3)** 60° **(4)** 90°

31. In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be:

DIRECT Class XII Part 2 · Ch 14 · **Page 342** · Exercise 14.6 Open PDF ↗

(1) 120 Hz **(2)** zero
(3) 30 Hz **(4)** 60 Hz

32. A body of mass 60 g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is:

DIRECT Class XI Part 1 · Ch 7 · **Page 134** · field = force per unit mass Open PDF ↗

(1) 180 N/kg **(2)** 0.05 N/kg
(3) 50 N/kg **(4)** 20 N/kg

33. A square loop of side 1 m and resistance 1 Ω is placed in a magnetic field of 0.5 T. If the plane of loop is perpendicular to the direction of magnetic field, the magnetic flux through the loop is:

DIRECT Class XII Part 1 · Ch 6 · **Page 156** · Phi=BA Open PDF ↗

(1) zero weber **(2)** 2 weber
(3) 0.5 weber **(4)** 1 weber

34. Let T_1 and T_2 be the energy of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is:

APPLIED Class XII Part 2 · Ch 12 · **Page 300** Open PDF ↗

(1) 9:4 **(2)** 1:4 **(3)** 4:1 **(4)** 4:9

35. A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is:

DIRECT Class XII Part 1 · Ch 4 · **Page 121** · B=mu0 n I Open PDF ↗

(1) 6.28×10^{-4} T **(2)** 6.28×10^{-2} T
(3) 12.56×10^{-2} T **(4)** 12.56×10^{-4} T

36. Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is:

APPLIED Class XI Part 2 · Ch 13 · **Page 271** Open PDF ↗

(1) 8 **(2)** 11 **(3)** 9 **(4)** 10

37. The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is:

APPLIED Class XI Part 2 · Ch 12 · **Page 246** Open PDF ↗

(1) 5.6 m^3 **(2)** 5.6×10^6 m^3
(3) 5.6×10^3 m^3 **(4)** 5.6×10^{-3} m^3

38. Two transparent media A and B are separated by a plane boundary. The speed of light in those media are 1.5×10^8 m/s and 2.0×10^8 m/s, respectively. The critical angle for a ray of light for these two media is:

APPLIED Class XII Part 2 · Ch 9 · **Page 229** Open PDF ↗

(1) $\tan^{-1}(0.750)$ **(2)** $\sin^{-1}(0.500)$
(3) $\sin^{-1}(0.750)$ **(4)** $\tan^{-1}(0.500)$

39. Match List-I with List-II:

List-I		*List-II*	
(a)	Gravitational constant (G)	(i)	$[L^2T^{-2}]$
(b)	Gravitational potential energy	(ii)	$[M^{-1}L^3T^{-2}]$
(c)	Gravitational potential	(iii)	$[LT^{-2}]$
(d)	Gravitational intensity	(iv)	$[ML^2T^{-2}]$

Choose the correct answer from the options given below:

APPLIED Class XI Part 1 · Ch 1 · **Page 7** Open PDF ↗

(1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

40. A series LCR circuit with inductance 10 H, capacitance 10 μF, resistance 50 Ω is connected to an ac source of voltage, $V = 200\sin(100t)$ volt. If the resonant frequency of the LCR circuit is ν_0 and the frequency of the ac source is ν, then:

APPLIED Class XII Part 1 · Ch 7 · **Page 188** Open PDF ↗

(1) $\nu = 100$ Hz; $\nu_0 = \dfrac{100}{\pi}$ Hz

(2) $\nu_0 = \nu = 50$ Hz

(3) $\nu_0 = \nu = \dfrac{50}{\pi}$ Hz

(4) $\nu_0 = \dfrac{50}{\pi}$ Hz, $\nu = 50$ Hz

41. A ball is projected with a velocity, 10 m s^{-1}, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be:

APPLIED Class XI Part 1 · Ch 3 · **Page 39** Open PDF ↗

(1) 10 m s^{-1} **(2)** Zero

(3) $5\sqrt{3}$ m s^{-1} **(4)** 5 m s^{-1}

42. From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is:

DIRECT Class XII Part 1 · Ch 4 · **Page 119** · Example 4.7 Open PDF ↗

(1) a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.

(2) uniform and remains constant for both the regions.

(3) a linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region.

(4) a linearly increasing function of distance r upto the boundary of the wire and then decreasing one with $1/r$ dependence for the outside region.

43. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The stretching of a spring is determined by the shear modulus of the material of the spring.

Reason (R): A coil spring of copper has more tensile strength than a steel spring of same dimensions.

In the light of the above statements, choose the most appropriate answer from the options given below:

APPLIED Class XI Part 2 · Ch 8 · **Page 172** Open PDF ↗

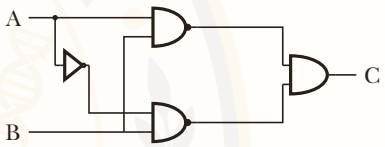
(1) (A) is false but (R) is true

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(4) (A) is true but (R) is false

44.



The truth table for the given logic circuit is:

DELETED PORTION digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(4)

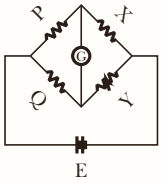
A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

45. A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is:

DIRECT Class XII Part 2 · Ch 13 · **Page 321** · Exercise 13.4 Open PDF ↗

(1) 25:16 **(2)** 1:1 **(3)** 4:5 **(4)** 5:4

46. A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistances P and Q:



APPLIED Class XII Part 1 · Ch 3 · **Page 101** Open PDF ↗

(1) do not play any significant role
(2) should be approximately equal to 2X
(3) should be approximately equal and are small
(4) should be very large and unequal

47. The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is:

DIRECT Class XI Part 1 · Ch 1 · **Page 5** · significant figures in a product Open PDF ↗

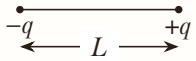
(1) 14×10^2 **(2)** 138×10^1
(3) 1382 **(4)** 1382.5

48. A big circular coil of 1000 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1}. If the vertical component of earth's magnetic field at that place is 2×10^{-5} T and electrical resistance of the coil is 12.56 Ω, then the maximum induced current in the coil will be:

APPLIED Class XII Part 1 · Ch 6 · **Page 171** Open PDF ↗

(1) 2 A **(2)** 0.25 A
(3) 1.5 A **(4)** 1 A

49. Two point charges $-q$ and $+q$ are placed at a distance of L, as shown in the figure.

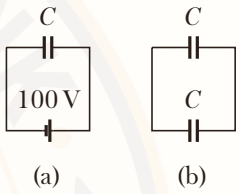


The magnitude of electric field intensity at a distance R ($R \gg L$) varies as:

DIRECT Class XII Part 1 · Ch 1 · **Page 25** · dipole field falls as 1/r^3 Open PDF ↗

(1) $\frac{1}{R^6}$ **(2)** $\frac{1}{R^2}$ **(3)** $\frac{1}{R^3}$ **(4)** $\frac{1}{R^4}$

50. A capacitor of capacitance $C = 900$ pF is charged fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance $C = 900$ pF as shown in figure (b). The electrostatic energy stored by the system (b) is:



DIRECT Class XII Part 1 · Ch 2 · **Page 75** · Example 2.10 Open PDF ↗

(1) 1.5×10^{-6} J **(2)** 4.5×10^{-6} J
(3) 3.25×10^{-6} J **(4)** 2.25×10^{-6} J

NEET 2022 · Answer Key

1 (3)	**2** (3)	**3** (1)	**4** (4)	**5** (4)	**6** (1)	**7** (3)	**8** (4)	**9** (1)	**10** (1)
11 (4)†	**12** (3)	**13** (4)	**14** (3)	**15** (2)	**16** (4)	**17** (1)	**18** (2)	**19** (4)	**20** (4)
21 (3)	**22** (3)	**23** (4)	**24** (4)	**25** (1)	**26** (1)	**27** (4)	**28** (2)	**29** (1)	**30** (4)
31 (4)	**32** (3)	**33** (3)	**34** (1)	**35** (3)	**36** (2)	**37** (1)	**38** (3)	**39** (3)	**40** (3)
41 (3)	**42** (4)	**43** (4)	**44** (4)†	**45** (4)	**46** (3)	**47** (1)	**48** (4)	**49** (3)	**50** (4)

Official NTA final key (Test Booklet Code R6); every answer was also solved and checked by the editor. (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness).

NEET 2022 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 50, 48 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**29**	60%
text, summary, table or figure	22	
NCERT solved Example	5	
NCERT Exercise	2	
Applied (concept in NCERT)	**19**	40%
Analysed	48	
Deleted portion (not analysed)	2	
Total	**50**	

Direct: Q 1, 2, 4, 5, 6, 7, 8, 9, 10, 13, 14, 15, 16, 19, 21, 22, 23, 24, 27, 28, 31, 32, 33, 35, 42, 45, 47, 49, 50.

From NCERT Examples: Q 13 (Example 2.5), 16 (Example 5.10), 22 (Example 6.11), 42 (Example 4.7), 50 (Example 2.10).

From NCERT Exercises: Q 31 (Exercise 14.6), 45 (Exercise 13.4).

2. Difficulty level

Easy	33
Moderate	15
Hard	0

The 2022 paper was easy: more than half the in-syllabus questions are direct NCERT statements, formulas, examples or exercises. Class XI: 20 questions, Class XII: 28 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	3	2	1
Motion in a Straight Line	XI	2	1	1
Motion in a Plane	XI	1	0	1
Laws of Motion	XI	1	0	1
Work, Energy & Power	XI	2	2	0
Rotational Motion	XI	3	3	0
Gravitation	XI	1	1	0
Mech. Prop. of Solids	XI	1	0	1
Mech. Prop. of Fluids	XI	2	1	1
Thermodynamics	XI	1	0	1
Kinetic Theory	XI	1	0	1
Oscillations	XI	1	0	1
Waves	XI	1	1	0
Electric Charges & Fields	XII	1	1	0
Electrostatic Potential & Capacitance	XII	3	3	0
Current Electricity	XII	4	2	2
Moving Charges & Magnetism	XII	4	4	0
Electromagnetic Induction	XII	2	1	1
Alternating Current	XII	2	1	1
Electromagnetic Waves	XII	2	2	0
Ray Optics	XII	3	1	2
Wave Optics	XII	1	0	1
Dual Nature	XII	2	1	1
Atoms	XII	1	0	1
Nuclei	XII	1	1	0
Semiconductor Electronics	XII	2	1	1

4. Teaching takeaways

- NCERT Examples 2.5, 2.10, 4.7, 5.10 and 6.11 and Exercises 13.4 and 14.6 appeared, several with the same numbers.
- Many questions test a single printed relation: $v = \sqrt{T/\mu}$, $P = V^2/R$, $B = \mu_0 nI$, $\Phi = BA$, $\lambda = h/p$.
- Only two questions were on deleted portions (β^+ decay and logic gates).
- Significant figures and the dimension table of Chapter 4 each gave a question.

5. Deleted portions

2 of the 50 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: alpha- and beta-decay (Q 11); digital electronics (logic gates) (Q 44).

NEET 2021
Physics Question Paper

12 September 2021 · Test Booklet Code M4 · Physics Q 1–50 (Section A: 1–35, Section B: 36–50, any 10 of 15) · 180 Marks

1. Consider the following statements (A) and (B) and identify the correct answer.

(A) A zener diode is connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

DELETED PORTION — Zener diode · removed from the 2026-27 NCERT; not analysed

(1) (A) and (B) both are correct.
(2) (A) and (B) both are incorrect.
(3) (A) is correct and (B) is incorrect.
(4) (A) is incorrect but (B) is correct.

2. A capacitor of capacitance 'C', is connected across an ac source of voltage V, given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would then be given by:

APPLIED — Class XII Part 1 · Ch 8 · **Page 203** Open PDF ↗

(1) $I_d = V_0 \omega C \cos \omega t$ **(2)** $I_d = \dfrac{V_0}{\omega C} \cos \omega t$

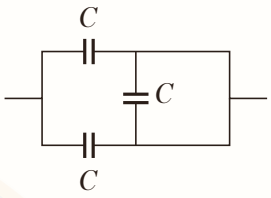
(3) $I_d = \dfrac{V_0}{\omega C} \sin \omega t$ **(4)** $I_d = V_0 \omega C \sin \omega t$

3. A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is:

DIRECT — Class XI Part 2 · Ch 13 · **Page 269** · K and U have period T/2 Open PDF ↗

(1) n **(2)** $2n$ **(3)** $3n$ **(4)** $4n$

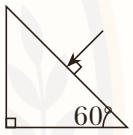
4. The equivalent capacitance of the combination shown in the figure is:



APPLIED — Class XII Part 1 · Ch 2 · **Page 72** Open PDF ↗

(1) $3C$ **(2)** $2C$ **(3)** $C/2$ **(4)** $3C/2$

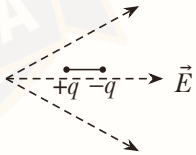
5. Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



APPLIED — Class XII Part 2 · Ch 9 · **Page 239** Open PDF ↗

(1) 60° **(2)** 30° **(3)** 45° **(4)** 90°

6. A dipole is placed in an electric field as shown. In which direction will it move?



APPLIED — Class XII Part 1 · Ch 1 · **Page 27** Open PDF ↗

(1) towards the left as its potential energy will increase.
(2) towards the right as its potential energy will decrease.
(3) towards the left as its potential energy will decrease.
(4) towards the right as its potential energy will increase.

7. Match Column-I and Column-II and choose the correct match from the given choices.

Column-I		*Column-II*	
(A)	Root mean square speed of gas molecules	(P)	$\frac{1}{3}nm\bar{v}^2$
(B)	Pressure exerted by ideal gas	(Q)	$\sqrt{\frac{3RT}{M}}$
(C)	Average kinetic energy of a molecule	(R)	$\frac{5}{2}RT$
(D)	Total internal energy of 1 mole of a diatomic gas	(S)	$\frac{3}{2}k_BT$

DIRECT Class XI Part 2 · Ch 12 · **Page 250** · vrms, P=(1/3)nm<v^2>, (3/2)kBT Open PDF ↗

(1) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)
(2) (A)-(Q), (B)-(R), (C)-(S), (D)-(P)
(3) (A)-(Q), (B)-(P), (C)-(S), (D)-(R)
(4) (A)-(R), (B)-(Q), (C)-(P), (D)-(S)

8. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be:

APPLIED Class XII Part 2 · Ch 9 · **Page 238** Open PDF ↗

(1) 25 **(2)** 15 **(3)** 50 **(4)** 30

9. If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) [F][A][T] **(2)** $[F][A][T^2]$
(3) $[F][A][T^{-1}]$ **(4)** $[F][A^{-1}][T]$

10. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C. The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is:

DIRECT Class XI Part 2 · Ch 10 · **Page 221** · Example 10.8 Open PDF ↗

(1) $\frac{13}{10}t$ **(2)** $\frac{13}{5}t$ **(3)** $\frac{10}{13}t$ **(4)** $\frac{5}{13}t$

11. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively:

APPLIED Class XI Part 1 · Ch 5 · **Page 78** Open PDF ↗

(1) $\frac{S}{4}, \frac{3gS}{2}$ **(2)** $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
(3) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$ **(4)** $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

12. The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be:

APPLIED Class XI Part 2 · Ch 9 · **Page 192** Open PDF ↗

(1) $\frac{Mg}{2}$ **(2)** Mg **(3)** $\frac{3}{2}Mg$ **(4)** $2Mg$

13. For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively?

APPLIED Class XII Part 1 · Ch 8 · **Page 207** Open PDF ↗

(1) $\hat{j}+\hat{k}, \hat{j}+\hat{k}$ **(2)** $-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$
(3) $\hat{j}+\hat{k}, -\hat{j}-\hat{k}$ **(4)** $-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

14. Column-I gives certain physical terms associated with flow of current through a metallic conductor. Column-II gives some mathematical relations involving electrical quantities. Match Column-I and Column-II with appropriate relations.

(A)	Drift Velocity	(P)	$\frac{m}{ne^2\rho}$
(B)	Electrical Resistivity	(Q)	nev_d
(C)	Relaxation Period	(R)	$\frac{eE}{m}\tau$
(D)	Current Density	(S)	$\frac{E}{J}$

DIRECT — Class XII Part 1 · Ch 3 · **Page 86** · vd, tau, rho, j relations Open PDF ↗

(1) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)
(2) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)
(3) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)
(4) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)

15. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is:

DIRECT — Class XI Part 2 · Ch 13 · **Page 267** · omega=sqrt(k/m) Open PDF ↗

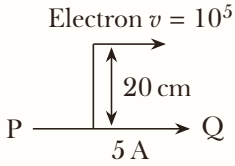
(1) 0.0628 s **(2)** 6.28 s
(3) 3.14 s **(4)** 0.628 s

16. A parallel plate capacitor has a uniform electric field '$\vec{E}$' in the space between the plates. If the distance between the plates is 'd' and the area of each plate is 'A', the energy stored in the capacitor is: (ε_0 = permittivity of free space)

DIRECT — Class XII Part 1 · Ch 2 · **Page 75** · energy density (1/2)eps0 E^2 Open PDF ↗

(1) $\frac{1}{2}\varepsilon_0 E^2$ **(2)** $\varepsilon_0 EAd$
(3) $\frac{1}{2}\varepsilon_0 E^2 Ad$ **(4)** $\frac{E^2 Ad}{\varepsilon_0}$

17. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



APPLIED — Class XII Part 1 · Ch 4 · **Page 118** Open PDF ↗

(1) 4×10^{-20} N **(2)** $8\pi \times 10^{-20}$ N
(3) $4\pi \times 10^{-20}$ N **(4)** 8×10^{-20} N

18. A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since:

DIRECT — Class XII Part 2 · Ch 9 · **Page 245** · light gathering and resolution Open PDF ↗

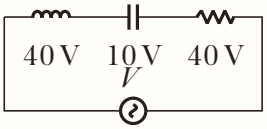
(1) a large aperture contributes to the quality and visibility of the images.
(2) a large area of the objective ensures better light gathering power.
(3) a large aperture provides a better resolution.
(4) all of the above.

19. A radioactive nucleus ${}^{A}_{Z}\mathrm{X}$ undergoes spontaneous decay in the sequence ${}^{A}_{Z}\mathrm{X} \rightarrow {}_{Z-1}\mathrm{B} \rightarrow {}_{Z-3}\mathrm{C} \rightarrow {}_{Z-2}\mathrm{D}$, where Z is the atomic number of element X. The possible decay particles in the sequence are:

DELETED PORTION — alpha- and beta-decay · removed from the 2026-27 NCERT; not analysed

(1) α, β^-, β^+ **(2)** α, β^+, β^-
(3) β^+, α, β^- **(4)** β^-, α, β^+

20. An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure. Potential difference across L, C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is:



APPLIED Class XII Part 1 · Ch 7 · **Page 187** Open PDF ↗

(1) $4\sqrt{2}\ \Omega$ (2) $5/\sqrt{2}\ \Omega$
(3) $4\ \Omega$ (4) $5\ \Omega$

21. The escape velocity from the Earth's surface is v. The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is:

APPLIED Class XI Part 1 · Ch 7 · **Page 136** Open PDF ↗

(1) v (2) $2v$ (3) $3v$ (4) $4v$

22. A screw gauge gives the following readings when used to measure the diameter of a wire: Main scale reading: 0 mm; Circular scale reading: 52 divisions. Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is:

APPLIED Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: screw gauge Open PDF ↗

(1) 0.52 cm (2) 0.026 cm
(3) 0.26 cm (4) 0.052 cm

23. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? ($g = 10\ m/s^2$)

APPLIED Class XI Part 1 · Ch 5 · **Page 83** Open PDF ↗

(1) 10.2 kW (2) 8.1 kW
(3) 12.3 kW (4) 7.0 kW

24. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\frac{S_n}{S_{n+1}}$ is:

DIRECT Class XI Part 1 · Ch 2 · **Page 20** · Example 2.5 Open PDF ↗

(1) $\frac{2n-1}{2n}$ (2) $\frac{2n-1}{2n+1}$
(3) $\frac{2n+1}{2n-1}$ (4) $\frac{2n}{2n-1}$

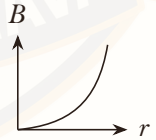
25. Polar molecules are the molecules:

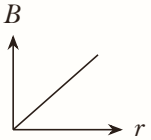
DIRECT Class XII Part 1 · Ch 1 · **Page 25** · polar molecules Open PDF ↗

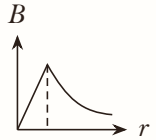
(1) having zero dipole moment.
(2) acquire a dipole moment only in the presence of electric field due to displacement of charges.
(3) acquire a dipole moment only when magnetic field is absent.
(4) having a permanent electric dipole moment.

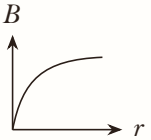
26. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by:

DIRECT Class XII Part 1 · Ch 4 · **Page 119** · Example 4.7 Open PDF ↗

(1) 

(2) 

(3) 

(4) 

27. In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs?

DELETED PORTION potentiometer · removed from the 2026-27 NCERT; not analysed

(1) 60 cm (2) 21.6 cm
(3) 64 cm (4) 62 cm

28. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is:

APPLIED Class XII Part 1 · Ch 2 · **Page 63** Open PDF ↗

(1) $\frac{R_1}{R_2}$ (2) $\frac{R_2}{R_1}$ (3) $\sqrt{\frac{R_1}{R_2}}$ (4) $\frac{R_1^2}{R_2^2}$

29. An electromagnetic wave of wavelength 'λ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then:

APPLIED Class XII Part 2 · Ch 11 · **Page 285** Open PDF ↗

(1) $\lambda = \left(\frac{2m}{hc}\right)\lambda_d^2$ (2) $\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$

(3) $\lambda = \left(\frac{2mc}{h}\right)\lambda_d^2$ (4) $\lambda = \left(\frac{2h}{mc}\right)\lambda_d^2$

30. The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

APPLIED Class XII Part 2 · Ch 14 · **Page 328** Open PDF ↗

(1) current in n-type = current in p-type.
(2) current in p-type > current in n-type.
(3) current in n-type > current in p-type.
(4) No current will flow in p-type, current will only flow in n-type.

31. The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be: $(h = 6.6 \times 10^{-34}\ \text{J s})$

DIRECT Class XII Part 2 · Ch 11 · **Page 284** · Example 11.1 Open PDF ↗

(1) 10^{18} (2) 10^{17} (3) 10^{16} (4) 10^{15}

32. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of:

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) $[M^2][L^{-1}][T^0]$ (2) $[M][L^{-1}][T^{-1}]$
(3) $[M][L^0][T^0]$ (4) $[M^2][L^{-2}][T^{-1}]$

33. The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25 Ω. What will be the effective resistance if they are connected in series?

APPLIED Class XII Part 1 · Ch 3 · **Page 98** Open PDF ↗

(1) 0.25 Ω (2) 0.5 Ω
(3) 1 Ω (4) 4 Ω

34. The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be:

DELETED PORTION radioactive decay law (half-life) · removed from the 2026-27 NCERT; not analysed

(1) 1/2 (2) $\frac{1}{2\sqrt{2}}$ (3) $\frac{2}{3}$ (4) $\frac{2}{3\sqrt{2}}$

35. A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is:

DIRECT Class XII Part 2 · Ch 13 · **Page 315** · 240 x 0.9 = 216 MeV Open PDF ↗

(1) 0.9 MeV (2) 9.4 MeV
(3) 804 MeV (4) 216 MeV

36. Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to:

DIRECT Class XII Part 1 · Ch 6 · **Page 167** · Example 6.8 Open PDF ↗

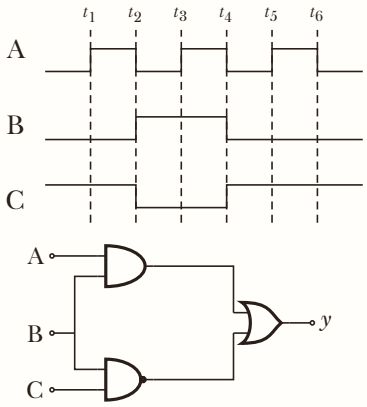
(1) $\frac{R_1}{R_2}$ (2) $\frac{R_2}{R_1}$ (3) $\frac{R_1^2}{R_2}$ (4) $\frac{R_2^2}{R_1}$

37. In the product $\vec{F} = q(\vec{v} \times \vec{B}) = q\vec{v} \times (B\hat{i} + B\hat{j} + B_0\hat{k})$. For $q = 1$ and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and $\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$. What will be the complete expression for $\vec{B}$?

APPLIED Class XII Part 1 · Ch 4 · **Page 109** Open PDF ↗

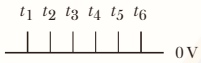
(1) $-8\hat{i} - 8\hat{j} - 6\hat{k}$ (2) $-6\hat{i} - 6\hat{j} - 8\hat{k}$
(3) $8\hat{i} + 8\hat{j} - 6\hat{k}$ (4) $6\hat{i} + 6\hat{j} - 8\hat{k}$

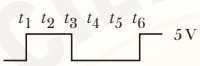
38. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y?

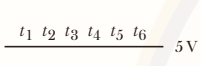


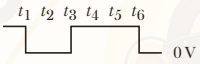
DELETED PORTION

digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) 

(2) 

(3) 

(4) 

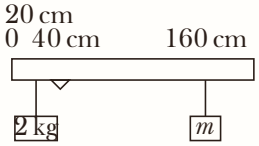
39. A series LCR circuit containing 5.0 H inductor, 80 μF capacitor and 40 Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be:

DELETED PORTION

sharpness of resonance (half-power frequencies) · removed from the 2026-27 NCERT; not analysed

(1) 25 rad/s and 75 rad/s
(2) 50 rad/s and 25 rad/s
(3) 46 rad/s and 54 rad/s
(4) 42 rad/s and 58 rad/s

40. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. (g = 10 m/s^2)



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(1) $\frac{1}{2}$ kg **(2)** $\frac{1}{3}$ kg **(3)** $\frac{1}{6}$ kg **(4)** $\frac{1}{12}$ kg

41. A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of:

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Class XII Part 2 · Ch 9 · **Page 238** Open PDF ↗

(1) 20 cm from the lens, it would be a real image.
(2) 30 cm from the lens, it would be a real image.
(3) 30 cm from the plane mirror, it would be a virtual image.
(4) 20 cm from the plane mirror, it would be a virtual image.

42. A car starts from rest and accelerates at 5 m/s^2. At t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at t = 6 s? (Take g = 10 m/s^2)

APPLIED

Class XI Part 1 · Ch 3 · **Page 37** Open PDF ↗

(1) 20 m/s, 5 m/s^2
(2) 20 m/s, 0
(3) $20\sqrt{2}$ m/s, 0
(4) $20\sqrt{2}$ m/s, 10 m/s^2

43. Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

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(1) 660 V **(2)** 1320 V
(3) 1520 V **(4)** 1980 V

44. A uniform conducting wire of length $12a$ and resistance 'R' is wound up as a current carrying coil in the shape of, (i) an equilateral triangle of side 'a', (ii) a square of side 'a'. The magnetic dipole moments of the coil in each case respectively are:

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(1) $\sqrt{3}Ia^2$ and $3Ia^2$ **(2)** $3Ia^2$ and Ia^2
(3) $3Ia^2$ and $4Ia^2$ **(4)** $4Ia^2$ and $3Ia^2$

45. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10\ \text{m/s}^2$) nearly:

APPLIED Class XI Part 1 · Ch 4 · **Page 55** Open PDF ↗

(1) 0 kg m/s **(2)** 4.2 kg m/s
(3) 2.1 kg m/s **(4)** 1.4 kg m/s

46. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit?

DIRECT Class XII Part 1 · Ch 7 · **Page 195** · Ip/Is=Ns/Np Open PDF ↗

(1) 0.2 A **(2)** 0.4 A **(3)** 2 A **(4)** 4 A

47. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution. If this particle were projected with the same speed at an angle 'θ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ, is then given by:

APPLIED Class XI Part 1 · Ch 3 · **Page 39** Open PDF ↗

(1) $\theta = \cos^{-1}\left(\dfrac{gT^2}{\pi^2 R}\right)^{1/2}$

(2) $\theta = \cos^{-1}\left(\dfrac{\pi^2 R}{gT^2}\right)^{1/2}$

(3) $\theta = \sin^{-1}\left(\dfrac{\pi^2 R}{gT^2}\right)^{1/2}$

(4) $\theta = \sin^{-1}\left(\dfrac{2gT^2}{\pi^2 R}\right)^{1/2}$

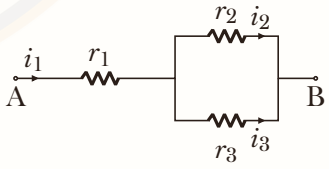
48. A particle of mass 'm' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth. (V_e = escape velocity). The maximum height above the surface reached by the particle is:

APPLIED Class XI Part 1 · Ch 7 · **Page 136** Open PDF ↗

(1) $R\left(\dfrac{k}{1-k}\right)^2$ **(2)** $R\left(\dfrac{k}{1+k}\right)^2$

(3) $\dfrac{R^2 k}{1+k}$ **(4)** $\dfrac{Rk^2}{1-k^2}$

49. Three resistors having resistances r_1, r_2 and r_3 are connected as shown in the given circuit. The ratio $\dfrac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is:



APPLIED Class XII Part 1 · Ch 3 · **Page 98** Open PDF ↗

(1) $\dfrac{r_1}{r_2 + r_3}$ **(2)** $\dfrac{r_2}{r_2 + r_3}$

(3) $\dfrac{r_1}{r_1 + r_2}$ **(4)** $\dfrac{r_2}{r_1 + r_3}$

50. From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times 'MR^2'. Then the value of 'K' is:

APPLIED Class XI Part 1 · Ch 6 · **Page 115** Open PDF ↗

(1) $\frac{3}{4}$ **(2)** $\frac{7}{8}$ **(3)** $\frac{1}{4}$ **(4)** $\frac{1}{8}$

NEET 2021 · Answer Key

1 (3)†	**2** (1)	**3** (2)	**4** (2)	**5** (1)	**6** (3)	**7** (3)	**8** (2)	**9** (2)	**10** (2)
11 (4)	**12** (1)	**13** (2)	**14** (1)	**15** (4)	**16** (3)	**17** (4)	**18** (4)	19 (3)†	**20** (4)*
21 (4)	**22** (4)	**23** (2)	**24** (2)	**25** (4)	**26** (3)	27 (1)†	**28** (2)	**29** (3)	**30** (3)
31 (3)	**32** (1)	**33** (4)	34 (2)†	**35** (4)	**36** (4)	**37** (2)	38 (3)†	39 (3)†	**40** (4)
41 (4)	**42** (4)	**43** (4)	**44** (1)	**45** (2)	**46** (1)	**47** (4)	**48** (4)	**49** (2)	**50** (1)

Official NTA final key (Test Booklet Code M4); every answer was also solved and checked by the editor. (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). *: see note.

Q20. Official key: 5 Ω, taking the given voltages as rms values (V_{rms} = 50 V, I_{rms} = 10 A).

NEET 2021 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 50, 44 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**14**	32%
text, summary, table or figure	9	
NCERT solved Example	5	
NCERT Exercise	0	
Applied (concept in NCERT)	**30**	68%
Analysed	44	
Deleted portion (not analysed)	6	
Total	**50**	

Direct: Q 3, 7, 10, 14, 15, 16, 18, 24, 25, 26, 31, 35, 36, 46. From NCERT Examples: Q 10 (Example 10.8), 24 (Example 2.5), 26 (Example 4.7), 31 (Example 11.1), 36 (Example 6.8).

2. Difficulty level

Easy	21
Moderate	21
Hard	2

Hard: Q 41, 47. The 2021 paper was easy to moderate; the harder questions are projectile, lens–mirror and escape-height applications. Class XI: 18 questions, Class XII: 26 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	3	0	3
Motion in a Straight Line	XI	1	1	0
Motion in a Plane	XI	2	0	2
Laws of Motion	XI	1	0	1
Work, Energy & Power	XI	2	0	2
Rotational Motion	XI	2	0	2
Gravitation	XI	2	0	2
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	1	1	0
Kinetic Theory	XI	1	1	0
Oscillations	XI	2	2	0
Electric Charges & Fields	XII	2	1	1
Electrostatic Potential & Capacitance	XII	4	1	3
Current Electricity	XII	3	1	2
Moving Charges & Magnetism	XII	4	1	3
Electromagnetic Induction	XII	1	1	0
Alternating Current	XII	2	1	1
Electromagnetic Waves	XII	2	0	2
Ray Optics	XII	4	1	3
Dual Nature	XII	2	1	1
Nuclei	XII	1	1	0
Semiconductor Electronics	XII	1	0	1

4. Teaching takeaways

- NCERT Examples 2.5, 4.7, 6.8, 10.8 and 11.1 each gave a question, some with the same numbers.
- The fission energy $240 \times 0.9 = 216$ MeV printed in Chapter 13 was asked directly.
- Six questions were on deleted portions (Zener, α/β decay, potentiometer, half-life, logic gates, half-power frequencies).
- Screw-gauge reading came from the Lab Manual.

5. Deleted portions

6 of the 50 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: Zener diode (Q 1); alpha- and beta-decay (Q 19); potentiometer (Q 27); radioactive decay law (half-life) (Q 34); digital electronics (logic gates) (Q 38); sharpness of resonance (half-power frequencies) (Q 39).

NEET 2020
Physics Question Paper

13 September 2020 · Test Booklet Code E1 · Physics Q 136–180 · 180 Marks

In this paper Physics is the last section; the original question numbers are kept.

136. For which one of the following, Bohr model is **not** valid?

DIRECT — Class XII Part 2 · Ch 12 · **Page 302** · Bohr model only for hydrogenic atoms Open PDF ↗

(1) Hydrogen atom
(2) Singly ionised helium atom (He^+)
(3) Deuteron atom
(4) Singly ionised neon atom (Ne^+)

137. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is: (c = speed of electromagnetic waves)

APPLIED — Class XII Part 1 · Ch 8 · **Page 207** · E and B carry equal energy Open PDF ↗

(1) $c : 1$ **(2)** 1:1 **(3)** $1 : c$ **(4)** $1 : c^2$

138. The Brewster's angle i_b for an interface should be:

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) $0° < i_b < 30°$ **(2)** $30° < i_b < 45°$
(3) $45° < i_b < 90°$ **(4)** $i_b = 90°$

139. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C. Its density is: ($R = 8.3\,J\,mol^{-1}K^{-1}$)

APPLIED — Class XI Part 2 · Ch 12 · **Page 246** Open PDF ↗

(1) $0.5\,kg/m^3$ **(2)** $0.2\,kg/m^3$
(3) $0.1\,kg/m^3$ **(4)** $0.02\,kg/m^3$

140. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, then the angle of incidence is nearly equal to:

APPLIED — Class XII Part 2 · Ch 9 · **Page 239** Open PDF ↗

(1) $\frac{A}{2\mu}$ **(2)** $\frac{2A}{\mu}$ **(3)** μA **(4)** $\frac{\mu A}{2}$

141. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is:

APPLIED — Class XI Part 2 · Ch 11 · **Page 236** · free expansion Open PDF ↗

(1) isothermal **(2)** adiabatic
(3) isochoric **(4)** isobaric

142. The energy equivalent of 0.5 g of a substance is:

DIRECT — Class XII Part 2 · Ch 13 · **Page 310** · Example 13.2 Open PDF ↗

(1) $4.5 \times 10^{16}\,J$ **(2)** $4.5 \times 10^{13}\,J$
(3) $1.5 \times 10^{13}\,J$ **(4)** $0.5 \times 10^{13}\,J$

143. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth?

DIRECT — Class XI Part 1 · Ch 7 · **Page 143** · Exercise 7.15 Open PDF ↗

(1) 48 N **(2)** 32 N **(3)** 30 N **(4)** 24 N

144. The solids which have the negative temperature coefficient of resistance are:

APPLIED · Class XII Part 1 · Ch 3 · **Page 91** Open PDF ↗

(1) metals
(2) insulators only
(3) semiconductors only
(4) insulators and semiconductors

145. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is:

DIRECT · Class XI Part 2 · Ch 13 · **Page 266** · a=-omega^2 x Open PDF ↗

(1) π rad **(2)** $\frac{3\pi}{2}$ rad
(3) $\frac{\pi}{2}$ rad **(4)** zero

146. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is:

APPLIED · Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: pitch = LC x divisions Open PDF ↗

(1) 0.01 mm **(2)** 0.25 mm
(3) 0.5 mm **(4)** 1.0 mm

147. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be:

DIRECT · Class XI Part 2 · Ch 14 · **Page 294** · Example 14.6 Open PDF ↗

(1) 523 Hz **(2)** 524 Hz
(3) 536 Hz **(4)** 537 Hz

148. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of:

DIRECT · Class XI Part 1 · Ch 6 · **Page 96** · two-particle CM Open PDF ↗

(1) 33 cm **(2)** 50 cm
(3) 67 cm **(4)** 80 cm

149. Find the torque about the origin when a force of $3\hat{j}$ N acts on a particle whose position vector is $2\hat{k}$ m.

DIRECT · Class XI Part 1 · Ch 6 · **Page 103** · vector product Open PDF ↗

(1) $6\hat{i}$ N m **(2)** $6\hat{j}$ N m
(3) $-6\hat{i}$ N m **(4)** $6\hat{k}$ N m

150. Light with an average flux of 20 W/cm^2 falls on a non-reflecting surface at normal incidence having surface area 20 cm^2. The energy received by the surface during time span of 1 minute is:

DELETED PORTION · intensity of EM waves · removed from the 2026-27 NCERT; not analysed

(1) 10×10^3 J **(2)** 12×10^3 J
(3) 24×10^3 J **(4)** 48×10^3 J

151. A spherical conductor of radius 10 cm has a charge of 3.2×10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere? $\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2\right)$

DIRECT · Class XII Part 1 · Ch 1 · **Page 35** · shell: field outside q/4 pi eps0 r^2 Open PDF ↗

(1) 1.28×10^4 N/C **(2)** 1.28×10^5 N/C
(3) 1.28×10^6 N/C **(4)** 1.28×10^7 N/C

152. In a certain region of space with volume 0.2 m^3, the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is:

DIRECT · Class XII Part 1 · Ch 2 · **Page 55** · E=-dV/dl Open PDF ↗

(1) zero **(2)** 0.5 N/C
(3) 1 N/C **(4)** 5 N/C

153. The increase in the width of the depletion region in a p-n junction diode is due to:

DIRECT · Class XII Part 2 · Ch 14 · **Page 336** · reverse bias widens depletion region Open PDF ↗

(1) forward bias only
(2) reverse bias only
(3) both forward bias and reverse bias
(4) increase in forward current

154. A 40 μF capacitor is connected to a 200 V, 50 Hz ac supply. The rms value of the current in the circuit is, nearly:

DIRECT — Class XII Part 1 · Ch 7 · **Page 185** · Example 7.4 Open PDF ↗

(1) 1.7 A **(2)** 2.05 A
(3) 2.5 A **(4)** 25.1 A

155. The mean free path for a gas, with molecular diameter d and number density n can be expressed as:

DIRECT — Class XI Part 2 · Ch 12 · **Page 254** · l=1/(sqrt2 n pi d^2) Open PDF ↗

(1) $\frac{1}{\sqrt{2}\,n\pi d}$ **(2)** $\frac{1}{\sqrt{2}\,n\pi d^2}$
(3) $\frac{1}{\sqrt{2}\,n^2\pi d^2}$ **(4)** $\frac{1}{\sqrt{2}\,n^2\pi^2 d^2}$

156. For transistor action, which of the following statements is **correct**?

DELETED PORTION — transistor · removed from the 2026-27 NCERT; not analysed

(1) Base, emitter and collector regions should have same doping concentrations.
(2) Base, emitter and collector regions should have same size.
(3) Both emitter junction as well as the collector junction are forward biased.
(4) The base region must be very thin and lightly doped.

157. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled?

APPLIED — Class XII Part 2 · Ch 11 · **Page 277** Open PDF ↗

(1) doubled **(2)** four times
(3) one-fourth **(4)** zero

158. When a uranium isotope $^{235}_{92}\mathrm{U}$ is bombarded with a neutron, it generates $^{89}_{36}\mathrm{Kr}$, three neutrons and:

DIRECT — Class XII Part 2 · Ch 13 · **Page 315** · U-235 fission equation Open PDF ↗

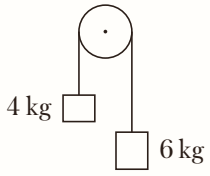
(1) $^{144}_{56}\mathrm{Ba}$ **(2)** $^{91}_{40}\mathrm{Zr}$ **(3)** $^{101}_{36}\mathrm{Kr}$ **(4)** $^{103}_{36}\mathrm{Kr}$

159. The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly:

DIRECT — Class XII Part 1 · Ch 2 · **Page 59** · 1 eV = 1.6e-19 J Open PDF ↗

(1) 6 **(2)** 0.6 **(3)** 0.06 **(4)** 0.006

160. Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is:



DIRECT — Class XI Part 1 · Ch 4 · **Page 70** · Exercise 4.16 Open PDF ↗

(1) g **(2)** $g/2$ **(3)** $g/5$ **(4)** $g/10$

161. A wire of length L, area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is:

DIRECT — Class XI Part 2 · Ch 8 · **Page 170** · Y=FL/(A dL) Open PDF ↗

(1) $\frac{MgL_1}{AL}$ **(2)** $\frac{Mg(L_1-L)}{AL}$
(3) $\frac{MgL}{AL_1}$ **(4)** $\frac{MgL}{A(L_1-L)}$

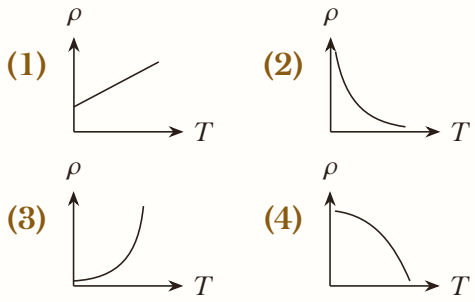
162. The average thermal energy for a monoatomic gas is: (k_B is Boltzmann constant and T, absolute temperature)

DIRECT — Class XI Part 2 · Ch 12 · **Page 250** · (3/2)kBT Open PDF ↗

(1) $\frac{1}{2}k_BT$ **(2)** $\frac{3}{2}k_BT$ **(3)** $\frac{5}{2}k_BT$ **(4)** $\frac{7}{2}k_BT$

163. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

DIRECT — Class XII Part 1 · Ch 3 · **Page 90** · resistivity of copper vs T (Fig 3.9) Open PDF ↗



164. The color code of a resistance is given below:



The values of resistance and tolerance, respectively, are:

DELETED PORTION — resistor colour code · removed from the 2026-27 NCERT; not analysed

(1) 470 kΩ, 5% **(2)** 47 kΩ, 10%
(3) 4.7 kΩ, 5% **(4)** 470 Ω, 5%

165. In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes:

APPLIED — Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

(1) double **(2)** half
(3) four times **(4)** one-fourth

166. The capacitance of a parallel plate capacitor with air as medium is 6 μF. With the introduction of a dielectric medium, the capacitance becomes 30 μF. The permittivity of the medium is: ($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2N^{-1}m^{-2}$)

DIRECT — Class XII Part 1 · Ch 2 · **Page 70** · eps=K eps0 Open PDF ↗

(1) $0.44 \times 10^{-13}\ C^2N^{-1}m^{-2}$
(2) $1.77 \times 10^{-12}\ C^2N^{-1}m^{-2}$
(3) $0.44 \times 10^{-10}\ C^2N^{-1}m^{-2}$
(4) $5.00\ C^2N^{-1}m^{-2}$

167. Dimensions of stress are:

DIRECT — Class XI Part 2 · Ch 8 · **Page 168** · stress unit Open PDF ↗

(1) $[MLT^{-2}]$ **(2)** $[ML^2T^{-2}]$
(3) $[ML^0T^{-2}]$ **(4)** $[ML^{-1}T^{-2}]$

168. Assume that light of wavelength 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is:

DELETED PORTION — resolving power of a telescope · removed from the 2026-27 NCERT; not analysed

(1) 3.66×10^{-7} rad **(2)** 1.83×10^{-7} rad
(3) 7.32×10^{-7} rad **(4)** 6.00×10^{-7} rad

169. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is:

APPLIED — Class XII Part 1 · Ch 7 · **Page 190** Open PDF ↗

(1) zero **(2)** 0.5 **(3)** 1.0 **(4)** −1.0

170. A short electric dipole has a dipole moment of 16×10^{-9} C m. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is: $\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N\,m^2/C^2\right)$

DIRECT — Class XII Part 1 · Ch 2 · **Page 50** · V=p cos theta/(4 pi eps0 r^2) Open PDF ↗

(1) 50 V **(2)** 200 V **(3)** 400 V **(4)** zero

171. An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1}. The permeability of the material of the rod is: ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

DIRECT — Class XII Part 1 · Ch 5 · **Page 146** · mu=mu0(1+chi) Open PDF ↗

(1) $2.4\pi \times 10^{-4}$ T m A^{-1}
(2) 8.0×10^{-5} T m A^{-1}
(3) $2.4\pi \times 10^{-5}$ T m A^{-1}
(4) $2.4\pi \times 10^{-7}$ T m A^{-1}

172. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is: ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

DIRECT Class XII Part 1 · Ch 4 · **Page 121** · B=mu0 n I Open PDF ↗

(1) 6.28×10^{-4} T (2) 3.14×10^{-4} T
(3) 6.28×10^{-5} T (4) 3.14×10^{-5} T

173. A charged particle having drift velocity of 7.5×10^{-4} m s^{-1} in an electric field of 3×10^{-10} V m^{-1}, has a mobility in m^{2}V^{-1}s^{-1} of:

DIRECT Class XII Part 1 · Ch 3 · **Page 88** · mu=vd/E Open PDF ↗

(1) 2.25×10^{15} (2) 2.5×10^{6}
(3) 2.5×10^{-6} (4) 2.25×10^{-15}

174. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5r_2$) through 1 K are in the ratio:

APPLIED Class XI Part 2 · Ch 10 · **Page 208** Open PDF ↗

(1) $\frac{27}{8}$ (2) $\frac{9}{4}$ (3) $\frac{3}{2}$ (4) $\frac{5}{3}$

175. An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is:

APPLIED Class XII Part 2 · Ch 11 · **Page 286** Open PDF ↗

(1) 10 V (2) 10^2 V (3) 10^3 V (4) 10^4 V

176. Taking into account of the significant figures, what is the value of 9.99 m – 0.0099 m?

DIRECT Class XI Part 1 · Ch 1 · **Page 5** · rounding in subtraction Open PDF ↗

(1) 9.9801 m (2) 9.98 m
(3) 9.980 m (4) 9.9 m

177. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: (g = 10 m/s^2)

DIRECT Class XI Part 1 · Ch 2 · **Page 17** · v^2=v0^2+2ax Open PDF ↗

(1) 360 m (2) 340 m
(3) 320 m (4) 300 m

178. A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of the water in the capillary is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is:

APPLIED Class XI Part 2 · Ch 9 · **Page 196** Open PDF ↗

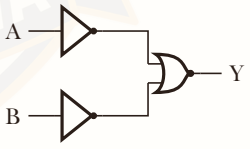
(1) 2.5 g (2) 5.0 g
(3) 10.0 g (4) 20.0 g

179. A resistance wire connected in the left gap of a metre bridge balances a 10 Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3:2. If the length of the resistance wire is 1.5 m, then the length of 1 Ω of the resistance wire is:

APPLIED Class XII Part 1 · Ch 3 · **Page 101** Open PDF ↗

(1) 1.0×10^{-2} m (2) 1.0×10^{-1} m
(3) 1.5×10^{-1} m (4) 1.5×10^{-2} m

180. For the logic circuit shown, the truth table is:



DELETED PORTION digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NEET 2020 · Answer Key

136 (4)	**137** (2)	**138** (3)†	**139** (2)	**140** (3)	**141** (2)	**142** (2)	**143** (2)	**144** (4)	**145** (1)
146 (3)	**147** (2)	**148** (3)	**149** (3)	**150** (3)†	**151** (2)	**152** (1)	**153** (2)	**154** (3)	**155** (2)
156 (4)†	**157** (4)	**158** (1)	**159** (3)	**160** (3)	**161** (4)	**162** (2)	**163** (3)	**164** (4)†	**165** (3)
166 (3)	**167** (4)	**168** (1)†	**169** (3)	**170** (2)	**171** (1)	**172** (1)	**173** (2)	**174** (1)	**175** (4)
176 (2)	**177** (4)	**178** (3)	**179** (2)	**180** (1)†					

Answers solved by the editor (the official final key of this year could not be obtained from the NTA/CBSE websites). (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness).

NEET 2020 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 39 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**26**	67%
text, summary, table or figure	21	
NCERT solved Example	3	
NCERT Exercise	2	
Applied (concept in NCERT)	**13**	33%
Analysed	39	
Deleted portion (not analysed)	6	
Total	**45**	

Direct: Q 136, 142, 143, 145, 147, 148, 149, 151, 152, 153, 154, 155, 158, 159, 160, 161, 162, 163, 166, 167, 170, 171, 172, 173, 176, 177.

From NCERT Examples: Q 142 (Example 13.2), 147 (Example 14.6), 154 (Example 7.4).

From NCERT Exercises: Q 143 (Exercise 7.15), 160 (Exercise 4.16).

2. Difficulty level

Easy	28
Moderate	11
Hard	0

The 2020 paper was the easiest of the eleven: most in-syllabus questions are single NCERT statements, formulas, examples or exercises. Class XI: 17 questions, Class XII: 22 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	2	1	1
Motion in a Straight Line	XI	1	1	0
Laws of Motion	XI	1	1	0
Rotational Motion	XI	2	2	0
Gravitation	XI	1	1	0
Mech. Prop. of Solids	XI	2	2	0
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	1	0	1
Thermodynamics	XI	1	0	1
Kinetic Theory	XI	3	2	1
Oscillations	XI	1	1	0
Waves	XI	1	1	0
Electric Charges & Fields	XII	1	1	0
Electrostatic Potential & Capacitance	XII	4	4	0
Current Electricity	XII	4	2	2
Moving Charges & Magnetism	XII	1	1	0
Magnetism & Matter	XII	1	1	0
Alternating Current	XII	2	1	1
Electromagnetic Waves	XII	1	0	1
Ray Optics	XII	1	0	1
Wave Optics	XII	1	0	1
Dual Nature	XII	2	0	2
Atoms	XII	1	1	0
Nuclei	XII	2	2	0
Semiconductor Electronics	XII	1	1	0

4. Teaching takeaways

- Exercise 7.15 (weight at height $R/2$), Exercise 4.16 (Atwood machine) and Examples 7.4, 13.2 and 14.6 were asked with new numbers.
- Twenty-six of the 39 in-syllabus questions are direct from the textbook.
- Six questions were on deleted portions (Brewster's law, EM intensity, transistor, colour code, telescope resolution, logic gates).
- Pitch of a screw gauge needs the Lab Manual.

5. Deleted portions

6 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: Brewster's law (Q 138); intensity of EM waves (Q 150); transistor (Q 156); resistor colour code (Q 164); resolving power of a telescope (Q 168); digital electronics (logic gates) (Q 180).

NEET 2019
Physics Question Paper

5 May 2019 · Test Booklet Code P1 · Physics Q 1–45 · 180 Marks

1. When a block of mass M is suspended by a long wire of length L, the length of the wire becomes $(L + l)$. The elastic potential energy stored in the extended wire is:

DIRECT — Class XI Part 2 · Ch 8 · **Page 174** · elastic PE = (1/2) F x extension Open PDF ↗

(1) Mgl **(2)** MgL
(3) $\frac{1}{2}Mgl$ **(4)** $\frac{1}{2}MgL$

2. A mass m is attached to a thin wire and whirled in a vertical circle. The wire is most likely to break when:

APPLIED — Class XI Part 1 · Ch 5 · **Page 79** · vertical circle (Example 5.7) Open PDF ↗

(1) the mass is at the highest point
(2) the wire is horizontal
(3) the mass is at the lowest point
(4) inclined at an angle of 60° from vertical

3. Ionized hydrogen atoms and α-particles with same momenta enters perpendicular to a constant magnetic field, B. The ratio of their radii of their paths $r_H : r_\alpha$ will be:

APPLIED — Class XII Part 1 · Ch 4 · **Page 112** Open PDF ↗

(1) 2:1 **(2)** 1:2 **(3)** 4:1 **(4)** 1:4

4. Body A of mass $4m$ moving with speed u collides with another body B of mass $2m$, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is:

APPLIED — Class XI Part 1 · Ch 5 · **Page 84** Open PDF ↗

(1) $\frac{1}{9}$ **(2)** $\frac{8}{9}$ **(3)** $\frac{4}{9}$ **(4)** $\frac{5}{9}$

5. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2°. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? ($\mu_{water} = 4/3$)

APPLIED — Class XII Part 2 · Ch 10 · **Page 267** Open PDF ↗

(1) 0.266° **(2)** 0.15°
(3) 0.05° **(4)** 0.1°

6. In which of the following devices, the eddy current effect is **not** used?

DELETED PORTION — eddy currents · removed from the 2026-27 NCERT; not analysed

(1) induction furnace
(2) magnetic braking in train
(3) electromagnet
(4) electric heater

7. A soap bubble, having radius of 1 mm, is blown from a detergent solution having a surface tension of 2.5×10^{-2} N/m. The pressure inside the bubble equals at a point Z_0 below the free surface of water in a container. Taking $g = 10\,m/s^2$, density of water = 10^3 kg/m^3, the value of Z_0 is:

APPLIED — Class XI Part 2 · Ch 9 · **Page 196** Open PDF ↗

(1) 100 cm **(2)** 10 cm
(3) 1 cm **(4)** 0.5 cm

8. Which colour of the light has the longest wavelength?

DIRECT — Class XII Part 1 · Ch 8 · **Page 210** · visible 700-400 nm Open PDF ↗

(1) red **(2)** blue **(3)** green **(4)** violet

9. A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it?

DELETED PORTION — kinetic energy of rolling · removed from the 2026-27 NCERT; not analysed

(1) 3 J **(2)** 30 kJ **(3)** 2 J **(4)** 1 J

10. The displacement of a particle executing simple harmonic motion is given by $y = A_0 + A\sin\omega t + B\cos\omega t$. Then the amplitude of its oscillation is given by:

DIRECT — Class XI Part 2 · Ch 13 · **Page 262** · A sin wt + B cos wt = D sin(wt+phi) Open PDF ↗

(1) $A_0 + \sqrt{A^2 + B^2}$ **(2)** $\sqrt{A^2 + B^2}$
(3) $\sqrt{A_0^2 + (A+B)^2}$ **(4)** $A + B$

11. Two similar thin equi-convex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1. When the space between the two lenses is filled with glycerin (which has the same refractive index ($\mu = 1.5$) as that of glass) then the equivalent focal length is F_2. The ratio $F_1 : F_2$ will be:

APPLIED — Class XII Part 2 · Ch 9 · **Page 238** Open PDF ↗

(1) 2:1 **(2)** 1:2 **(3)** 2:3 **(4)** 3:4

12. Increase in temperature of a gas filled in a container would lead to:

DIRECT — Class XI Part 2 · Ch 12 · **Page 250** · KE proportional to T Open PDF ↗

(1) increase in its mass
(2) increase in its kinetic energy
(3) decrease in its pressure
(4) decrease in intermolecular distance

13. An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is, (nearly): ($m_e = 9 \times 10^{-31}$ kg)

APPLIED — Class XII Part 2 · Ch 11 · **Page 286** Open PDF ↗

(1) 12.2×10^{-13} m **(2)** 12.2×10^{-12} m
(3) 12.2×10^{-14} m **(4)** 12.2 nm

14. A copper rod of 88 cm and an aluminium rod of unknown length have their increase in length independent of increase in temperature. The length of aluminium rod is: ($\alpha_{Cu} = 1.7 \times 10^{-5}\ K^{-1}$ and $\alpha_{Al} = 2.2 \times 10^{-5}\ K^{-1}$)

APPLIED — Class XI Part 2 · Ch 10 · **Page 205** Open PDF ↗

(1) 6.8 cm **(2)** 113.9 cm
(3) 88 cm **(4)** 68 cm

15. Pick the **wrong** answer in the context with rainbow.

DELETED PORTION — rainbow · removed from the 2026-27 NCERT; not analysed

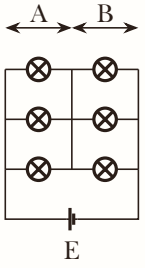
(1) When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.
(2) The order of colours is reversed in the secondary rainbow.
(3) An observer can see a rainbow when his front is towards the sun.
(4) Rainbow is a combined effect of dispersion, refraction and reflection of sunlight.

16. A body weighs 200 N on the surface of the earth. How much will it weigh half way down to the centre of the earth?

DIRECT — Class XI Part 1 · Ch 7 · **Page 143** · Exercise 7.16 Open PDF ↗

(1) 150 N **(2)** 200 N **(3)** 250 N **(4)** 100 N

17. Six similar bulbs are connected as shown in the figure with a DC source of emf E, and zero internal resistance. The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be:



APPLIED — Class XII Part 1 · Ch 3 · **Page 92** Open PDF ↗

(1) 4:9 **(2)** 9:4 **(3)** 1:2 **(4)** 2:1

18. For a p-type semiconductor, which of the following statements is **true**?

DIRECT — Class XII Part 2 · Ch 14 · **Page 342** · Exercise 14.2 Open PDF ↗

(1) Electrons are the majority carriers and trivalent atoms are the dopants.
(2) Holes are the majority carriers and trivalent atoms are the dopants.
(3) Holes are the majority carriers and pentavalent atoms are the dopants.
(4) Electrons are the majority carriers and pentavalent atoms are the dopants.

19. Average velocity of a particle executing SHM in one complete vibration is:

APPLIED — Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $\frac{A\omega}{2}$ **(2)** $A\omega$ **(3)** $\frac{A\omega^2}{2}$ **(4)** zero

20. The unit of thermal conductivity is:

DIRECT — Class XI Part 2 · Ch 10 · **Page 214** · unit of K Open PDF ↗

(1) $\mathrm{J\,m\,K^{-1}}$ **(2)** $\mathrm{J\,m^{-1}K^{-1}}$
(3) $\mathrm{W\,m\,K^{-1}}$ **(4)** $\mathrm{W\,m^{-1}K^{-1}}$

21. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is:

APPLIED — Class XI Part 1 · Ch 6 · **Page 117** Open PDF ↗

(1) 2×10^{-6} N m **(2)** 2×10^{-3} N m
(3) 12×10^{-4} N m **(4)** 2×10^{6} N m

22. A force $F = 20 + 10y$ acts on a particle in y-direction where F is in newton and y in meter. Work done by this force to move the particle from $y = 0$ to $y = 1$ m is:

DIRECT — Class XI Part 1 · Ch 5 · **Page 75** · W = integral of F dx Open PDF ↗

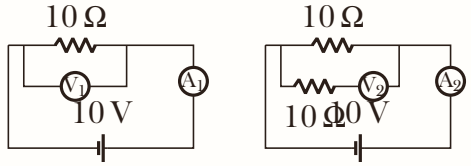
(1) 30 J **(2)** 5 J **(3)** 25 J **(4)** 20 J

23. Which of the following acts as a circuit protection device?

APPLIED — Class XII Part 1 · Ch 3 · **Page 92** · fuse (heating effect) Open PDF ↗

(1) conductor **(2)** inductor
(3) switch **(4)** fuse

24. In the circuits shown below, the readings of the voltmeters and the ammeters will be:



Circuit 1 Circuit 2

APPLIED — Class XII Part 1 · Ch 3 · **Page 83** Open PDF ↗

(1) $V_2 > V_1$ and $i_1 = i_2$
(2) $V_1 = V_2$ and $i_1 > i_2$
(3) $V_1 = V_2$ and $i_1 = i_2$
(4) $V_2 > V_1$ and $i_1 > i_2$

25. A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre:

DIRECT — Class XII Part 1 · Ch 1 · **Page 36** · shell: zero inside, kq/r^2 outside Open PDF ↗

(1) increases as r increases for $r < R$ and for $r > R$
(2) zero as r increases for $r < R$, decreases as r increases for $r > R$
(3) zero as r increases for $r < R$, increases as r increases for $r > R$
(4) decreases as r increases for $r < R$ and for $r > R$

26. At a point A on the earth's surface the angle of dip, $\delta = +25°$. At a point B on the earth's surface the angle of dip, $\delta = -25°$. We can interpret that:

DELETED PORTION — Earth's magnetism (angle of dip) · removed from the 2026-27 NCERT; not analysed

(1) A and B are both located in the northern hemisphere.
(2) A is located in the southern hemisphere and B is located in the northern hemisphere.
(3) A is located in the northern hemisphere and B is located in the southern hemisphere.
(4) A and B are both located in the southern hemisphere.

27. The total energy of an electron in an atom in an orbit is −3.4 eV. Its kinetic and potential energies are, respectively:

DIRECT Class XII Part 2 · Ch 12 · **Page 296** · K=-E, U=2E Open PDF ↗

(1) −3.4 eV, −3.4 eV **(2)** −3.4 eV, −6.8 eV
(3) 3.4 eV, −6.8 eV **(4)** 3.4 eV, 3.4 eV

28. In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?

DIRECT Class XII Part 2 · Ch 9 · **Page 229** · i = ic gives r = 90 deg Open PDF ↗

(1) 180°
(2) 0°
(3) equal to angle of incidence
(4) 90°

29. The work done to raise a mass m from the surface of the earth to a height h, which is equal to the radius of the earth, is:

APPLIED Class XI Part 1 · Ch 7 · **Page 134** Open PDF ↗

(1) mgR **(2)** $2mgR$
(3) $\frac{1}{2}mgR$ **(4)** $\frac{3}{2}mgR$

30. When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be:

APPLIED Class XI Part 1 · Ch 2 · **Page 17** Open PDF ↗

(1) $1 : \sqrt{2}$ **(2)** $\sqrt{2} : 1$
(3) $1 : \sqrt{3}$ **(4)** $1 : 2\sqrt{3}$

31. α-particle consists of:

DIRECT Class XII Part 2 · Ch 13 · **Page 314** · alpha = helium nucleus Open PDF ↗

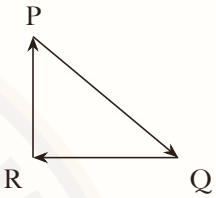
(1) 2 protons and 2 neutrons only
(2) 2 electrons, 2 protons and 2 neutrons
(3) 2 electrons and 4 protons only
(4) 2 protons only

32. The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by:

APPLIED Class XI Part 1 · Ch 3 · **Page 34** · Example 3.3 setting Open PDF ↗

(1) 30° west **(2)** 0°
(3) 60° west **(4)** 45° west

33. A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will:



DIRECT Class XI Part 1 · Ch 4 · **Page 58** · forces forming a closed triangle Open PDF ↗

(1) increase
(2) decrease
(3) remain constant
(4) change according to the smallest force $\overrightarrow{QR}$

34. Two particles A and B are moving in uniform circular motion in concentric circles of radii r_A and r_B with speed v_A and v_B respectively. Their time period of rotation is the same. The ratio of angular speed of A to that of B will be:

DIRECT Class XI Part 1 · Ch 3 · **Page 42** · omega = 2 pi/T Open PDF ↗

(1) $r_A : r_B$ **(2)** $v_A : v_B$
(3) $r_B : r_A$ **(4)** 1:1

35. A block of mass 10 kg is in contact against the inner wall of a hollow cylindrical drum of radius 1 m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis, will be: ($g = 10\,m/s^2$)

APPLIED Class XI Part 1 · Ch 4 · **Page 63** Open PDF ↗

(1) $\sqrt{10}$ rad/s **(2)** $\frac{10}{2\pi}$ rad/s
(3) 10 rad/s **(4)** 10π rad/s

36. Two parallel infinite line charges with linear charge densities $+\lambda$ C/m and $-\lambda$ C/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges?

APPLIED — Class XII Part 1 · Ch 1 · **Page 34** Open PDF ↗

(1) zero **(2)** $\frac{2\lambda}{\pi\varepsilon_0 R}$ N/C

(3) $\frac{\lambda}{\pi\varepsilon_0 R}$ N/C **(4)** $\frac{\lambda}{2\pi\varepsilon_0 R}$ N/C

37. Two point charges A and B, having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes:

APPLIED — Class XII Part 1 · Ch 1 · **Page 8** Open PDF ↗

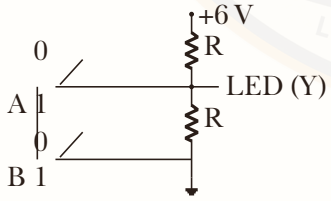
(1) F **(2)** $\frac{9F}{16}$ **(3)** $\frac{16F}{9}$ **(4)** $\frac{4F}{3}$

38. A small hole of area of cross-section 2 mm^2 is present near the bottom of a fully filled open tank of height 2 m. Taking $g = 10\ \text{m/s}^2$, the rate of flow of water through the open hole would be nearly:

DIRECT — Class XI Part 2 · Ch 9 · **Page 189** · v=sqrt(2gh) Open PDF ↗

(1) $12.6 \times 10^{-6}\ \text{m}^3/\text{s}$
(2) $8.9 \times 10^{-6}\ \text{m}^3/\text{s}$
(3) $2.23 \times 10^{-6}\ \text{m}^3/\text{s}$
(4) $6.4 \times 10^{-6}\ \text{m}^3/\text{s}$

39.



The **correct** Boolean operation represented by the circuit diagram drawn is:

DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) AND **(2)** OR
(3) NAND **(4)** NOR

40. In which of the following processes, heat is neither absorbed nor released by a system?

DIRECT — Class XI Part 2 · Ch 11 · **Page 234** · Table 11.2 Open PDF ↗

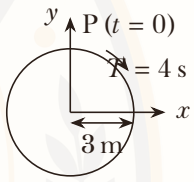
(1) isothermal **(2)** adiabatic
(3) isobaric **(4)** isochoric

41. A 800 turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field 5×10^{-5} T. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s, the emf induced in the coil will be:

DIRECT — Class XII Part 1 · Ch 6 · **Page 158** · emf = -N dPhi/dt Open PDF ↗

(1) 2 V **(2)** 0.2 V
(3) 2×10^{-3} V **(4)** 0.02 V

42. The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the fig.



y-projection of the radius vector of rotating particle P is:

DIRECT — Class XI Part 2 · Ch 13 · **Page 265** · reference circle Open PDF ↗

(1) $y(t) = -3\cos 2\pi t$, where y in m
(2) $y(t) = 4\sin\left(\frac{\pi t}{2}\right)$, where y in m
(3) $y(t) = 3\cos\left(\frac{3\pi t}{2}\right)$, where y in m
(4) $y(t) = 3\cos\left(\frac{\pi t}{2}\right)$, where y in m

43. A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3 V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively:

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(1) zero, 60 μA **(2)** 60 μA, 60 μA
(3) 60 μA, zero **(4)** zero, zero

44. In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement X, where $X = \dfrac{A^2B^{1/2}}{C^{1/3}D^3}$, will be:

DIRECT

Class XI Part 1 · Ch 1 · **Page 6** · relative errors add
Open PDF ↗

(1) $\left(\dfrac{3}{13}\right)\%$ **(2)** 16%

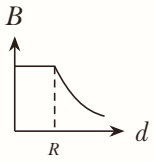
(3) −10% **(4)** 10%

45. A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field, B with the distance, d, from the centre of the conductor, is **correctly** represented by the figure:

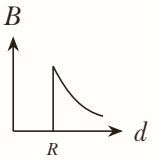
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Class XII Part 1 · Ch 4 · **Page 119** · Example 4.7
Open PDF ↗

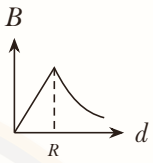
(1)



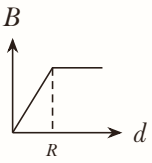
(2)



(3)



(4)



NEET 2019 · Answer Key

1 (3)	**2** (3)	**3** (1)	**4** (2)	**5** (2)	**6** (4)†	**7** (3)	**8** (1)	**9** (1)†	**10** (2)
11 (2)	**12** (2)	**13** (2)	**14** (4)	**15** (3)†	**16** (4)	**17** (2)	**18** (2)	**19** (4)	**20** (4)
21 (1)	**22** (3)	**23** (4)	**24** (3)	**25** (2)	**26** (3)†	**27** (3)	**28** (4)	**29** (3)	**30** (3)
31 (1)	**32** (1)	**33** (3)	**34** (4)	**35** (3)	**36** (3)	**37** (2)	**38** (1)	**39** (3)†	**40** (2)
41 (4)	**42** (4)	**43** (2)	**44** (2)	**45** (3)					

Answers solved by the editor (the official final key of this year could not be obtained from the NTA/CBSE websites). (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness).

NEET 2019 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 40 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**20**	50%
text, summary, table or figure	17	
NCERT solved Example	1	
NCERT Exercise	2	
Applied (concept in NCERT)	**20**	50%
Analysed	40	
Deleted portion (not analysed)	5	
Total	**45**	

Direct: Q 1, 8, 10, 12, 16, 18, 20, 22, 25, 27, 28, 31, 33, 34, 38, 40, 41, 42, 44, 45.

From NCERT Examples: Q 45 (Example 4.7).

From NCERT Exercises: Q 16 (Exercise 7.16), 18 (Exercise 14.2).

2. Difficulty level

Easy	20
Moderate	20
Hard	0

The 2019 paper was easy to moderate, with a balance of direct NCERT facts and short two-step applications. Class XI: 22 questions, Class XII: 18 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	1	1	0
Motion in a Straight Line	XI	1	0	1
Motion in a Plane	XI	2	1	1
Laws of Motion	XI	2	1	1
Work, Energy & Power	XI	3	1	2
Rotational Motion	XI	1	0	1
Gravitation	XI	2	1	1
Mech. Prop. of Solids	XI	1	1	0
Mech. Prop. of Fluids	XI	2	1	1
Thermal Prop. of Matter	XI	2	1	1
Thermodynamics	XI	1	1	0
Kinetic Theory	XI	1	1	0
Oscillations	XI	3	2	1
Electric Charges & Fields	XII	3	1	2
Current Electricity	XII	3	0	3
Moving Charges & Magnetism	XII	2	1	1
Electromagnetic Induction	XII	1	1	0
Electromagnetic Waves	XII	2	1	1
Ray Optics	XII	2	1	1
Wave Optics	XII	1	0	1
Dual Nature	XII	1	0	1
Atoms	XII	1	1	0
Nuclei	XII	1	1	0
Semiconductor Electronics	XII	1	1	0

4. Teaching takeaways

- Exercise 7.16 (weight half way down) and Exercise 14.2 (p-type carriers) were asked directly.
- Example 4.7 (field inside and outside a thick wire) appeared as a graph question.
- Five questions were on deleted portions (eddy currents, rolling, rainbow, angle of dip, logic gates).
- Percentage error, Torricelli's speed of efflux and $\varepsilon = -N\, d\Phi/dt$ gave one-line marks.

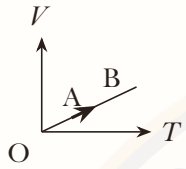
5. Deleted portions

5 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: eddy currents (Q 6); kinetic energy of rolling (Q 9); rainbow (Q 15); Earth's magnetism (angle of dip) (Q 26); digital electronics (logic gates) (Q 39).

NEET 2018
Physics Question Paper

6 May 2018 · Test Booklet Code AA · Physics Q 1–45 · 180 Marks

1. The volume (V) of a monatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to state B, is



APPLIED · Class XI Part 2 · Ch 11 · **Page 235** Open PDF ↗

(1) $\frac{1}{3}$ **(2)** $\frac{2}{3}$ **(3)** $\frac{2}{5}$ **(4)** $\frac{2}{7}$

2. The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is

APPLIED · Class XI Part 2 · Ch 14 · **Page 292** Open PDF ↗

(1) 12.5 cm **(2)** 8 cm
(3) 13.2 cm **(4)** 16 cm

3. At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere? (Given: Mass of oxygen molecule $m = 2.76\times10^{-26}$ kg, Boltzmann's constant $k_B = 1.38 \times 10^{-23}\,\text{J K}^{-1}$)

APPLIED · Class XI Part 2 · Ch 12 · **Page 250** Open PDF ↗

(1) 5.016×10^4 K **(2)** 8.360×10^4 K
(3) 2.508×10^4 K **(4)** 1.254×10^4 K

4. The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is

DIRECT · Class XI Part 2 · Ch 11 · **Page 238** · eta=1-T2/T1 Open PDF ↗

(1) 6.25% **(2)** 20% **(3)** 26.8% **(4)** 12.5%

5. A carbon resistor of $(47 \pm 4.7)\,\text{k}\Omega$ is to be marked with rings of different colours for its identification. The colour code sequence will be

DELETED PORTION · resistor colour code · removed from the 2026-27 NCERT; not analysed

(1) Yellow – Green – Violet – Gold
(2) Yellow – Violet – Orange – Silver
(3) Violet – Yellow – Orange – Silver
(4) Green – Orange – Violet – Gold

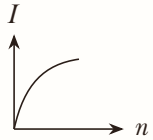
6. A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes $10I$. The value of 'n' is

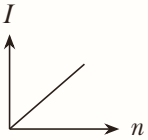
APPLIED · Class XII Part 1 · Ch 3 · **Page 95** Open PDF ↗

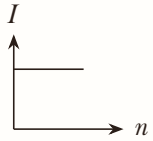
(1) 20 **(2)** 11 **(3)** 10 **(4)** 9

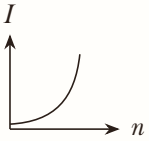
7. A battery consists of a variable number 'n' of identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n?

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(1) 

(2) 

(3) 

(4) 

8. Unpolarised light is incident from air on a plane surface of a material of refractive index 'μ'. At a particular angle of incidence 'i', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation?

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) $i = \sin^{-1}\left(\frac{1}{\mu}\right)$

(2) Reflected light is polarised with its electric vector perpendicular to the plane of incidence

(3) Reflected light is polarised with its electric vector parallel to the plane of incidence

(4) $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

9. In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 Å and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20°. To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to

APPLIED — Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

(1) 2.1 mm **(2)** 1.9 mm
(3) 1.8 mm **(4)** 1.7 mm

10. An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of

DIRECT — Class XII Part 2 · Ch 9 · **Page 245** · light gathering and resolution Open PDF ↗

(1) large focal length and large diameter
(2) large focal length and small diameter
(3) small focal length and large diameter
(4) small focal length and small diameter

11. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is

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(1) 2 : −1 **(2)** 1 : −1 **(3)** 1:1 **(4)** 1 : −2

12. An electron of mass m with an initial velocity $\vec{V} = V_0\hat{i}$ ($V_0 > 0$) enters an electric field $\vec{E} = -E_0\hat{i}$ (E_0 = constant > 0) at $t = 0$. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is

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(1) $\lambda_0 t$ **(2)** $\lambda_0\left(1 + \frac{eE_0}{mV_0}t\right)$

(3) $\frac{\lambda_0}{\left(1 + \frac{eE_0}{mV_0}t\right)}$ **(4)** λ_0

13. For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the disintegration of 450 nuclei is

DELETED PORTION — radioactive decay law (half-life) · removed from the 2026-27 NCERT; not analysed

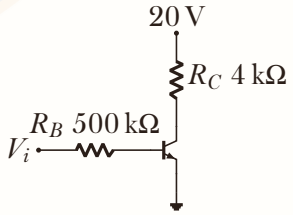
(1) 30 **(2)** 10 **(3)** 20 **(4)** 15

14. When the light of frequency $2\nu_0$ (where ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is

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(1) 4:1 **(2)** 1:4 **(3)** 1:2 **(4)** 2:1

15. In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0$ and $V_{CE} = 0$. The values of I_B, I_C and β are given by



DELETED PORTION — transistor · removed from the 2026-27 NCERT; not analysed

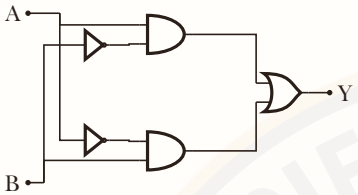
(1) $I_B = 20\ \mu A$, $I_C = 5$ mA, $\beta = 250$
(2) $I_B = 25\ \mu A$, $I_C = 5$ mA, $\beta = 200$
(3) $I_B = 40\ \mu A$, $I_C = 10$ mA, $\beta = 250$
(4) $I_B = 40\ \mu A$, $I_C = 5$ mA, $\beta = 125$

16. In a p-n junction diode, change in temperature due to heating

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(1) does not affect resistance of p-n junction
(2) affects only forward resistance
(3) affects only reverse resistance
(4) affects the overall V–I characteristics of p-n junction

17. In the combination of the following gates the output Y can be written in terms of inputs A and B as



DELETED PORTION digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) $\overline{A \cdot B} + A \cdot B$ **(2)** $A \cdot \bar{B} + \bar{A} \cdot B$
(3) $\overline{A \cdot B}$ **(4)** $\overline{A + B}$

18. An em wave is propagating in a medium with a velocity $\vec{V} = V\hat{i}$. The instantaneous oscillating electric field of this em wave is along $+y$ axis. Then the direction of oscillating magnetic field of the em wave will be along

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(1) $-y$ direction **(2)** $+z$ direction
(3) $-z$ direction **(4)** $-x$ direction

19. The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is

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(1) 30° **(2)** 45° **(3)** 60° **(4)** zero

20. An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be

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(1) 30 cm towards the mirror
(2) 36 cm away from the mirror
(3) 30 cm away from the mirror
(4) 36 cm towards the mirror

21. The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance

DIRECT Class XII Part 1 · Ch 6 · **Page 169** · U=LI^2/2 Open PDF ↗

(1) 1.389 H **(2)** 138.88 H
(3) 0.138 H **(4)** 13.89 H

22. An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

DIRECT Class XII Part 1 · Ch 1 · **Page 17** · Example 1.7 Open PDF ↗

(1) 10 times greater **(2)** 5 times greater
(3) smaller **(4)** equal

23. The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

DELETED PORTION force between capacitor plates · removed from the 2026-27 NCERT; not analysed

(1) proportional to the square root of the distance between the plates.
(2) linearly proportional to the distance between the plates.
(3) independent of the distance between the plates.
(4) inversely proportional to the distance between the plates.

24. A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is

DIRECT Class XI Part 2 · Ch 14 · **Page 299** · Exercise 14.15 Open PDF ↗

(1) 350 m/s **(2)** 339 m/s
(3) 330 m/s **(4)** 300 m/s

25. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is

DIRECT Class XI Part 2 · Ch 13 · **Page 266** · a=-omega^2 x Open PDF ↗

(1) 2 s **(2)** π s **(3)** 2π s **(4)** 1 s

26. A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is

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(1) 14.76 A **(2)** 5.98 A
(3) 7.14 A **(4)** 11.32 A

27. A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from

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(1) the lattice structure of the material of the rod
(2) the magnetic field
(3) the current source
(4) the induced electric field due to the changing magnetic field

28. An inductor 20 mH, a capacitor 100 μF and a resistor 50 Ω are connected in series across a source of emf, $V = 10 \sin 314t$. The power loss in the circuit is

APPLIED Class XII Part 1 · Ch 7 · **Page 190** Open PDF ↗

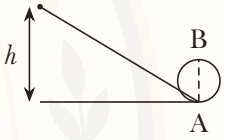
(1) 2.74 W **(2)** 0.43 W
(3) 0.79 W **(4)** 1.13 W

29. Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is

DIRECT Class XII Part 1 · Ch 4 · **Page 130** · current and voltage sensitivity Open PDF ↗

(1) 250 Ω **(2)** 25 Ω **(3)** 40 Ω **(4)** 500 Ω

30. A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter $AB = D$. The height h is equal to



APPLIED Class XI Part 1 · Ch 5 · **Page 79** · Example 5.7 setting Open PDF ↗

(1) $\frac{7}{5}D$ **(2)** D **(3)** $\frac{3}{2}D$ **(4)** $\frac{5}{4}D$

31. Three objects, A: (a solid sphere), B: (a thin circular disk) and C: (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation

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(1) $W_B > W_A > W_C$ **(2)** $W_A > W_B > W_C$
(3) $W_C > W_B > W_A$ **(4)** $W_A > W_C > W_B$

32. A moving block having mass m, collides with another stationary block having mass $4m$. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v, then the value of coefficient of restitution (e) will be

DELETED PORTION — coefficient of restitution · removed from the 2026-27 NCERT; not analysed

(1) 0.8 **(2)** 0.25 **(3)** 0.5 **(4)** 0.4

33. Which one of the following statements is *incorrect*?

DIRECT — Class XI Part 1 · Ch 4 · **Page 60** · mu is dimensionless Open PDF ↗

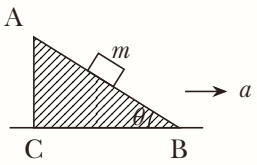
(1) Frictional force opposes the relative motion.
(2) Limiting value of static friction is directly proportional to normal reaction.
(3) Rolling friction is smaller than sliding friction.
(4) Coefficient of sliding friction has dimensions of length.

34. A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\vec{E}$. Due to the force $q\vec{E}$, its velocity increases from 0 to 6 m/s in one second duration. At that instant the direction of the field is reversed. The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 seconds are respectively

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(1) 1 m/s, 3.5 m/s **(2)** 1 m/s, 3 m/s
(3) 2 m/s, 4 m/s **(4)** 1.5 m/s, 3 m/s

35. A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and θ for the block to remain stationary on the wedge is



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(1) $a = g\cos\theta$ **(2)** $a = \dfrac{g}{\sin\theta}$
(3) $a = \dfrac{g}{\operatorname{cosec}\theta}$ **(4)** $a = g\tan\theta$

36. The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at $(2, 0, -3)$, about the point $(2, -2, -2)$, is given by

APPLIED — Class XI Part 1 · Ch 6 · **Page 106** Open PDF ↗

(1) $-7\hat{i} - 8\hat{j} - 4\hat{k}$ **(2)** $-4\hat{i} - \hat{j} - 8\hat{k}$
(3) $-8\hat{i} - 4\hat{j} - 7\hat{k}$ **(4)** $-7\hat{i} - 4\hat{j} - 8\hat{k}$

37. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of –0.004 cm, the correct diameter of the ball is

APPLIED — Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: screw gauge zero error Open PDF ↗

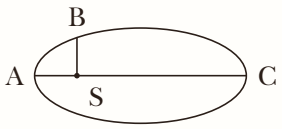
(1) 0.053 cm **(2)** 0.525 cm
(3) 0.521 cm **(4)** 0.529 cm

38. A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere?

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(1) Rotational kinetic energy
(2) Moment of inertia
(3) Angular velocity
(4) Angular momentum

39. The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A, K_B and K_C, respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then



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(1) $K_B < K_A < K_C$ **(2)** $K_A > K_B > K_C$
(3) $K_A < K_B < K_C$ **(4)** $K_B > K_A > K_C$

40. If the mass of the Sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is *not* correct?

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(1) Time period of a simple pendulum on the Earth would decrease.
(2) Walking on the ground would become more difficult.
(3) Raindrops will fall faster.
(4) 'g' on the Earth will not change.

41. A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is

DELETED PORTION — kinetic energy of rolling · removed from the 2026-27 NCERT; not analysed

(1) 10:7 **(2)** 5:7 **(3)** 7:10 **(4)** 2:5

42. A small sphere of radius 'r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to

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(1) r^5 **(2)** r^2 **(3)** r^3 **(4)** r^4

43. The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_0. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. The value of n is

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(1) $\frac{256}{81}$ **(2)** $\frac{4}{3}$ **(3)** $\frac{3}{4}$ **(4)** $\frac{81}{256}$

44. Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3A. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

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(1) 4F **(2)** 6F **(3)** 9F **(4)** F

45. A sample of 0.1 g of water at 100°C and normal pressure (1.013×10^5 N m^{-2}) requires 54 cal of heat energy to convert to steam at 100°C. If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample, is

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(1) 42.2 J **(2)** 208.7 J
(3) 104.3 J **(4)** 84.5 J

NEET 2018 · Answer Key

1 (3)	**2** (3)	**3** (2)	**4** (3)	5 (2)†	**6** (3)	**7** (3)	8 (2)†	**9** (2)	**10** (1)
11 (2)	**12** (3)	13 (3)†	**14** (3)	15 (4)†	**16** (4)	17 (2)†	**18** (2)	**19** (2)	**20** (2)
21 (4)	**22** (3)	23 (3)†	**24** (2)	**25** (2)	**26** (4)	**27** (3)	**28** (3)	**29** (1)	**30** (4)
31 (3)	32 (2)†	**33** (4)	**34** (2)	**35** (4)	**36** (4)	**37** (4)	**38** (4)	**39** (2)	**40** (4)
41 (2)†	**42** (1)	**43** (1)	**44** (3)	**45** (2)					

Answers solved by the editor (the official final key of this year could not be obtained from the NTA/CBSE websites). (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness).

NEET 2018 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 37 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**11**	30%
text, summary, table or figure	8	
NCERT solved Example	2	
NCERT Exercise	1	
Applied (concept in NCERT)	**26**	70%
Analysed	37	
Deleted portion (not analysed)	8	
Total	**45**	

Direct: Q 4, 10, 11, 21, 22, 24, 25, 29, 33, 38, 39.
From NCERT Examples: Q 22 (Example 1.7), 39 (Example 7.1).
From NCERT Exercises: Q 24 (Exercise 14.15).

2. Difficulty level

Easy	11
Moderate	25
Hard	1

Hard: Q 28. The 2018 paper was moderate: many questions need two or three steps, and eight were on topics now deleted. Class XI: 20 questions, Class XII: 17 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	1	0	1
Motion in a Straight Line	XI	1	0	1
Laws of Motion	XI	2	1	1
Work, Energy & Power	XI	1	0	1
Rotational Motion	XI	3	1	2
Gravitation	XI	2	1	1
Mech. Prop. of Solids	XI	1	0	1
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	1	0	1
Thermodynamics	XI	3	1	2
Kinetic Theory	XI	1	0	1
Oscillations	XI	1	1	0
Waves	XI	2	1	1
Electric Charges & Fields	XII	1	1	0
Current Electricity	XII	2	0	2
Moving Charges & Magnetism	XII	2	1	1
Magnetism & Matter	XII	1	0	1
Electromagnetic Induction	XII	1	1	0
Alternating Current	XII	1	0	1
Electromagnetic Waves	XII	1	0	1
Ray Optics	XII	3	1	2
Wave Optics	XII	1	0	1
Dual Nature	XII	2	0	2
Atoms	XII	1	1	0
Semiconductor Electronics	XII	1	0	1

4. Teaching takeaways

- Exercise 14.15 (resonance tube) and Examples 1.7 and 7.1 were asked again.
- Eight questions were on deleted portions (colour code, Brewster's law, half-life, transistor, logic gates, force between plates, restitution, rolling).
- Mechanics questions in this paper are multi-step (wedge, toy car, loop-the-loop).
- Galvanometer sensitivities and $\frac{1}{2}LI^2$ gave quick direct marks.

5. Deleted portions

8 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: resistor colour code (Q 5); Brewster's law (Q 8); radioactive decay law (half-life) (Q 13); transistor (Q 15); digital electronics (logic gates) (Q 17); force between capacitor plates (Q 23); coefficient of restitution (Q 32); kinetic energy of rolling (Q 41).

NEET 2017
Physics Question Paper

7 May 2017 · Test Booklet Code B · Physics Q 136–180 · 180 Marks

In this paper Physics is the last section; the original question numbers are kept.

136. Two astronauts are floating in gravitational free space after having lost contact with their spaceship. The two will:

APPLIED — Class XI Part 1 · Ch 7 · **Page 129** · law of gravitation Open PDF ↗

(1) move towards each other.
(2) move away from each other.
(3) will become stationary.
(4) keep floating at the same distance between them.

137. Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly:

APPLIED — Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

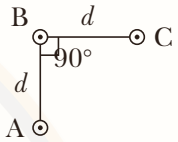
(1) 1.59 **(2)** 1.69 **(3)** 1.78 **(4)** 1.25

138. In a common emitter transistor amplifier the audio signal voltage across the collector is 3 V. The resistance of collector is 3 kΩ. If current gain is 100 and the base resistance is 2 kΩ, the voltage and power gain of the amplifier is:

DELETED PORTION — transistor (CE amplifier) · removed from the 2026-27 NCERT; not analysed

(1) 15 and 200 **(2)** 150 and 15000
(3) 20 and 2000 **(4)** 200 and 1000

139. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I' along the same direction is shown in Fig. Magnitude of force per unit length on the middle wire 'B' is given by:



APPLIED — Class XII Part 1 · Ch 4 · **Page 123** Open PDF ↗

(1) $\frac{2\mu_0 i^2}{\pi d}$ **(2)** $\frac{\sqrt{2}\mu_0 i^2}{\pi d}$
(3) $\frac{\mu_0 i^2}{\sqrt{2}\pi d}$ **(4)** $\frac{\mu_0 i^2}{2\pi d}$

140. The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is:

APPLIED — Class XII Part 2 · Ch 11 · **Page 285** · thermal KE 3kT/2 Open PDF ↗

(1) $\frac{h}{\sqrt{3mkT}}$ **(2)** $\frac{2h}{\sqrt{3mkT}}$
(3) $\frac{2h}{\sqrt{mkT}}$ **(4)** $\frac{h}{\sqrt{mkT}}$

141. Radioactive material 'A' has decay constant '8λ' and material 'B' has decay constant 'λ'. Initially they have same number of nuclei. After what time, the ratio of number of nuclei of material 'B' to that 'A' will be $\frac{1}{e}$?

DELETED PORTION — radioactive decay law (decay constant) · removed from the 2026-27 NCERT; not analysed

(1) $\frac{1}{7\lambda}$ **(2)** $\frac{1}{8\lambda}$ **(3)** $\frac{1}{9\lambda}$ **(4)** $\frac{1}{\lambda}$

142. The given electrical network is equivalent to:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) OR gate (2) NOR gate
(3) NOT gate (4) AND gate

143. The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length, its new resistance will be:

APPLIED — Class XII Part 1 · Ch 3 · **Page 84** · R=rho l/A, volume constant Open PDF ↗

(1) $\frac{R}{n}$ (2) n^2R (3) $\frac{R}{n^2}$ (4) nR

144. Two cars moving in opposite directions approach each other with speed of 22 m/s and 16.5 m/s respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is [velocity of sound 340 m/s]:

DELETED PORTION — Doppler effect · removed from the 2026-27 NCERT; not analysed

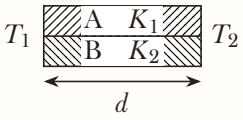
(1) 361 Hz (2) 411 Hz
(3) 448 Hz (4) 350 Hz

145. In an electromagnetic wave in free space the root mean square value of the electric field is $E_{rms} = 6\,V/m$. The peak value of the magnetic field is:

APPLIED — Class XII Part 1 · Ch 8 · **Page 212** · E0/B0=c Open PDF ↗

(1) 2.83×10^{-8} T (2) 0.70×10^{-8} T
(3) 4.23×10^{-8} T (4) 1.41×10^{-8} T

146. Two rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2. The thermal conductivity of the composite rod will be:



APPLIED — Class XI Part 2 · Ch 10 · **Page 216** · rods side by side (parallel) Open PDF ↗

(1) $\frac{3(K_1+K_2)}{2}$ (2) K_1+K_2
(3) $2(K_1+K_2)$ (4) $\frac{K_1+K_2}{2}$

147. Which one of the following represents forward bias diode?

APPLIED — Class XII Part 2 · Ch 14 · **Page 334** Open PDF ↗

(1) -4 V — -3 V
(2) -2 V — $+2$ V
(3) 3 V — 5 V
(4) 0 V — -2 V

148. A 250-turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 μA and subjected to a magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is:

APPLIED — Class XII Part 1 · Ch 5 · **Page 139** · U=-m.B, W=2mB Open PDF ↗

(1) 4.55 μJ (2) 2.3 μJ
(3) 1.15 μJ (4) 9.1 μJ

149. If θ_1 and θ_2 be the apparent angles of dip observed in two vertical planes at right angles to each other, then the true angle of dip θ is given by:

DELETED PORTION — Earth's magnetism (angle of dip) · removed from the 2026-27 NCERT; not analysed

(1) $\tan^2\theta = \tan^2\theta_1 + \tan^2\theta_2$
(2) $\cot^2\theta = \cot^2\theta_1 - \cot^2\theta_2$
(3) $\tan^2\theta = \tan^2\theta_1 - \tan^2\theta_2$
(4) $\cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$

150. A potentiometer is an accurate and versatile device to make electrical measurements of E.M.F. because the method involves:

DELETED PORTION — potentiometer · removed from the 2026-27 NCERT; not analysed

(1) potential gradients
(2) a condition of no current flow through the galvanometer
(3) a combination of cells, galvanometer and resistances
(4) cells

151. The ratio of resolving powers of an optical microscope for two wavelengths $\lambda_1 = 4000$ Å and $\lambda_2 = 6000$ Å is:

DELETED PORTION — resolving power of microscope · removed from the 2026-27 NCERT; not analysed

(1) 9:4 **(2)** 3:2 **(3)** 16:81 **(4)** 8:27

152. A thin prism having refracting angle 10° is made of glass of refractive index 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination produces dispersion without deviation. The refracting angle of second prism should be:

APPLIED — Class XII Part 2 · Ch 9 · **Page 238** · thin prism delta=(mu-1)A Open PDF ↗

(1) 6° **(2)** 8° **(3)** 10° **(4)** 4°

153. A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system:

APPLIED — Class XII Part 1 · Ch 2 · **Page 76** · energy stored in a capacitor Open PDF ↗

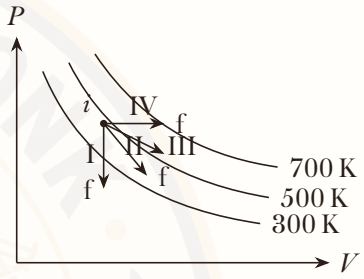
(1) decreases by a factor of 2
(2) remains the same
(3) increases by a factor of 2
(4) increases by a factor of 4

154. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be:

APPLIED — Class XI Part 1 · Ch 3 · **Page 46** · relative velocity Open PDF ↗

(1) $\dfrac{t_1 t_2}{t_2 - t_1}$ **(2)** $\dfrac{t_1 t_2}{t_2 + t_1}$
(3) $t_1 - t_2$ **(4)** $\dfrac{t_1 + t_2}{2}$

155. Thermodynamic processes are indicated in the following diagram.



Match the following:

P.	Process I	a.	Adiabatic
Q.	Process II	b.	Isobaric
R.	Process III	c.	Isochoric
S.	Process IV	d.	Isothermal

APPLIED — Class XI Part 2 · Ch 11 · **Page 232** Open PDF ↗

(1) P→c, Q→a, R→d, S→b
(2) P→c, Q→d, R→b, S→a
(3) P→d, Q→b, R→a, S→c
(4) P→a, Q→c, R→d, S→b

156. One end of string of length l is connected to a particle of mass 'm' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed 'v', the net force on the particle (directed towards center) will be (T represents the tension in the string):

DIRECT — Class XI Part 1 · Ch 4 · **Page 69** · Exercise 4.4 Open PDF ↗

(1) $T + \dfrac{mv^2}{l}$ **(2)** $T - \dfrac{mv^2}{l}$
(3) Zero **(4)** T

157. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?

APPLIED
Class XI Part 2 · Ch 14 · **Page 293** · odd harmonics
Open PDF ↗

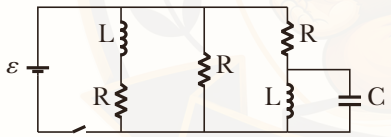
(1) 20 Hz **(2)** 30 Hz
(3) 40 Hz **(4)** 10 Hz

158. The photoelectric threshold wavelength of silver is 3250×10^{-10} m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536×10^{-10} m is: (Given $h = 4.14 \times 10^{-15}$ eV s and $c = 3 \times 10^{8}$ m s^{-1})

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Class XII Part 2 · Ch 11 · **Page 278** Open PDF ↗

(1) $\approx 0.6 \times 10^{6}$ m s^{-1}
(2) $\approx 61 \times 10^{3}$ m s^{-1}
(3) $\approx 0.3 \times 10^{6}$ m s^{-1}
(4) $\approx 6 \times 10^{5}$ m s^{-1}

159. Figure shows a circuit that contains three identical resistors with resistance $R = 9.0\ \Omega$ each, two identical inductors with inductance $L = 2.0$ mH each, and an ideal battery with emf $\varepsilon = 18$ V. The current 'i' through the battery just after the switch closed is,



APPLIED
Class XII Part 1 · Ch 6 · **Page 167** · inductor opposes sudden change Open PDF ↗

(1) 0.2 A **(2)** 2 A
(3) 0 ampere **(4)** 2 mA

160. A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be:

APPLIED
Class XI Part 2 · Ch 10 · **Page 219** · Stefan-Boltzmann law Open PDF ↗

(1) 450 **(2)** 1000 **(3)** 1800 **(4)** 225

161. A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

DIRECT
Class XI Part 1 · Ch 6 · **Page 126** · Exercise 6.13
Open PDF ↗

(1) 0.25 rad/s^2 **(2)** 25 rad/s^2
(3) 5 m/s^2 **(4)** 25 m/s^2

162. Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2. They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is:

APPLIED
Class XI Part 1 · Ch 6 · **Page 122** Open PDF ↗

(1) $\frac{1}{4}I(\omega_1 - \omega_2)^2$ **(2)** $I(\omega_1 - \omega_2)^2$
(3) $\frac{1}{8}(\omega_1 - \omega_2)^2$ **(4)** $\frac{1}{2}I(\omega_1 + \omega_2)^2$

163. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:

APPLIED
Class XI Part 2 · Ch 12 · **Page 252** · equipartition
Open PDF ↗

(1) $15RT$ **(2)** $9RT$
(3) $11RT$ **(4)** $4RT$

164. The bulk modulus of a spherical object is 'B'. If it is subjected to uniform pressure 'p', the fractional decrease in radius is:

APPLIED
Class XI Part 2 · Ch 8 · **Page 173** · bulk modulus
Open PDF ↗

(1) $\frac{B}{3p}$ **(2)** $\frac{3p}{B}$ **(3)** $\frac{p}{3B}$ **(4)** $\frac{p}{B}$

165. Which of the following statements are correct?

(a) Centre of mass of a body always coincides with the centre of gravity of the body.

(b) Centre of mass of a body is the point at which the total gravitational torque on the body is zero.

(c) A couple on a body produce both translational and rotational motion in a body.

(d) Mechanical advantage greater than one means that small effort can be used to lift a large load.

DIRECT

Class XI Part 1 · Ch 6 · **Page 112** · centre of gravity and mechanical advantage Open PDF ↗

(1) (a) and (b) **(2)** (b) and (c)

(3) (c) and (d) **(4)** (b) and (d)

166. The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then:

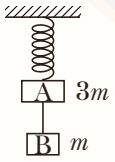
DIRECT

Class XI Part 1 · Ch 7 · **Page 133** · g(h) and g(d) Open PDF ↗

(1) $d = 1$ km **(2)** $d = \frac{3}{2}$ km

(3) $d = 2$ km **(4)** $d = \frac{1}{2}$ km

167. Two blocks A and B of masses $3m$ and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively:



APPLIED

Class XI Part 1 · Ch 4 · **Page 59** · spring force Open PDF ↗

(1) $\frac{g}{3}, g$ **(2)** g, g **(3)** $\frac{g}{3}, \frac{g}{3}$ **(4)** $g, \frac{g}{3}$

168. The ratio of wavelengths of the last line of Balmer series and the last line of Lyman series is:

APPLIED

Class XII Part 2 · Ch 12 · **Page 291** · series limits Open PDF ↗

(1) 1 **(2)** 4 **(3)** 0.5 **(4)** 2

169. A beam of light from a source L is incident normally on a plane mirror fixed at a certain distance x from the source. The beam is reflected back as a spot on a scale placed just above the source L. When the mirror is rotated through a small angle θ, the spot of the light is found to move through a distance y on the scale. The angle θ is given by:

APPLIED

Class XII Part 2 · Ch 9 · **Page 222** · laws of reflection Open PDF ↗

(1) $\frac{y}{x}$ **(2)** $\frac{x}{2y}$ **(3)** $\frac{x}{y}$ **(4)** $\frac{y}{2x}$

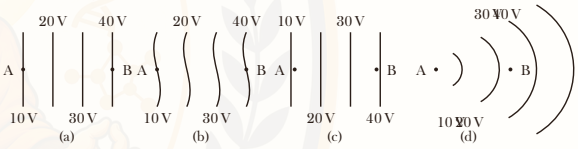
170. A spring of force constant k is cut into lengths of ratio 1:2:3. They are connected in series and the new force constant is k'. Then they are connected in parallel and force constant is k''. Then $k' : k''$ is:

APPLIED

Class XI Part 2 · Ch 13 · **Page 267** · Example 13.6 setting Open PDF ↗

(1) 1:9 **(2)** 1:11 **(3)** 1:14 **(4)** 1:6

171. The diagrams below show regions of equipotentials.



A positive charge is moved from A to B in each diagram.

APPLIED

Class XII Part 1 · Ch 2 · **Page 48** · work independent of path Open PDF ↗

(1) In all the four cases the work done is the same.

(2) Minimum work is required to move q in figure (a).

(3) Maximum work is required to move q in figure (b).

(4) Maximum work is required to move q in figure (c).

172. Suppose the charge of a proton and an electron differ slightly. One of them is $-e$, the other is $(e + \Delta e)$. If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance d (much greater than atomic size) apart is zero, then Δe is of the order of [Given mass of hydrogen $m_h = 1.67 \times 10^{-27}$ kg]

APPLIED Class XII Part 1 · Ch 1 · **Page 9** Open PDF ↗

(1) 10^{-23} C **(2)** 10^{-37} C
(3) 10^{-47} C **(4)** 10^{-20} C

173. A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]:

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ **(2)** $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$
(3) $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$ **(4)** $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

174. Two Polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1. The intensity of transmitted light through P_2 is:

APPLIED Class XII Part 2 · Ch 10 · **Page 271** Open PDF ↗

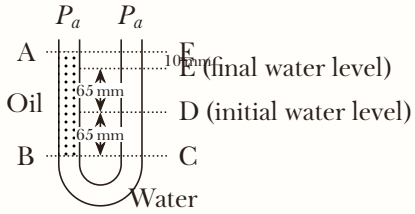
(1) $\frac{I_0}{4}$ **(2)** $\frac{I_0}{8}$ **(3)** $\frac{I_0}{16}$ **(4)** $\frac{I_0}{2}$

175. A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is:

APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $\frac{\sqrt{5}}{2\pi}$ **(2)** $\frac{4\pi}{\sqrt{5}}$ **(3)** $\frac{2\pi}{\sqrt{3}}$ **(4)** $\frac{\sqrt{5}}{\pi}$

176. A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is:



APPLIED Class XI Part 2 · Ch 9 · **Page 184** · equal pressure at the same level Open PDF ↗

(1) 425 kg m^{-3} **(2)** 800 kg m^{-3}
(3) 928 kg m^{-3} **(4)** 650 kg m^{-3}

177. Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take 'g' constant with a value 10 m/s^2. The work done by the (i) gravitational force and the (ii) resistive force of air is:

DIRECT Class XI Part 1 · Ch 5 · **Page 73** · Example 5.2 Open PDF ↗

(1) (i) 1.25 J (ii) −8.25 J
(2) (i) 100 J (ii) 8.75 J
(3) (i) 10 J (ii) −8.75 J
(4) (i) −10 J (ii) −8.25 J

178. The x and y coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$ respectively, where x and y are in meters and t in seconds. The acceleration of the particle at $t = 2$ s is:

APPLIED Class XI Part 1 · Ch 3 · **Page 37** Open PDF ↗

(1) 5 m/s^2 **(2)** −4 m/s^2
(3) −8 m/s^2 **(4)** 0

179. A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2\ \Omega$, the total charge flowing through the coil during this time is:

APPLIED

Class XII Part 1 · Ch 6 · **Page 157** · Q = N delta-phi / R Open PDF ↗

(1) 16 μC **(2)** 32 μC
(3) 16π μC **(4)** 32π μC

180. A carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is:

APPLIED

Class XI Part 2 · Ch 11 · **Page 236** · refrigerator Open PDF ↗

(1) 90 J **(2)** 99 J **(3)** 100 J **(4)** 1 J

NEET 2017 · Answer Key

136 (1)	**137** (3)	**138** (2)†	**139** (3)	**140** (1)	**141** (1)*†	**142** (2)†	**143** (2)	**144** (3)†	**145** (1)
146 (4)	**147** (4)	**148** (4)	**149** (4)†	**150** (2)†	**151** (2)†	**152** (1)	**153** (1)	**154** (2)	**155** (1)
156 (4)	**157** (1)	**158** (1)*	**159** (2)*	**160** (3)	**161** (2)	**162** (1)	**163** (3)	**164** (3)	**165** (4)
166 (3)	**167** (1)	**168** (2)	**169** (4)	**170** (2)	**171** (1)	**172** (2)	**173** (4)	**174** (2)	**175** (2)
176 (3)	**177** (3)	**178** (2)	**179** (2)	**180** (1)					

Answers solved by the editor (the official final key of this year could not be obtained from the NTA/CBSE websites). (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). *: see note.

Q141. $N_B/N_A = e^{7\lambda t}$, so the ratio is e (not $1/e$) at $t = 1/(7\lambda)$; the intended answer is (1).

Q158. $K_{max} \approx 1.08$ eV gives $v \approx 6.2\times10^5$ m/s; options (1) and (4) are the same value.

Q159. The capacitor in the printed figure is not mentioned in the question; with only the three resistors and two inductors, just the middle resistor carries current at $t = 0$: $i = 18/9 = 2$ A.

NEET 2017 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 38 questions are in the present syllabus; percentages are of these.

Direct from NCERT	5	13%
text, summary, table or figure	2	
NCERT solved Example	1	
NCERT Exercise	2	
Applied (concept in NCERT)	**33**	87%
Analysed	38	
Deleted portion (not analysed)	7	
Total	**45**	

Direct: Q 156, 161, 165, 166, 177.
From NCERT Examples: Q 177 (Example 5.2).
From NCERT Exercises: Q 156 (Exercise 4.4), 161 (Exercise 6.13).

2. Difficulty level

Easy	17
Moderate	18
Hard	3

Hard: Q 172, 173, 179. The 2017 paper was moderate and long: most questions need a formula and one or two steps, and seven were on topics now deleted. Class XI: 21 questions, Class XII: 17 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Units & Measurement	XI	1	0	1
Motion in a Plane	XI	2	0	2
Laws of Motion	XI	2	1	1
Work, Energy & Power	XI	1	1	0
Rotational Motion	XI	3	2	1
Gravitation	XI	2	1	1
Mech. Prop. of Solids	XI	1	0	1
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	2	0	2
Thermodynamics	XI	2	0	2
Kinetic Theory	XI	1	0	1
Oscillations	XI	2	0	2
Waves	XI	1	0	1
Electric Charges & Fields	XII	1	0	1
Electrostatic Potential & Capacitance	XII	2	0	2
Current Electricity	XII	1	0	1
Moving Charges & Magnetism	XII	1	0	1
Magnetism & Matter	XII	1	0	1
Electromagnetic Induction	XII	2	0	2
Electromagnetic Waves	XII	1	0	1
Ray Optics	XII	2	0	2
Wave Optics	XII	2	0	2
Dual Nature	XII	2	0	2
Atoms	XII	1	0	1
Semiconductor Electronics	XII	1	0	1

4. Teaching takeaways

- Exercise 4.4 (string on a peg), Exercise 6.13 (rope on a hollow cylinder) and Example 5.2 (raindrop) were asked almost word for word.
- Seven questions were on deleted portions (transistor amplifier, decay constants, logic gates, Doppler effect, angle of dip, potentiometer, resolving power).
- 33 of the 38 in-syllabus questions are applied: they use an NCERT formula in a new setting (springs cut in a ratio, U-tube with oil, coil in a solenoid).
- Two questions have printing slips (Q158 options (1) and (4) are equal; Q159 figure has an extra capacitor); see the key notes.

5. Deleted portions

7 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: transistor (CE amplifier) (Q 138); radioactive decay law (decay constant) (Q 141); digital electronics (logic gates) (Q 142); Doppler effect (Q 144); Earth's magnetism (angle of dip) (Q 149); potentiometer (Q 150); resolving power of microscope (Q 151).

NEET 2016
Physics Question Paper

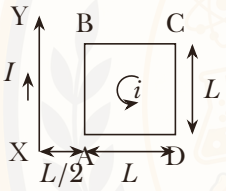
NEET Phase I · 1 May 2016 · Test Booklet Code A · Physics Q 1–45 · 180 Marks

1. From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre?

DELETED PORTION — parallel axis theorem · removed from the 2026-27 NCERT; not analysed

(1) $15MR^2/32$ **(2)** $13MR^2/32$
(3) $11MR^2/32$ **(4)** $9MR^2/32$

2. A square loop ABCD carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be:



APPLIED — Class XII Part 1 · Ch 4 · **Page 123** · force between parallel currents Open PDF ↗

(1) $\dfrac{2\mu_0 Ii}{3\pi}$ **(2)** $\dfrac{\mu_0 Ii}{2\pi}$
(3) $\dfrac{2\mu_0 IiL}{3\pi}$ **(4)** $\dfrac{\mu_0 IiL}{2\pi}$

3. The magnetic susceptibility is negative for:

DIRECT — Class XII Part 1 · Ch 5 · **Page 147** · diamagnetic: chi negative Open PDF ↗

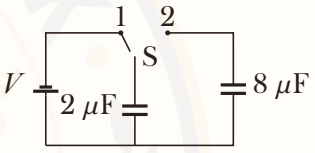
(1) diamagnetic material only
(2) paramagnetic material only
(3) ferromagnetic material only
(4) paramagnetic and ferromagnetic materials

4. A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 m s^{-1}. Then, the frequency of sound that the observer hears in the echo reflected from the cliff is: (Take velocity of sound in air = 330 m s^{-1})

DELETED PORTION — Doppler effect · removed from the 2026-27 NCERT; not analysed

(1) 765 Hz **(2)** 800 Hz
(3) 838 Hz **(4)** 885 Hz

5.



A capacitor of 2 μF is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its stored energy dissipated is:

APPLIED — Class XII Part 1 · Ch 2 · **Page 76** · energy shared between capacitors Open PDF ↗

(1) 0% **(2)** 20% **(3)** 75% **(4)** 80%

6. In a diffraction pattern due to a single slit of width 'a', the first minimum is observed at an angle 30° when light of wavelength 5000 Å is incident on the slit. The first secondary maximum is observed at an angle of:

APPLIED — Class XII Part 2 · Ch 10 · **Page 267** · single slit Open PDF ↗

(1) $\sin^{-1}\left(\dfrac{1}{4}\right)$ **(2)** $\sin^{-1}\left(\dfrac{2}{3}\right)$
(3) $\sin^{-1}\left(\dfrac{1}{2}\right)$ **(4)** $\sin^{-1}\left(\dfrac{3}{4}\right)$

7. At what height from the surface of earth the gravitation potential and the value of g are $-5.4 \times 10^7\,\mathrm{J\,kg^{-2}}$ and $6.0\,\mathrm{m\,s^{-2}}$ respectively? Take the radius of earth as 6400 km:

APPLIED Class XI Part 1 · Ch 7 · **Page 134** · V=-GM/r, g=GM/r^2 Open PDF ↗

(1) 2600 km **(2)** 1600 km
(3) 1400 km **(4)** 2000 km

8. Out of the following options which one can be used to produce a propagating electromagnetic wave?

DIRECT Class XII Part 1 · Ch 8 · **Page 205** · accelerated charges radiate Open PDF ↗

(1) A charge moving at constant velocity
(2) A stationary charge
(3) A chargeless particle
(4) An accelerating charge

9. Two identical charged spheres suspended from a common point by two massless strings of lengths l, are initially at a distance d ($d \ll l$) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity v. Then v varies as a function of the distance x between the spheres, as:

APPLIED Class XII Part 1 · Ch 1 · **Page 9** Open PDF ↗

(1) $v \propto x^{1/2}$ **(2)** $v \propto x$
(3) $v \propto x^{-1/2}$ **(4)** $v \propto x^{-1}$

10. A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2. The ratio λ_2/λ_1 is:

APPLIED Class XI Part 2 · Ch 14 · **Page 285** · v=sqrt(T/mu) Open PDF ↗

(1) $\sqrt{\dfrac{m_1}{m_2}}$ **(2)** $\sqrt{\dfrac{m_1+m_2}{m_2}}$
(3) $\sqrt{\dfrac{m_2}{m_1}}$ **(4)** $\sqrt{\dfrac{m_1+m_2}{m_1}}$

11. A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is: (Take 1 cal = 4.2 Joules)

APPLIED Class XI Part 2 · Ch 11 · **Page 236** · refrigerator Open PDF ↗

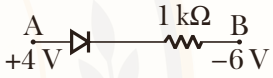
(1) 2.365 W **(2)** 23.65 W
(3) 236.5 W **(4)** 2365 W

12. An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is:

APPLIED Class XI Part 2 · Ch 14 · **Page 293** · odd harmonics Open PDF ↗

(1) 66.7 cm **(2)** 100 cm
(3) 150 cm **(4)** 200 cm

13. Consider the junction diode as ideal. The value of current flowing through AB is:



APPLIED Class XII Part 2 · Ch 14 · **Page 334** · forward bias Open PDF ↗

(1) 0 A **(2)** 10^{-2} A
(3) 10^{-1} A **(4)** 10^{-3} A

14. The charge flowing through a resistance R varies with time t as $Q = at - bt^2$, where a and b are positive constants. The total heat produced in R is:

APPLIED Class XII Part 1 · Ch 3 · **Page 91** · P=I^2R Open PDF ↗

(1) $\dfrac{a^3R}{6b}$ **(2)** $\dfrac{a^3R}{3b}$ **(3)** $\dfrac{a^3R}{2b}$ **(4)** $\dfrac{a^3R}{b}$

15. A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1, at wavelength 500 nm is U_2 and that at 1000 nm is U_3. Wien's constant, $b = 2.88 \times 10^6$ nm K. Which of the following is correct?

APPLIED Class XI Part 2 · Ch 10 · **Page 218** · Wien's law Open PDF ↗

(1) $U_1 = 0$ **(2)** $U_3 = 0$
(3) $U_1 > U_2$ **(4)** $U_2 > U_1$

16. Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good?

APPLIED — Class XI Part 2 · Ch 10 · **Page 207** · linear expansion Open PDF ↗

(1) $\alpha_1 l_2 = \alpha_2 l_1$ **(2)** $\alpha_1 l_2^2 = \alpha_2 l_1^2$
(3) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ **(4)** $\alpha_1 l_1 = \alpha_2 l_2$

17. A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800 Ω is connected in the collector circuit and the voltage drop across it is 0.8 V. If the current amplification factor is 0.96 and the input resistance of the circuit is 192 Ω, the voltage gain and the power gain of the amplifier will respectively be:

DELETED PORTION — transistor (CE amplifier) · removed from the 2026-27 NCERT; not analysed

(1) 4, 3.84 **(2)** 3.69, 3.84
(3) 4, 4 **(4)** 4, 3.69

18. The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is $d = 5\lambda$, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance $D = 10d$?

APPLIED — Class XII Part 2 · Ch 10 · **Page 266** Open PDF ↗

(1) I_0 **(2)** $\frac{I_0}{4}$ **(3)** $\frac{3}{4}I_0$ **(4)** $\frac{I_0}{2}$

19. A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad s^{-2}. Its net acceleration in m s^{-2} at the end of 2.0 s is approximately:

APPLIED — Class XI Part 1 · Ch 6 · **Page 117** · rotational kinematics, a_t and a_c Open PDF ↗

(1) 8.0 **(2)** 7.0 **(3)** 6.0 **(4)** 3.0

20. An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is: (c being velocity of light)

APPLIED — Class XII Part 2 · Ch 11 · **Page 285** Open PDF ↗

(1) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$ **(2)** $\left(\frac{E}{2m}\right)^{1/2}$
(3) $c(2mE)^{1/2}$ **(4)** $\frac{1}{c}\left(\frac{2m}{E}\right)^{1/2}$

21. A disk and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?

DELETED PORTION — kinetic energy of rolling · removed from the 2026-27 NCERT; not analysed

(1) Disk
(2) Sphere
(3) Both reach at the same time
(4) Depends on their masses

22. The angle of incidence for a ray of light at a refracting surface of a prism is 45°. The angle of prism is 60°. If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are:

APPLIED — Class XII Part 2 · Ch 9 · **Page 240** · minimum deviation Open PDF ↗

(1) 45°; $\frac{1}{\sqrt{2}}$ **(2)** 30°; $\sqrt{2}$
(3) 45°; $\sqrt{2}$ **(4)** 30°; $\frac{1}{\sqrt{2}}$

23. When an α-particle of mass 'm' moving with velocity 'v' bombards on a heavy nucleus of charge 'Ze', its distance of closest approach from the nucleus depends on m as:

APPLIED — Class XII Part 2 · Ch 12 · **Page 295** · distance of closest approach Open PDF ↗

(1) $\frac{1}{m}$ **(2)** $\frac{1}{\sqrt{m}}$ **(3)** $\frac{1}{m^2}$ **(4)** m

24. A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to 8×10^{-4} J by the end of the second revolution after the beginning of the motion?

APPLIED Class XI Part 1 · Ch 3 · **Page 42** Open PDF ↗

(1) $0.1\ \text{m/s}^2$ **(2)** $0.15\ \text{m/s}^2$
(3) $0.18\ \text{m/s}^2$ **(4)** $0.2\ \text{m/s}^2$

25. The molecules of a given mass of a gas have r.m.s. velocity of 200 m s^{-1} at 27°C and 1.0×10^5 N m^{-2} pressure. When the temperature and pressure of the gas are respectively, 127°C and 0.05×10^5 N m^{-2}, the r.m.s. velocity of its molecules in m s^{-1} is:

APPLIED Class XI Part 2 · Ch 12 · **Page 252** · v_rms proportional to sqrt(T) Open PDF ↗

(1) $100\sqrt{2}$ **(2)** $\dfrac{400}{\sqrt{3}}$
(3) $\dfrac{100\sqrt{2}}{3}$ **(4)** $\dfrac{100}{3}$

26. A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B', at radial distances $\dfrac{a}{2}$ and $2a$ respectively, from the axis of the wire is:

DIRECT Class XII Part 1 · Ch 4 · **Page 119** · Example 4.7 Open PDF ↗

(1) $\dfrac{1}{4}$ **(2)** $\dfrac{1}{2}$ **(3)** 1 **(4)** 4

27. A particle moves so that its position vector is given by $\vec{r} = \cos\omega t\,\hat{x} + \sin\omega t\,\hat{y}$, where ω is a constant. Which of the following is true?

APPLIED Class XI Part 1 · Ch 3 · **Page 42** · uniform circular motion Open PDF ↗

(1) Velocity and acceleration both are perpendicular to $\vec{r}$.
(2) Velocity and acceleration both are parallel to $\vec{r}$.
(3) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin.
(4) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin.

28. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?

DIRECT Class XI Part 1 · Ch 5 · **Page 79** · Example 5.7 Open PDF ↗

(1) $\sqrt{gR}$ **(2)** $\sqrt{2gR}$ **(3)** $\sqrt{3gR}$ **(4)** $\sqrt{5gR}$

29. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\dfrac{V}{4}$. The threshold wavelength for the metallic surface is:

APPLIED Class XII Part 2 · Ch 11 · **Page 278** Open PDF ↗

(1) 4λ **(2)** 5λ **(3)** $\dfrac{5}{2}\lambda$ **(4)** 3λ

30. A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then:

APPLIED Class XI Part 2 · Ch 11 · **Page 233** · adiabatic curve is steeper Open PDF ↗

(1) Compressing the gas isothermally will require more work to be done.
(2) Compressing the gas through adiabatic process will require more work to be done.
(3) Compressing the gas isothermally or adiabatically will require the same amount of work.
(4) Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.

31. A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two cells are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the positive end of the wire in the two cases. The ratio of emf's is:

DELETED PORTION potentiometer · removed from the 2026-27 NCERT; not analysed

(1) 5:1 **(2)** 5:4 **(3)** 3:4 **(4)** 3:2

32. A astronomical telescope has objective and eye-piece of focal lengths 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance:

APPLIED Class XII Part 2 · Ch 9 · **Page 243** Open PDF ↗

(1) 37.3 cm **(2)** 46.0 cm
(3) 50.0 cm **(4)** 54.0 cm

33. Two non-mixing liquids of densities ρ and $n\rho$ ($n > 1$) are put in a container. The height of each liquid is h. A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL ($p < 1$) in the denser liquid. The density d is equal to:

APPLIED Class XI Part 2 · Ch 9 · **Page 184** · upthrust from both liquids Open PDF ↗

(1) $\{1 + (n + 1)p\}\rho$ **(2)** $\{2 + (n + 1)p\}\rho$
(3) $\{2 + (n - 1)p\}\rho$ **(4)** $\{1 + (n - 1)p\}\rho$

34. To get output 1 for the following circuit, the correct choice for the input is:



DELETED PORTION digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) A = 0, B = 1, C = 0
(2) A = 1, B = 0, C = 0
(3) A = 1, B = 1, C = 0
(4) A = 1, B = 0, C = 1

35. A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is: [Latent heat of ice is 3.4×10^5 J/kg and $g = 10$ N/kg]

APPLIED Class XI Part 2 · Ch 10 · **Page 212** · latent heat Open PDF ↗

(1) 34 km **(2)** 544 km
(3) 136 km **(4)** 68 km

36. The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is:

APPLIED Class XI Part 1 · Ch 7 · **Page 136** · v_e=sqrt(2GM/R) Open PDF ↗

(1) 1:2 **(2)** $1 : 2\sqrt{2}$
(3) 1:4 **(4)** $1 : \sqrt{2}$

37. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is:

APPLIED Class XI Part 1 · Ch 3 · **Page 33** Open PDF ↗

(1) 0° **(2)** 90° **(3)** 45° **(4)** 180°

38. Given the value of Rydberg constant is 10^7 m^{-1}, the wave number of the last line of the Balmer series in hydrogen spectrum will be:

APPLIED Class XII Part 2 · Ch 12 · **Page 291** · Balmer formula Open PDF ↗

(1) 0.025×10^4 m^{-1} **(2)** 0.5×10^7 m^{-1}
(3) 0.25×10^7 m^{-1} **(4)** 2.5×10^7 m^{-1}

39. A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F} = (2t\hat{i} + 3t^2\hat{j})$ N, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y axis. What power will be developed by the force at the time t?

APPLIED Class XI Part 1 · Ch 5 · **Page 83** · P=F.v Open PDF ↗

(1) $(2t^2 + 3t^3)$ W **(2)** $(2t^2 + 4t^4)$ W
(3) $(2t^3 + 3t^4)$ W **(4)** $(2t^3 + 3t^5)$ W

40. An inductor 20 mH, a capacitor 50 μF and a resistor 40 Ω are connected in series across a source of emf $V = 10 \sin 340t$. The power loss in A.C. circuit is:

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(1) 0.51 W **(2)** 0.67 W
(3) 0.76 W **(4)** 0.89 W

41. If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is:

APPLIED

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(1) $\frac{3}{2}A + 4B$ **(2)** $3A + 7B$

(3) $\frac{3}{2}A + \frac{7}{3}B$ **(4)** $\frac{A}{2} + \frac{B}{3}$

42. A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is:

DIRECT

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(1) 4 H **(2)** 3 H **(3)** 2 H **(4)** 1 H

43. A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal capacitor C:

DIRECT

Class XII Part 1 · Ch 7 · **Page 183** · average power in a capacitor is zero Open PDF ↗

(1) Current $I(t)$, lags voltage $V(t)$ by 90°.

(2) Over a full cycle the capacitor C does not consume any energy from the voltage source.

(3) Current $I(t)$ is in phase with voltage $V(t)$.

(4) Current $I(t)$ leads voltage $V(t)$ by 180°.

44. Match the corresponding entries of column 1 with column 2. [Where m is the magnification produced by the mirror]

Column 1		*Column 2*	
(A)	$m = -2$	(a)	Convex mirror
(B)	$m = -\frac{1}{2}$	(b)	Concave mirror
(C)	$m = +2$	(c)	Real image
(D)	$m = +\frac{1}{2}$	(d)	Virtual image

APPLIED

Class XII Part 2 · Ch 9 · **Page 226** · sign of magnification Open PDF ↗

(1) A→b and c; B→b and c; C→b and d; D→a and d

(2) A→a and c; B→a and d; C→a and b; D→c and d

(3) A→a and d; B→b and c; C→b and d; D→b and c

(4) A→c and d; B→b and d; C→b and c; D→a and d

45. A car is negotiating a curved road of radius R. The road is banked at an angle θ. The coefficient of friction between the tyres of the car and the road is μ_s. The maximum safe velocity on this road is:

DIRECT

Class XI Part 1 · Ch 4 · **Page 64** · banked road Open PDF ↗

(1) $\sqrt{gR^2 \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

(2) $\sqrt{gR \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

(3) $\sqrt{\frac{g}{R} \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

(4) $\sqrt{\frac{g}{R^2} \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

NEET 2016 · Answer Key

1 (2)†	**2** (1)	**3** (1)	4 (3)†	**5** (4)	**6** (4)	**7** (1)	**8** (4)	**9** (3)	**10** (2)
11 (3)	**12** (3)	**13** (2)	**14** (1)	**15** (4)	**16** (4)	17 (1)†	**18** (4)	**19** (1)	**20** (1)
21 (2)†	**22** (2)	**23** (1)	**24** (1)	**25** (2)	**26** (3)	**27** (3)	**28** (4)	**29** (4)	**30** (2)
31 (4)†	**32** (4)	**33** (4)	34 (4)†	**35** (3)	**36** (2)	**37** (2)	**38** (3)	**39** (4)	**40** (1)*
41 (3)	**42** (4)	**43** (2)	**44** (1)	**45** (2)					

Answers solved by the editor (the official final key of this year could not be obtained from the NTA/CBSE websites). (1)–(4) are the printed options. Grey†: deleted portion (answer given for completeness). *: see note.

Q40. The exact value is about 0.46 W; (1) 0.51 W is the nearest option and is the intended answer.

NEET 2016 · Paper Analysis

1. Direct vs Applied

Checked question by question against the four 2026-27 NCERT books (XI Part 1 & 2, XII Part 1 & 2). Out of 45, 39 questions are in the present syllabus; percentages are of these.

Direct from NCERT	**7**	18%
text, summary, table or figure	5	
NCERT solved Example	2	
NCERT Exercise	0	
Applied (concept in NCERT)	**32**	82%
Analysed	39	
Deleted portion (not analysed)	6	
Total	**45**	

Direct: Q 3, 8, 26, 28, 42, 43, 45.

From NCERT Examples: Q 26 (Example 4.7), 28 (Example 5.7).

2. Difficulty level

Easy	19
Moderate	17
Hard	3

Hard: Q 9, 14, 18. The 2016 paper (Phase I) was moderate to hard: many questions combine two ideas, and six were on topics now deleted. Class XI: 19 questions, Class XII: 20 (analysed questions only). (Difficulty is the editor's assessment.)

3. Chapter-wise distribution

Chapter	Cl.	Qs	Dir	App
Motion in a Straight Line	XI	1	0	1
Motion in a Plane	XI	3	0	3
Laws of Motion	XI	1	1	0
Work, Energy & Power	XI	2	1	1
Rotational Motion	XI	1	0	1
Gravitation	XI	2	0	2
Mech. Prop. of Fluids	XI	1	0	1
Thermal Prop. of Matter	XI	3	0	3
Thermodynamics	XI	2	0	2
Kinetic Theory	XI	1	0	1
Waves	XI	2	0	2
Electric Charges & Fields	XII	1	0	1
Electrostatic Potential & Capacitance	XII	1	0	1
Current Electricity	XII	1	0	1
Moving Charges & Magnetism	XII	2	1	1
Magnetism & Matter	XII	1	1	0
Electromagnetic Induction	XII	1	1	0
Alternating Current	XII	2	1	1
Electromagnetic Waves	XII	1	1	0
Ray Optics	XII	3	0	3
Wave Optics	XII	2	0	2
Dual Nature	XII	2	0	2
Atoms	XII	2	0	2
Semiconductor Electronics	XII	1	0	1

4. Teaching takeaways

- Examples 4.7 (field inside and outside a thick wire) and 5.7 (vertical loop, $\sqrt{5gR}$) gave direct marks.
- Six questions were on deleted portions (parallel axis theorem, Doppler effect, transistor amplifier, rolling, potentiometer, logic gates).
- 32 of the 39 in-syllabus questions are applied; the hard ones (leaking charges, $Q = at - bt^2$, YDSE in front of a slit) need a short derivation.
- Circular motion was asked four times (tangential acceleration, $\vec{r} = \cos\omega t\,\hat{x} + \sin\omega t\,\hat{y}$, vertical loop, banked road).

5. Deleted portions

6 of the 45 questions are on topics removed from the 2026-27 NCERT (rationalised syllabus). They are marked Deleted Portion and are not analysed: parallel axis theorem (Q 1); Doppler effect (Q 4); transistor (CE amplifier) (Q 17); kinetic energy of rolling (Q 21); potentiometer (Q 31); digital electronics (logic gates) (Q 34).

NEET 2016–2026 · Summary

Every question of the eleven papers was checked one by one against the four 2026-27 NCERT Physics books (Class XI Part 1 & 2, Class XII Part 1 & 2). Direct: the answer is stated in the textbook (text, summary, table or figure), or the question is an NCERT solved Example or Exercise (same or changed numbers), or it needs only one printed formula. Applied: the concept is in the textbook but the question needs several steps or a new situation. Deleted portion: the topic was removed from the rationalised syllabus; such questions are not analysed. Percentages are of the analysed questions.

Year	Direct				Applied	Analysed	Deleted	Direct
	Total	Text	Example	Exercise			portion	%
2026	**16**	15	0	1	27	43	2	37%
2025	**11**	9	2	0	30	41	4	27%
2024	**21**	18	2	1	23	44	6	48%
2023	**23**	19	2	2	21	44	6	52%
2022	**29**	22	5	2	19	48	2	60%
2021	**14**	9	5	0	30	44	6	32%
2020	**26**	21	3	2	13	39	6	67%
2019	**20**	17	1	2	20	40	5	50%
2018	**11**	8	2	1	26	37	8	30%
2017	**5**	2	1	2	33	38	7	13%
2016	**7**	5	2	0	32	39	6	18%
Total	**183**	145	25	13	274	457	**58**	40%

Of the 515 questions, **457** are within the present syllabus. Of these, **183** (40%) are direct from the NCERT textbook and 274 (60%) are applications of NCERT concepts. **58** questions (11% of all 515) are from deleted portions; 38 of them are in the papers of 2016–2021, set before the rationalised syllabus, and only 6 in 2025 and 2026.

Deleted portions asked, 2016–2026 (number of questions):

Digital electronics (logic gates) (11);
Brewster's law (4);
Radioactive decay law (half-life, mean life, activity) (4);
Transistor (incl. amplifier) (4);
Alpha- and beta-decay (3);
Parallel axis theorem (3);
Potentiometer (3);
Resistor colour code (3);
Rolling motion (3);
Zener diode (3);
Doppler effect (2);
Earth's magnetism (incl. angle of dip) (2);
Resolving power (telescope, microscope) (2);
Coefficient of restitution (1);
Eddy currents (1);
Force between capacitor plates (1);
Intensity of EM waves (1);
Magnetic field of a finite straight wire (1);
Oscillation period of a magnetic needle (1);
Rainbow (1);
Sharpness of resonance (Q-factor) (1);
Solar cell (1);
Types of errors (1);
Velocity selector (1).

The 15 Most Repeated Topics

Every in-syllabus question of 2016–2026 was given one topic label while the answer keys were solved. The table counts the questions on each topic over the eleven papers (deleted portions left out; ties are ordered by the number of years in which the topic appeared). Years in bold are the years the topic was asked.

#	Topic	NCERT chapter	Qs	Years	2016 → 2026
1	Simple harmonic motion	Oscillations (XI)	**17**	10	16 **17 18 19 20 21 22 23 24 25 26**
2	Kinetic theory of gases	Kinetic Theory (XI)	**15**	11	**16 17 18 19 20 21 22 23 24 25 26**
3	Rotational dynamics (torque, moment of inertia)	Rotational Motion (XI)	**15**	10	**16 17 18 19 20 21 22 23 24 25** 26
4	Kinematics in one dimension	Motion in a Straight Line (XI)	**13**	10	**16** 17 **18 19 20 21 22 23 24 25 26**
5	Bohr model of hydrogen atom	Atoms (XII)	**12**	10	**16 17 18 19 20** 21 **22 23 24 25 26**
6	Measurement and errors	Units & Measurement (XI)	**12**	9	16 17 **18 19 20 21 22 23 24 25 26**
7	AC circuits (reactance, LCR, phase)	Alternating Current (XII)	**12**	8	**16** 17 18 19 **20 21 22 23 24 25 26**
8	Circular motion	Laws of Motion (XI)	**12**	7	**16 17 18 19** 20 21 22 23 **24 25 26**
9	Work-energy theorem and power	Work, Energy & Power (XI)	**11**	9	**16 17** 18 **19** 20 **21 22 23 24 25 26**
10	Electromagnetic waves (nature, E/B = c)	Electromagnetic Waves (XII)	**10**	10	**16 17 18** 19 **20 21 22 23 24 25 26**
11	Dimensional analysis	Units & Measurement (XI)	**10**	6	16 **17** 18 19 20 **21 22** 23 **24 25 26**
12	de Broglie waves	Dual Nature (XII)	**9**	9	**16 17 18 19 20 21 22** 23 **24 25** 26
13	Photoelectric effect	Dual Nature (XII)	**9**	8	**16 17 18** 19 **20** 21 **22 23** 24 **25 26**
14	Thermodynamic processes	Thermodynamics (XI)	**9**	8	**16 17 18 19 20** 21 **22** 23 **24** 25 **26**
15	Young's double-slit interference	Wave Optics (XII)	**9**	8	**16 17 18 19 20** 21 **22 23 24** 25 26

These 15 topics carry **175** of the 457 in-syllabus questions (38%), about 16 questions in every paper. Nine of them are Class XI topics and 6 are Class XII topics: NEET weighs the two years almost equally.

Questions asked again, almost word for word (in the present syllabus)

2016 Q26, 2019 Q45, 2021 Q26, 2022 Q42	field inside and outside a thick wire (Example 4.7)
2019 Q16, 2020 Q143, 2025 Q32	weight of a body inside or above the earth (Exercises 7.15, 7.16)
2016 Q38, 2017 Q168, 2023 Q11	series limits of the hydrogen spectrum
2021 Q24 ≈ 2022 Q13	Galileo's law of odd numbers (Example 2.5)
2018 Q10 ≈ 2021 Q18	telescope objective: large focal length and large aperture
2018 Q11 ≈ 2019 Q27	kinetic and total energy in a Bohr orbit ($K = -E$)
2020 Q148 ≈ 2022 Q14	centre of mass of two masses on a light rod
2020 Q154 ≈ 2024 Q38	current through a capacitor on 50 Hz ac (Example 7.4)
2020 Q155 ≈ 2026 Q2	mean free path $l = 1/(\sqrt{2}\, n\pi d^2)$

How to Prepare for NEET Physics

Every point below comes from the question-by-question analysis of the eleven papers in this book.

1. What the papers show

- **40%** of the in-syllabus questions (183 of 457) are direct from the NCERT textbook; the rest (274) apply NCERT concepts.
- 25 questions are NCERT solved Examples and 13 are NCERT Exercises, often with the same numbers.
- Class XII gives **52%** of the questions, Class XI 48%.
- Heaviest chapters (questions in 11 years): Current Electricity (30), Ray Optics (27), Electrostatic Potential & Capacitance (26), Dual Nature (22), Moving Charges & Magnetism (22), Rotational Motion (22).
- Difficulty: 230 easy, 212 moderate, only 15 hard. NEET rewards clear concepts and careful arithmetic more than tricks.
- 58 old questions are on deleted topics, but only 6 in 2025 and 2026. The rationalised NCERT is the syllabus now.
- Questions do come back (see the previous page), so the old papers are worth solving fully.

2. What to study from

1. **NCERT Physics 2026-27** (XI Parts 1–2, XII Parts 1–2): the main book. Read every line, including the boxes, tables, figures, Summary and Points to Ponder.
2. **NCERT solved Examples and Exercises**: work all of them by hand. They are the single biggest source of numerical questions.
3. **NCERT Exemplar Physics**: for the applied questions; the MCQ and short-answer sections are enough.
4. **NCERT Lab Manual**: NEET 2020, 2021, 2025 and 2026 asked screw-gauge, vernier and optical-bench questions from it.
5. **NEET papers 2016–2026** (this book), solved under time, with the answer keys.
6. One standard reference for extra MCQ practice only after NCERT is complete. Do not start a new book in the last month.

3. How to work

- Make a one-page formula and fact sheet for each chapter while reading NCERT. Revise from these sheets.
- Keep an error notebook: every wrong or guessed question, the reason, and the right idea. Read it every week.
- Practise the graph, circuit and ray-diagram questions; about one question in 5 has a figure (104 of 515).
- Learn units, dimensions, standard values (e, h, c, m_e, 13.6 eV, 1 u = 931.5 MeV) and the ratio and proportion short cuts; they save time in nearly every paper.
- Check every old question against the present syllabus; skip the ones marked Deleted Portion.

4. Plan of action

Phase	Work
1. Foundation	NCERT chapter by chapter, Class XII (about half the marks): text, Examples, Exercises, formula sheet.
2. Class XI	The same for Class XI (the other half), with more time on Units and Measurement, Rotational Motion, Thermodynamics, Kinetic Theory and Oscillations.
3. Topic drills	Solve this book topic by topic; start with the 15 most repeated topics. Then Exemplar MCQs on the same topics.
4. Full papers	One complete NEET Physics section every two days: 45 questions in 50 minutes. Check with the key and update the error notebook.
5. Last 2 weeks	Only revision: formula sheets, NCERT Summaries, error notebook, and a paper on alternate days. No new material.

5. In the examination hall

- NEET gives about one minute per question over all three subjects; aim to finish Physics in 50 minutes. First pass: answer every direct NCERT question at once (about a third of the paper).
- Second pass: the numericals; leave any question that takes more than two minutes and come back.
- Each correct answer gives 4 marks and each wrong answer costs 1. Answer a doubtful question only when two options can be ruled out.
- Fill the OMR as you go, not at the end.

6. A weekly rhythm

Five days of NCERT and practice, one day of a timed paper, one day of revision and the error notebook. A steady 2–3 hours of Physics a day, kept up for months, beats long sessions before the exam.